Fluctuations of sample eigenvalues in a spiked covariance model with Haar-rotated spikes: a transition at $p^2 \asymp n$

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November 10, 2026

Abstract

We study the outlying eigenvalues of a sample covariance matrix whose population covariance is itself random. The population covariance is $\Sigma_p = A_p + O_p B_p O_p^* + \sigma^2 I_p$, where A_p and B_p are deterministic positive semidefinite matrices of fixed ranks r and s, and where O_p is Haar distributed on the orthogonal group in the real case, and on the unitary group in the complex case. Conditionally on O_p, we observe n independent centered Gaussian vectors with covariance Σ_p, and we build the sample covariance matrix $\widehat{\Sigma}_p$ from these vectors. The fluctuations of the spike eigenvalues of $\widehat{\Sigma}_p$ come from two different sources: the Haar matrix O_p, which displaces the population spikes at the scale $1/p$, and the sampling, which acts at the scale $n^{-1/2}$. Comparing these two scales amounts to comparing p^2 with n, so the ratio which governs the fluctuations is p^2/n, while in the classical spiked model it is the aspect ratio p/n. We give the limiting law of each spike in four regimes where $n, p \to \infty$. When $p/n \to c > 0$, and when $p/n \to 0$ with $p^2/n \to \infty$, the sampling term dominates, the limit does not depend on O_p and is Gaussian, as in the classical spiked model. When $p^2/n \to 0$, the Haar randomness dominates, and the limit is a weighted sum of independent χ_1^2 variables in the real case, and of independent exponential variables in the complex case. When $p^2/n \to \tau \in (0, \infty)$, the two terms are of the same order, and the limit is the convolution of Gaussian and χ_1^2 or Exponential variables. We then apply these results to a linear mixed model with a continuous batch effect, in which the Haar assumption is what makes the separation of the signal and of the batch possible, and we build a test for the value of a single spike.

2020 Mathematics Subject Classification. Primary 60B20; Secondary 60F05, 62H10.

Keywords. Spiked covariance model; Haar random matrices; eigenvalue fluctuations; central limit theorem; low-aspect-ratio regime; spectral inference.

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1 Introduction

Classical spiked covariance estimation theory treats the population covariance matrix as deterministic. In many situations, however, a part of the covariance structure is itself produced by a random mechanism: a latent environment, a nuisance effect, or a random effect drawn once and then shared by the observations. The population covariance then carries deterministic spikes and randomly oriented spikes at the same time. In this paper, we study sample covariance matrices whose population covariance has randomly rotated spikes. In this situation, we have to estimate both population randomness and sampling randomness. To handle such situations, we develop a fluctuation theory for such models.

In our model the classical aspect ratio p/n alone no longer determines the limiting behavior. In the low-aspect-ratio regime $p/n \to 0$, the finer ratio p^2/n, which does not appear in the classical spiked model, turns out to be the relevant parameter.

The reason is that a finite rank spike leaves the limiting spectral distribution unchanged, so that first-order free probability cannot detect it. The appropriate framework is cyclic-monotone independence, introduced in [13], which is designed precisely to describe the discrete part of the spectrum of the model with a Haar-rotated finite rank perturbation. The fluctuations of such discrete eigenvalues were subsequently computed in [12] for polynomials in Haar unitary and finite rank matrices. The present paper extends this line of results to the sample covariance setting and reveals a fluctuation transition governed by p^2/n. Depending on their relative size, the limiting law is Gaussian, non-Gaussian, or the convolution of the two.

Building on this framework, we compute the fluctuations of the outlying eigenvalues in each of the above regimes. A key feature of our setting is that the fluctuation comes not only from the sampling variability of the entries of the data matrix, but also from the randomness of the rotation carried by the spikes. These two sources are asymptotically independent, and their relative weight is governed by p^2/n: they are of the same order only when $p^2/n \to \tau \in (0, \infty)$. This transition cannot be seen in [12], where there is no sampling step, nor in the classical spiked model, where the population matrix is deterministic. Quantifying this transition is the

main contribution of this paper.

1.1 The classical spiked covariance model

In this work we study a spiked covariance model. The spiked covariance model is a small-rank perturbation of a multiple of the identity. It was introduced by Johnstone [19] and has become one of the standard models of high-dimensional statistics; the books [1, 37] give a complete account.

We recall foundational results of random matrix theory, which are the starting point of our work. Let Σ_p be a deterministic $p \times p$ covariance matrix whose eigenvalues are $\sigma^2+\theta_1, \dots, \sigma^2+\theta_K$ and σ^2 with multiplicity $p-K$, where K is fixed and $\theta_1 > \cdots > \theta_K > 0$ are called the spikes. Let $\widehat{\Sigma}_p$ be the sample covariance matrix built from n independent centered real or circular complex Gaussian vectors with covariance Σ_p, and let $\widehat{\lambda}_{p,1} \geq \cdots \geq \widehat{\lambda}_{p,p}$ be its eigenvalues, which we call the sample eigenvalues. Assume the proportional regime $p/n \to c > 0$. Four important results describe the behavior of these eigenvalues in the proportional regime.

First, most of the eigenvalues do not stay near σ^2. The empirical distribution of the eigenvalues of $\widehat{\Sigma}_p$ converges to the Marčenko–Pastur law [29]. When $c \leq 1$ this law is absolutely continuous and spread over the interval $[\sigma^2(1-\sqrt{c})^2, \sigma^2(1+\sqrt{c})^2]$, which is called the bulk.

Second, a spike is visible only if it is large enough. Baik and Silverstein [6] showed that there is a threshold at $\sigma^2\sqrt{c}$ such that for $\ell \leq K$,

$$\widehat{\lambda}_{p,\ell} \xrightarrow{\text{a.s.}} \begin{cases} \phi_c(\theta_\ell) := (\sigma^2+\theta_\ell)\Big(1+\frac{c\,\sigma^2}{\theta_\ell}\Big) & \text{if } \theta_\ell > \sigma^2\sqrt{c}, \\ \sigma^2(1+\sqrt{c})^2 & \text{if } \theta_\ell \leq \sigma^2\sqrt{c}. \end{cases}$$

Notice that $\phi_c(\theta_\ell) > \sigma^2 + \theta_\ell$ so the sample eigenvalue is pushed up if $\theta_\ell > \sigma^2\sqrt{c}$. If not $\widehat{\lambda}_{p,\ell}$ converges to the upper edge of the bulk and carries no information on θ_ℓ. The transition itself was discovered earlier, for complex data, by Baik, Ben Arous, Péché [5]: the threshold is named after them and is called the BBP transition. A spike above the threshold is called supercritical.

Third, the fluctuations of a simple supercritical spike are real Gaussian. Bai and Yao [2, 3] proved that

$$\sqrt{n}\big(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big) \xrightarrow{\text{law}} \mathcal{N}\big(0, \kappa\, V_c(\theta_\ell)\big), \qquad V_c(\theta) := (\sigma^2+\theta)^2\Big(1-\frac{c\,\sigma^4}{\theta^2}\Big),$$

where $\kappa = 2$ for real data and $\kappa = 1$ for complex data. Fourth, the eigenvalues that stay in the bulk behave in a completely different way. The largest of them converges to the upper bulk edge $\sigma^2(1+\sqrt{c})^2$, and it fluctuates at the scale $n^{-2/3}$; its limit law is the Tracy–Widom law, of order one for real Gaussian data [19, 35] and of order two for complex Gaussian data [17, 34].

These four facts hold in the proportional regime, when p and n are comparable. In the disproportional regime, i.e. when $p/n \to 0$, the behavior of eigenvalues is quite different. For instance, in the simplest case where there is no spike and $\Sigma_p = \sigma^2 I_p$, the empirical distribution of the rescaled eigenvalues converges to a semi-circle law, and not to the Marčenko–Pastur law [4]. Moreover, when Σ_p is a spiked covariance, as $p/n \to 0$, all spikes become supercritical and there is no BBP transition. In particular, instead, $\forall \ell \leq K$,

$$\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_\ell,$$

which comes from the standard fact $\|\widehat{\Sigma}_p - \Sigma_p\| = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$: this is for instance a consequence of [21, Corollary 2].

We study both the cases $p/n \to c > 0$ and $p/n \to 0$, and we will show that our model behaves as above in the proportional regime, but not in the disproportional one. The reason

lies in the definition of the model, which we give now. Related central limit theorems for spiked sample eigenvalues without assuming convergence of the aspect ratio were obtained by Liu et al. [28]. For our conditional argument, we establish the required triangular-array result in Proposition 4.6.

1.2 The model studied in this paper

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, and set $\kappa = 2$ if $\mathbb{F} = \mathbb{R}$ and $\kappa = 1$ if $\mathbb{F} = \mathbb{C}$. Let $r, s \geq 1$ be fixed integers, and let

$$\alpha_1, \dots, \alpha_r, \qquad \beta_1, \dots, \beta_s$$

be strictly positive numbers, all distinct. For $p \geq r + s$ let A_p and B_p be deterministic positive semidefinite matrices of rank r and s respectively, whose nonzero eigenvalues are $\alpha_1, \dots, \alpha_r$ for A_p and $\beta_1, \dots, \beta_s$ for B_p, let $\sigma^2 > 0$ be fixed, and let O_p be a random matrix distributed according to the Haar measure on the orthogonal group $O(p)$ if $\mathbb{F} = \mathbb{R}$, and on the unitary group $U(p)$ if $\mathbb{F} = \mathbb{C}$ We define formally these groups in Section 2.2. The population covariance matrix is

$$\Sigma_p = A_p + O_p B_p O_p^* + \sigma^2 I_p, \qquad \sigma^2 > 0, \tag{1}$$

where O_p^* denotes the conjugate transposed matrix of O_p when O_p is unitary, and the transposed matrix when O_p is orthogonal. The Haar measure is defined formally in Section 2.2. Informally, O_p is picked uniformly at random among all orthogonal, respectively unitary, matrices of size p. After O_p is drawn, we observe i.i.d. centered Gaussian vectors with covariance Σ_p conditionally on O_p

$$x_{p,1}, \dots, x_{p,n} \mid O_p \overset{\text{i.i.d.}}{\sim} \mathcal{N}_{\mathbb{F}}(0, \Sigma_p).$$

We write $\mathcal{N}_{\mathbb{F}}$ for the Gaussian law when $\mathbb{F} = \mathbb{R}$, and for the circular complex Gaussian law with covariance Σ when $\mathbb{F} = \mathbb{C}$. From these vectors, we form the sample covariance matrix

$$\widehat{\Sigma}_p = \frac{1}{n} \sum_{k=1}^{n} x_{p,k} x_{p,k}^*.$$

We write $\lambda_{p,\ell}$ for a spike eigenvalue of the population covariance matrix Σ_p, and $\widehat{\lambda}_{p,\ell}$ for the corresponding spike eigenvalue of $\widehat{\Sigma}_p$. We consider these eigenvalues to be sorted, i.e. $\lambda_{p,1} \geq \cdots \geq \lambda_{p,p}$ and $\widehat{\lambda}_{p,1} \geq \cdots \geq \widehat{\lambda}_{p,p}$. We write $\theta_1 > \cdots > \theta_{r+s} > 0$ for the sorted elements of $\{\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s\}$, and we call them the spikes of the model.

Three comments explain why this model is worth studying.

(i) The matrix Σ_p is the sum of two perturbations of $\sigma^2 I_p$, differing by their eigenvectors. Let $a_1, \dots, a_r$ be orthonormal eigenvectors of A_p associated with $\alpha_1, \dots, \alpha_r$, and let $b_1, \dots, b_s$ be orthonormal eigenvectors of B_p associated with $\beta_1, \dots, \beta_s$. The eigenvectors of $O_p B_p O_p^*$ are the vectors $O_p b_1, \dots, O_p b_s$, so they are random and uniformly oriented. Indeed, when O_p is Haar distributed, $(O_p b_1, \dots, O_p b_s)$ is uniformly distributed over all orthonormal families of s vectors of $\mathbb{F}^p$. This is the simplest way to write two small-rank perturbations that are uniformly oriented with respect to each other. Hence, this model is relevant in applications where the data carries an unwanted structure along directions which are unknown, and which have no particular relation with the directions of interest; this happens for instance when the unwanted structure comes from an unpredictable measurement issue during the collection of the data. We discuss such applications in Section 1.5.

(ii) Haar conjugation is a standard source of asymptotic freeness [36]. For sums of randomly rotated matrices, this framework describes limiting empirical spectral distributions through free additive convolution. In our fixed-rank setting, however, the empirical spectral distributions of

A_p and $O_pB_pO_p^*$ both converge to δ_0. The free additive convolution of these limiting measures is therefore also δ_0, which does not determine the finitely many nonzero eigenvalues.

The relevant effect of Haar rotation here is that the two spike subspaces become asymptotically orthogonal. Consequently, the nonzero eigenvalues of $A_p + O_pB_pO_p^*$ converge to the combined set of strengths $\{\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s\}$; see Proposition 2.3. Cyclic-monotone independence [13] provides a framework for studying such discrete spectral limits. For finite-rank Haar unitary models with simple limiting eigenvalues, fluctuation formulas were obtained in [12]. Our work adds a Gaussian sampling step and studies how sampling fluctuations combine with the fluctuations of the population spikes.

(iii) Finally, and this is the point that separates our model from the classical spiked model, the population covariance matrix Σ_p is itself random. In Johnstone's model, Σ_p is a fixed matrix and the only randomness comes from the n samples. Here there are two independent sources of randomness: the matrix O_p, drawn once, and the n sample vectors. The eigenvalue $\lambda_{p,\ell}$ of Σ_p is therefore itself a random variable, and the sample eigenvalue $\widehat{\lambda}_{p,\ell}$ fluctuates around a random target. This is what makes the problem different, and we describe it in the next section.

1.3 Two sources of randomness

By fluctuations, we refer to the asymptotic distribution of the rescaled difference between the sample eigenvalue $\widehat{\lambda}_{p,\ell}$ and its deterministic limit. In the classical spiked model, the exact scaling rate and limiting law were identified by Bai and Yao [2, 3]: the normalization is $\sqrt{n}$ and the limit is Gaussian. In our model, while the classical Gaussian fluctuations remain valid for $p/n \to c > 0$, the regime $p/n \to 0$ yields several distinct fluctuation regimes depending on how fast n grows relative to p. These unusual fluctuations come from the two sources of randomness in the model.

The Haar fluctuation. Even before any sample is drawn, the eigenvalue $\lambda_{p,\ell}$ of Σ_p is random and does not equal $\sigma^2 + \theta_\ell$. The difference depends on the overlaps between the two spike subspaces,

$$c_{ij}^{(p)} = a_i^* O_p b_j, \qquad 1 \le i \le r, \quad 1 \le j \le s.$$

Because O_p is Haar distributed, O_pb_j is uniformly distributed on the unit sphere of $\mathbb{F}^p$, so $\mathbb{E}[(O_pb_j)(O_pb_j)^*] = I_p/p$ by invariance. Hence $c_{ij}^{(p)}$ has mean zero and second moment $1/p$, and it is of size $p^{-1/2}$.

The spike eigenvalues of Σ_p depend on O_p only through these rs overlaps. Indeed, $A_p + O_pB_pO_p^*$ has rank $r+s$, and its range is spanned by the $r+s$ vectors $a_1, \dots, a_r, O_pb_1, \dots, O_pb_s$, which are almost surely linearly independent because of O_p being Haar. Taking these vectors as a basis of the range, the matrix of the restriction has the α_i and the β_j on its diagonal, and entries proportional to the $c_{ij}^{(p)}$ outside of it; see Collins et al. [12]. If all the overlaps were zero, this matrix would be $\mathrm{diag}(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s)$, whose eigenvalues are the θ_ℓ, and they are simple because the θ_ℓ are distinct. A simple eigenvalue is an analytic function of the entries of a matrix, so each $\lambda_{p,\ell}$ is an analytic function of the overlaps in a neighbourhood of the origin. This function has no linear term. Indeed, b_j is only defined up to a sign in the real case, and up to a phase in the complex case, and this change multiplies the j-th column of $(c_{ij}^{(p)})_{ij}$ by -1, respectively by $\mathrm{e}^{\mathrm{i}\varphi}$, while it changes neither B_p nor Σ_p. Its constant term is its value at zero overlap, that is, $\sigma^2 + \theta_\ell$. So the leading term of $\lambda_{p,\ell} - \sigma^2 - \theta_\ell$ is quadratic in the overlaps, and it is of size $1/p$ and not $p^{-1/2}$, as each overlap is of size $p^{-1/2}$.

Proposition 2.3, which is a specialization of the result of Collins et al. [12] and an extension

to the orthogonal case, characterizes the fluctuation. For instance, when $\theta_\ell = \alpha_i$,

$$p\big(\lambda_{p,\ell} - \sigma^2 - \theta_\ell\big) \xrightarrow{\text{law}} W_\ell = \sum_{j=1}^{s} \frac{\alpha_i \beta_j}{\alpha_i - \beta_j} \, |g_{ij}|^2, \tag{2}$$

where the g_{ij} are the limits in law of the $\sqrt{p}\, c_{ij}^{(p)}$, and are independent standard Gaussian variables, real or circular complex. In particular the $|g_{ij}|^2$ are i.i.d. χ_1^2 variables if $\mathbb{F} = \mathbb{R}$, and i.i.d. exponential Exp(1) variables if $\mathbb{F} = \mathbb{C}$. A similar result holds when $\theta_\ell = \beta_k$ for some $k \leq s$, see Proposition 2.3 for details.

The sample fluctuation. Conditionally on $\mathcal{F}_p = \sigma(O_p)$, the matrix Σ_p is fixed and the only randomness left is the one of the samples. We are then in a spiked model with spikes $\lambda_{p,\ell} - \sigma^2$, and the Bai–Yao CLT [3] presented in Section 1.1 suggests that for any $\ell \leq r+s$, $\widehat{\lambda}_{p,\ell}$ fluctuates at the speed $n^{-1/2}$ around

$$\phi_{p/n}\big(\lambda_{p,\ell} - \sigma^2\big) = \lambda_{p,\ell} + \frac{p}{n} \, \frac{\sigma^2 \lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2},$$

and that the law of the fluctuation is Gaussian. However, let us emphasize that the Bai–Yao CLT is stated for fixed spikes, while here the spike $\lambda_{p,\ell}$ changes with p, so this reasoning is heuristic. We nevertheless follow this heuristic. Writing $\phi_{p/n}(\theta) = \sigma^2 + \theta + p\sigma^2/n + \frac{p\sigma^4}{n\theta}$, one has the identity

$$\phi_{p/n}\big(\lambda_{p,\ell} - \sigma^2\big) - \phi_{p/n}(\theta_\ell) = \big(\lambda_{p,\ell} - \sigma^2 - \theta_\ell\big)\big(1 - \frac{p\,\sigma^4}{n(\lambda_{p,\ell} - \sigma^2)\,\theta_\ell}\big) = \mathrm{O}_{\mathbb{P}}\left(\frac{1}{p}\right), \tag{3}$$

which is a $\mathrm{O}_{\mathbb{P}}(1/p)$ because it corresponds to the non-scaled Haar fluctuation which scales at p by (2), multiplied by a bounded term when $\ell \leq r+s$. Therefore we can write the informal, heuristic decomposition

$$\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell) = \underbrace{\Big(\widehat{\lambda}_{p,\ell} - \phi_{p/n}\big(\lambda_{p,\ell} - \sigma^2\big)\Big)}_{\mathrm{O}_{\mathbb{P}}(1/\sqrt{n}),\ \text{sample fluctuation}} + \underbrace{\Big(\phi_{p/n}\big(\lambda_{p,\ell} - \sigma^2\big) - \phi_{p/n}(\theta_\ell)\Big)}_{\mathrm{O}_{\mathbb{P}}(1/p),\ \text{Haar fluctuation}}. \tag{4}$$

When $p/n \to c > 0$, the Haar fluctuation is of order $1/p$, hence of order $1/n$, which is negligible compared to the sample fluctuation, of order $1/\sqrt{n}$. In this case the Haar randomness does not impact the limit, and the sample eigenvalues of our model behave as in the Bai–Yao theorem, because the spike $\lambda_{p,\ell} - \sigma^2$ converges fast enough to θ_ℓ. This is Theorem 3.5.

When $p/n \to 0$, the Bai–Yao CLT does not apply, since it assumes $c > 0$. However, we prove in Proposition 4.6 a specialized CLT for a deterministic spiked model with $p/n \to 0$. Heuristically, Proposition 4.6 suggests that $\hat{\lambda}_{p,\ell}$ still fluctuates around $\phi_{p/n}\big(\lambda_{p,\ell} - \sigma^2\big)$, as in the case where $c > 0$. Hence, the identity (3) still holds when $c = 0$ and the two terms keep the same orders. The behaviour of $\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)$ is therefore decided by the size of $1/\sqrt{n}$ compared to $1/p$.

A transition at $p^2 \asymp n$. The two fluctuations have different sizes: the Haar one is of size $1/p$, the sample one of size $1/\sqrt{n}$, and they are independent. To obtain a limit which is neither zero nor divergent, one has to multiply by the inverse of the larger of the two, that is, by $\min(p, \sqrt{n})$. Comparing $1/p$ with $n^{-1/2}$ amounts to comparing p^2 with n, so the relevant parameter is p^2/n and not p/n. We stress this: the ratio p/n does not distinguish the three disproportional regimes, since it tends to zero in all three. p^2/n it the ratio that governs the behavior of fluctuations, and, as we will see, the limiting law is different for $p^2/n \to \infty$, $p^2/n \to \tau \in (0, \infty)$, and $p^2/n \to 0$.

No single normalization covers the three disproportional regimes. If $p^2/n \to 0$ then $\sqrt{n}/p \to \infty$, so $\sqrt{n}(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ diverges in probability. If $p^2/n \to \infty$, the same happens with the

scaling p. Both work only when $p^2/n \to \tau \in (0, \infty)$, and they then differ by the factor $\sqrt{\tau}$. In what follows, we work with the normalization p when τ is finite, because it is the only one of the two which gives a single formula for the whole range $\tau \in [0, \infty)$, and with $\sqrt{n}$ when $p^2/n \to \infty$.

1.4 Main results

We describe the fluctuations of $\widehat{\lambda}_{p,\ell}$, for each fixed spike θ_ℓ, in four regimes of (p, n). Recall from Section 1.1 that

$$\phi_c(\theta) = (\sigma^2 + \theta)\Big(1 + \frac{c\,\sigma^2}{\theta}\Big), \qquad V_c(\theta) = (\sigma^2 + \theta)^2\Big(1 - \frac{c\,\sigma^4}{\theta^2}\Big),$$

so that $\phi_0(\theta) = \sigma^2 + \theta$ and $V_0(\theta) = (\sigma^2 + \theta)^2$. The centering is the same function in the four regimes: it is $\phi_{p/n}(\theta_\ell)$ in all cases. Let $G_\ell \sim \mathcal{N}(0, 1)$ be independent of W_ℓ. The results are collected in Table 1.

Table 1: The four regimes, at a fixed spike θ_ℓ. Here $\kappa = 2$ if $\mathbb{F} = \mathbb{R}$ and $\kappa = 1$ if $\mathbb{F} = \mathbb{C}$; the variable W_ℓ is the Haar fluctuation of Section 1.3 and $G_\ell \sim \mathcal{N}(0, 1)$ is independent of it. In the first row we assume in addition that θ_ℓ is supercritical, that is, $\theta_\ell > \sigma^2\sqrt{c}$; in the three other rows this holds for p large, since the threshold $\sigma^2\sqrt{p/n}$ tends to zero. The word "diverges" means that the absolute value of the scaled fluctuation tends to infinity in probability, so that no subsequence of it converges in distribution; see Proposition 3.6.

Regime	Scaling by $\sqrt{n}$, and limit	Scaling by p, and limit
$p/n \to c > 0$	$\sqrt{n}\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)) \overset{\text{law}}{\to} \sqrt{\kappa V_c(\theta_\ell)}\,G_\ell$	$p\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ diverges
$p/n \to 0$, $p^2/n \to \infty$	$\sqrt{n}\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)) \overset{\text{law}}{\to} \sqrt{\kappa V_0(\theta_\ell)}\,G_\ell$	$p\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ diverges
$p^2/n \to \tau \in (0, \infty)$	$\sqrt{n}\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)) \overset{\text{law}}{\to} \sqrt{\kappa V_0(\theta_\ell)}\,G_\ell + \tau^{-1/2} W_\ell$	$p\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)) \overset{\text{law}}{\to} W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}\,G_\ell$
$p^2/n \to 0$	$\sqrt{n}\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ diverges	$p\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)) \overset{\text{law}}{\to} W_\ell$

The first row covers the proportional regime $p/n \to c > 0$: in this regime, for a supercritical spike θ_ℓ, the classical Bai–Yao CLT holds in our Haar spiked model the same way as in the classical deterministic spiked model of [19]. The second row covers the regime $p/n \to 0$ and $p^2/n \to \infty$; in this regime, $\sqrt{n}\,(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ converges to the same gaussian distribution $\sqrt{\kappa V_0(\theta_\ell)}\,G_\ell$ as in the proportional regime (setting $c = 0$ in the first row yields the same result as in the second row). This is because in both regimes, the Haar fluctuation is of size $1/p$, which is much smaller than the sample fluctuation of size $1/\sqrt{n}$; the Haar matrix leaves no trace in the limit and only the sample fluctuation impacts the limiting distribution. However, we keep the two rows separated because the two regimes differ in terms of proof techniques that we use to show the results of Table 1 and in terms of behavior of the rest of the spectrum. In particular, we show in Theorem 3.5 that the Marcenko–Pastur bulk, the BBP phase transition, the Tracy–Widom edge properties still hold in the proportional regime.

The three last rows cover the three possible cases in the disproportional regime $p/n \to 0$. Each column has one term which does not change, and one term which appears at $p^2/n \to \tau \in (0, \infty)$. Under the scaling by $\sqrt{n}$, the Gaussian term $\sqrt{\kappa V_0(\theta_\ell)}G_\ell$ is the same in the second and the third row, and it does not depend on the rate at which p/n goes to zero; what appears in the third row is the Haar fluctuation W_ℓ, scaled by $\tau^{-1/2}$. Under the scaling by p, the variable W_ℓ is the same in the third and the fourth row, and what appears in the third row is the Gaussian term, scaled by $\sqrt{\tau}$. So the mixed regime $p^2/n \to \tau \in (0, \infty)$ is the only case where the two fluctuations of Section 1.3 have the same size, and the only one where both appear in the limit.

Notice that the limit law in the mixed regime is a convolution of gaussian and exponential (if $\mathbb{F} = \mathbb{C}$) or chi-squared (if $\mathbb{F} = \mathbb{R}$) variables. Indeed, the variable W_ℓ comes from the Haar

matrix, the variable G_ℓ comes from the sample vectors, these two sources are independent before the limit, and they stay independent in the limit.

Each column also interpolates towards its neighbour: under the scaling by p, letting $\tau \to 0$ in the third row gives the fourth row, and under the scaling by $\sqrt{n}$, letting $\tau \to \infty$ makes the coefficient $\tau^{-1/2}$ vanish and gives the second row. We come back to this in Section 3.4. Let us emphasize that the fourth row is the case $\tau = 0$ of the third, but we display it separately for two reasons: the limiting law is different, since it loses its Gaussian part and reduces to W_ℓ alone; and the scaling by $\sqrt{n}$, which still gives a limit in the third row, diverges here, so the two rows do not merge under that scaling.

We state these results formally in Section 3, once the model and the labeling of the eigenvalues have been set up in Section 2, and we prove them in Section 4 and in the appendices.

1.5 An application: linear mixed models with a continuous batch effect

Our model describes a real-world problem and allows us to relax a constraint in linear mixed models. Linear mixed models (LMM) are used when the data come in batches [24, 32], i.e., in groups of observations which share something: samples processed on the same day, measurements taken during the same time window,... One observes n vectors of $\mathbb{F}^p$, split into G batches, and writes each sample k as

$$x_{p,k} = s_{p,k} + m_{p,z_k} + \varepsilon_{p,k},$$

where $z_k \in [G]$ is the batch of sample k, and the three terms are independent centered Gaussian vectors: the effect of interest $s_{p,k} \sim \mathcal{N}(0, A_p)$, drawn independently for each sample; the batch effect $m_{p,g} \sim \mathcal{N}(0, \Gamma_p)$, drawn once per batch and shared by all the samples of that batch; and the measurement noise $\varepsilon_{p,k} \sim \mathcal{N}(0, \sigma^2 I_p)$. The eigenvalues of A_p and of Γ_p are what one wants to estimate. The covariance of one sample is $A_p + \Gamma_p + \sigma^2 I_p$, so it does not separate A_p from Γ_p. What makes the separation possible is that two samples $k \neq k'$ of the same batch share the same batch effect, while their other terms are independent, which gives

$$\mathbb{E}\big[x_{p,k} x_{p,k'}^*\big] = \Gamma_p.$$

One therefore estimates Γ_p by averaging $x_{p,k} x_{p,k'}^*$ over the pairs of samples in the same batch and recovers A_p by subtraction.

A limit of the classical LMM is that it requires batches of size at least two, which does not hold when the batch effect varies slightly from one sample to the next instead of being shared by a group: for instance a slow drift of an instrument between two measurements. We then call the batch effect continuous, each batch is of size one, and there is no pair to average over: the only available object is the sample covariance matrix $\widehat{\Sigma}_p$, which estimates the sum $A_p + \Gamma_p + \sigma^2 I_p$, so A_p and Γ_p cannot be recovered.

Our model keeps the covariance structure of the classical LMM, allows a single point per batch, and adds in return one assumption on Γ_p: its eigenvectors have no preferred orientation, and are unrelated to those of A_p. As explained in Section 1.2, this is the Haar assumption, that is, $\Gamma_p = O_p B_p O_p^*$, and the model becomes (1). It is what makes the separation of A_p and Γ_p possible again: the eigenvectors of A_p and Γ_p are asymptotically orthogonal, and the spikes of Σ_p are close to the union of the α_i and the β_j rather than to a mixture of them. In practice, this assumption requires that the batch effect be caused by a mechanism that is unrelated to the effect of interest, and it fails when the two are confounded.

Batch effects of this kind are well documented in genomics [27, 18], where $x_{p,k}$ collects the expression levels of p genes for one biological sample. Typically, $p \approx 10^3$–10^4 and $n \approx 10^2$, so $p/n \to c > 0$ in this setup. In industrial quality control, $x_{p,k}$ collects p measured characteristics of one manufactured item, and the batch effect is the distortion added by the measuring instrument, which changes from one item to another. Typically, $p \approx 10^1$ and $n \approx 10^3$, so $p/n \to 0$

while $p^2/n \to \tau < \infty$. The four regimes of Table 1 therefore correspond to different application areas.

Section 3.7 develops the industrial quality control application, states the assumptions under which the model is reasonable, discusses its limits, and derives a test for a spike from Table 1.

1.6 Related work

Our work is related to spiked covariance models, Haar matrix models, and sample covariance matrices with random populations.

The spiked covariance model with a deterministic population matrix. The proportional regime is by now classical [19, 5, 6, 31, 2, 3], with a comprehensive treatment provided by [1, 37]. Bloemendal et al. [10] relax the assumption of a converging p/n ratio. They establish large deviation bounds on outlier locations and eigenvector overlaps under the weaker condition that p and n are polynomially related, in a generalization of the spiked model. The disproportional regime where $p, n \to \infty$ with $p/n \to 0$ is also central to this model. Johnstone and Lu [20] showed that the leading sample principal component is consistent if and only if $p/n \to 0$ in their asymptotic model. In the same regime, Donoho and Feldman [15] derived optimal shrinkage rules. Liu et al. [28] proved a CLT for spiked sample eigenvalues without assuming convergence of p/n, subject to a condition ensuring that the spikes remain distant. These works consider deterministic population matrices. In our model, such results are relevant after conditioning on the Haar matrix; we prove the precise low-aspect-ratio statement needed for our conditional argument in Proposition 4.6.

The spiked invariant model. The matrix $\Sigma_p = A_p + O_p B_p O_p^* + \sigma^2 I_p$ in (1) is a sum of matrices, one of which is conjugated by a Haar matrix. Models involving polynomials of diagonal matrices and Haar unitary/orthogonal matrices have been extensively studied. Voiculescu [36] established that the global limiting spectral distribution of such additive models is governed by free convolution. Bao et al. [7] obtained quantitative convergence rates for the empirical spectral distribution in additive Haar models. For the multiplicative version of this model involving matrices A_p, B_p, O_p, Ding and Ji [14] established the limits and optimal convergence rates for both the outlier eigenvalues and their corresponding eigenvectors. For fixed-rank perturbations, the normalized limiting spectral distribution does not describe the individual spikes. Collins et al. [13] studied the discrete limiting spectrum in a compact-operator framework. Fluctuations for finite-rank polynomial models in the Haar unitary case, under a simplicity assumption on the limiting eigenvalues, were computed in [12]. These results provide the background for the population fluctuation analysis in our model.

Fluctuations of outliers. Benaych-Georges et al. [8] studied extreme eigenvalues of additive finite-rank deformations of deterministic matrices, with extensions to random matrix ensembles including Wishart matrices. In their orthonormalised Gaussian setting, the perturbation has the form $O_p B_p O_p^*$. In our model, this Haar deformation enters the population covariance before the conditional Gaussian sampling step. Non-Gaussian outlier limits are also known for deformed Wigner matrices [11], where they can depend on the entry distribution. Here the non-Gaussian component comes from the Haar-induced population spike shifts.

Random population covariance matrices. Sample covariance matrices with random population covariance have also been studied. Silverstein [33] established almost sure convergence of the empirical spectral distribution in this setting. Kwak, Lee and Park [23] studied extremal eigenvalues for random diagonal population matrices, while Lee and Li [26] proved a CLT for linear spectral statistics in a random diagonal population model. Our focus is on individual sample outliers generated by fixed-rank Haar perturbations of a scalar covariance matrix.

Our contribution. For fixed ranks and pairwise distinct fixed spikes, our results describe

the interaction between Haar-induced population spike shifts of order p^{-1} and Gaussian sampling fluctuations of order $n^{-1/2}$. We center each sample outlier at $\phi_{p/n}(\theta_\ell)$, which includes the deterministic sampling bias. In the low-aspect-ratio regime $p/n \to 0$, the ratio p^2/n determines the fluctuation regime. When p^2/n tends to zero, a positive finite limit, or infinity, the appropriately normalized fluctuation of each fixed sample outlier has a Haar limit, a convolution of the Haar and Gaussian laws, or a Gaussian limit, respectively. The Haar limit is a linear combination of independent chi-square variables with one degree of freedom in the real case, and independent unit-mean exponential variables in the complex case. The Haar and Gaussian components of the mixed limit are independent. We also recover the Gaussian fluctuation law for each supercritical spike in the proportional regime.

1.7 Organization of the paper

The rest of the paper is organized as follows. Section 2 states the model, sets the notation, and recalls the results on Haar matrices and on the classical spiked model that we use as inputs. It also fixes the labeling of the eigenvalues and defines the asymptotic regimes. Section 3 contains the main results. We first treat the disproportional regime $p/n \to 0$, which is the main object of the paper. In the disproportional regime the ratio p^2/n separates three cases: the Haar-dominated regime $p^2/n \to 0$, the mixed regime $p^2/n \to \tau \in (0, \infty)$, both treated in Theorem 3.1, and the sample-dominated regime $p^2/n \to \infty$, treated in Theorem 3.4. We then treat the proportional regime $p/n \to c > 0$, where the classical Bai–Yao CLT applies. Next, we state the results on the two scalings, give a rank-one example, and present applications of these results and a spectral test. Section 4 proves these results. All the proofs share one first step, the reduction to a deterministic spiked model conditionally on the Haar matrix; after this step each proof is specific to its regime. Appendix A collects the facts used several times in the paper, and the more technical proofs are given in Appendices B and C.

2 Model, notation and preliminaries

In this section, we recall the model introduced in Section 1.2, we introduce important definitions and conventions, we state important results on Haar matrices and we formalize the foundational theorems of random matrix theory mentioned in Section 1.1.

2.1 The model

Throughout the paper we work over a field $\mathbb{F}$, which is either $\mathbb{R}$ or $\mathbb{C}$, fixed once and for all. The choice of $\mathbb{F}$ ties together three objects: the entries of the matrices, the group from which the Haar matrix is drawn, and the law of the samples. We use the following terminology.

- In the real case, $\mathbb{F} = \mathbb{R}$: the matrices A_p, B_p have real entries, O_p is Haar distributed on the orthogonal group $O(p)$, and the samples are real Gaussian vectors.
- In the complex case, $\mathbb{F} = \mathbb{C}$: the matrices A_p, B_p have complex entries, O_p is Haar distributed on the unitary group $U(p)$, and the samples are circular complex Gaussian vectors.

The groups $O(p)$ and $U(p)$, the Haar measure, the circular complex Gaussian law, and the constant κ ($\kappa = 2$ in the real case, $\kappa = 1$ in the complex case) are defined in Section 2.2. We write M^* for the transpose in the real case and the conjugate transpose in the complex case.

Fix two integers $r, s \geq 1$ and two families of strictly positive numbers

$$\alpha_1, \dots, \alpha_r, \qquad \beta_1, \dots, \beta_s,$$

which we assume to be all distinct, and fix a number $\sigma^2 > 0$. For every $p \geq r+s$, let A_p and B_p be deterministic positive semidefinite $p \times p$ matrices over $\mathbb{F}$, of rank r and s respectively, whose nonzero eigenvalues are $\alpha_1, \dots, \alpha_r$ for A_p and $\beta_1, \dots, \beta_s$ for B_p. Let O_p be Haar distributed on the orthogonal group $O(p)$ when $\mathbb{F} = \mathbb{R}$, and on the unitary group $U(p)$ when $\mathbb{F} = \mathbb{C}$. The population covariance matrix is

$$\Sigma_p = A_p + O_p B_p O_p^* + \sigma^2 I_p. \tag{5}$$

Conditionally on O_p, we observe n independent centered Gaussian vectors with covariance Σ_p,

$$x_{p,1}, \dots, x_{p,n} \mid O_p \overset{\text{i.i.d.}}{\sim} \mathcal{N}_{\mathbb{F}}(0, \Sigma_p),$$

where $\mathcal{N}_{\mathbb{F}}(0, \Sigma_p)$ is the centered real Gaussian law with covariance Σ_p when $\mathbb{F} = \mathbb{R}$, and the centered circular complex Gaussian law with covariance Σ_p when $\mathbb{F} = \mathbb{C}$. From these vectors we form the sample covariance matrix

$$\widehat{\Sigma}_p = \frac{1}{n} \sum_{k=1}^{n} x_{p,k} x_{p,k}^*.$$

We order the eigenvalues of Σ_p and of $\widehat{\Sigma}_p$ decreasingly,

$$\lambda_{p,1} \geq \cdots \geq \lambda_{p,p}, \qquad \widehat{\lambda}_{p,1} \geq \cdots \geq \widehat{\lambda}_{p,p},$$

and we write $\theta_1 > \cdots > \theta_{r+s} > 0$ for the elements of $\{\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s\}$ sorted decreasingly. We call the θ_ℓ the spikes of the model. As shown in Section 2.3, the matrix $A_p + O_p B_p O_p^*$ has rank $r+s$ almost surely, so Σ_p has $r+s$ eigenvalues different from σ^2, called the spike eigenvalues, and the eigenvalue σ^2 with multiplicity $p-r-s$. The precise correspondence between a population spike eigenvalue $\lambda_{p,\ell}$ and the associated sample spike eigenvalue $\widehat{\lambda}_{p,\ell}$ is fixed in Section 2.6.

2.2 Groups, Haar measure, and conventions

For $p \geq 1$ we write

$$O(p) = \{Q \in M_p(\mathbb{R}) : Q^\top Q = I_p\}, \qquad U(p) = \{Q \in M_p(\mathbb{C}) : Q^* Q = I_p\},$$

and we denote by $O(p)$ the unitary (orthogonal) group when $\mathbb{F} = \mathbb{R}$, and the group $U(p)$ when $\mathbb{F} = \mathbb{C}$. All the matrices O_p, $p \geq 1$, are defined on the same probability space, so that almost sure statements about the sequence $(\lambda_{p,\ell})_{p \geq 1}$ make sense. Since our results are convergences in law, they do not depend on the joint law chosen for the O_p, and the reader may take them independent.

O_p and U_p are compact groups. On such a group there is a unique Borel probability measure which is invariant under left multiplication by a fixed element of the group; it is called the Haar measure, and compactness implies that it is also invariant under right multiplication and under inversion. A random matrix is Haar distributed when its law is this measure. This is the precise meaning of the informal description given in Section 1.2: no orientation is more likely than another, and a Haar orthogonal (resp.unitary) matrix is a matrix sampled uniformly on $O(p)$ (resp. $U(p)$).

In the real case, M^* denotes the transpose $M^\top$; in the complex case, it denotes the conjugate transpose. We use the same notation in both cases.

A standard real Gaussian variable is $\mathcal{N}(0,1)$. A standard circular complex Gaussian variable is $z = x + \mathrm{i}y$, where x and y are independent $\mathcal{N}(0, 1/2)$; hence $\mathbb{E}|z|^2 = 1$ and $|z|^2 \sim \mathrm{Exp}(1)$. Recall from Section 1.2 that

$$\kappa = \begin{cases} 2, & \text{in the real case,} \\ 1, & \text{in the complex case.} \end{cases} \tag{6}$$

With these conventions, a standard real or complex Gaussian variable z satisfies $\mathrm{Var}(|z|^2-1)=\kappa$.

We write $\mathcal{F}_p=\sigma(O_p)$ for the sigma-algebra generated by the Haar matrix. When we condition on $\mathcal{F}_p$, the notation refers to the remaining Gaussian randomness.

If X_p and X are random variables, then $X_p \xrightarrow{a.s.} X$ means almost sure convergence, $X_p \xrightarrow{\mathbb{P}} X$ means convergence in probability, and $X_p \xrightarrow{\text{law}} X$ means convergence in distribution. If $a_p>0$, then $X_p=o_{\mathbb{P}}(a_p)$ means $X_p/a_p \xrightarrow{\mathbb{P}} 0$, and $X_p=\mathrm{O}_{\mathbb{P}}(a_p)$ means that $\{X_p/a_p\}_{p\geq 1}$ is stochastically bounded; that is, for any $\epsilon>0$, there exists a constant $M>0$ such that $\sup_{p\geq 1}\mathbb{P}(|X_p|/a_p>M)<\epsilon$. We say that a sequence of events Ω_p holds with high probability when $\mathbb{P}(\Omega_p)\to 1$. We also use uniform versions of this notation: if I is a fixed compact interval, then $\sup_{t\in I}|X_p(t)|=o_{\mathbb{P}}(a_p)$ means that the non-negative random variable on the left, divided by a_p, converges to zero in probability.

Finally, we say that a sequence X_p of real random variables diverges in probability, and we write $|X_p| \xrightarrow{\mathbb{P}} \infty$, when $\mathbb{P}(|X_p|\leq M)\to 0$ for every constant $M>0$. By Lemma A.19, such a sequence is not stochastically bounded, and no subsequence of it converges in distribution.

2.3 Reduction to the diagonal case and facts on Σ_p

Remark 2.1 (Reduction to the diagonal case). In the rest of the paper, we may and do assume without loss of generality that

$$A_p=\mathrm{diag}(\alpha_1,\dots,\alpha_r,0,\dots,0), \qquad B_p=\mathrm{diag}(\beta_1,\dots,\beta_s,0,\dots,0).$$

Indeed, write $A_p=UCU^*$ and $B_p=VDV^*$, where C and D are the two diagonal matrices above, and where $U,V\in U(p)$ are deterministic. Put $\widetilde{O}_p=U^*O_pV$ and $\widetilde{x}_{p,k}=U^*x_{p,k}$. By the left and right invariance of the Haar measure, $\widetilde{O}_p$ is again Haar distributed, and, conditionally on O_p, the vectors $\widetilde{x}_{p,1},\dots,\widetilde{x}_{p,n}$ are i.i.d. with law $\mathcal{N}_{\mathbb{F}}(0,\widetilde{\Sigma}_p)$, where

$$\widetilde{\Sigma}_p := U^*\Sigma_pU = C+\widetilde{O}_pD\widetilde{O}_p^*+\sigma^2I_p.$$

Thus $(\widetilde{O}_p,\widetilde{x}_{p,1},\dots,\widetilde{x}_{p,n})$ has exactly the law of the data of the diagonal model. Moreover, for each realization of O_p and of the samples,

$$\frac{1}{n}\sum_{k=1}^{n}\widetilde{x}_{p,k}\widetilde{x}_{p,k}^*=U^*\widehat{\Sigma}_pU,$$

and a conjugation by U does not change eigenvalues. Hence the vector $(\lambda_{p,\ell},\widehat{\lambda}_{p,\ell})_{\ell\leq r+s}$ is the same random variable in the two models, and its law is the same for every p. Only the nonzero eigenvalues of A_p and B_p enter our results.

From now on, A_p and B_p are the two diagonal matrices of Remark 2.1. In particular, the eigenvectors of A_p are $e_1,\dots,e_r$, and the eigenvectors of $O_pB_pO_p^*$ are the first s columns of O_p.

Both A_p and $O_pB_pO_p^*$ are positive semidefinite, so Σ_p is positive definite, and $\|\Sigma_p\|_{\mathrm{op}}\leq \sigma^2+\max_i\alpha_i+\max_j\beta_j$ for every p. By Proposition 2.3, the spikes $\lambda_{p,\ell}-\sigma^2$ converge in probability to the distinct positive limits θ_ℓ, $\ell=1,\dots,r+s$; hence, with probability tending to one, Σ_p has $r+s$ eigenvalues different from σ^2, which we call the spike eigenvalues, and the eigenvalue σ^2 with multiplicity at least $p-r-s$.

2.4 Two properties of Haar matrices

The following result is the fixed-block Gaussian limit for Haar matrices. In the real case it is a direct consequence of Jiang [16, Theorem 1], which gives convergence in total variation for blocks of size $r\times s$ with r,s fixed. The complex case is stated in Collins et al. [12, Lemma 2.1].

Proposition 2.2 (Fixed-block limit for Haar matrices)**.** *Let r, s be fixed, and let $O_p = (o_{ij}^{(p)})$ be Haar distributed on $O(p)$ or $U(p)$. Then*

$$\big(\sqrt{p}\, o_{ij}^{(p)}\big)_{1\le i\le r,\, 1\le j\le s} \xrightarrow{\text{law}} (g_{ij})_{1\le i\le r,\, 1\le j\le s},$$

where the entries of (g_{ij}) are i.i.d. standard real Gaussian variables in the real case, and i.i.d. standard circular complex Gaussian variables in the complex case.

The next result gives the Haar fluctuation of the population spike eigenvalues, at the scale $1/p$. The complex case is a specialization of a result shown by Collins et al. [12, Theorem 1.2 and 1.3]. We prove the result for the real case in Appendix B, using the same fixed-block perturbation computation as in the complex case, and then using the real case of Proposition 2.2.

Proposition 2.3 (Population spike fluctuations at scale $1/p$)**.** *Recall that the α_i and β_j are all distinct. For each $\ell \in \{1, \dots, r+s\}$,*

$$\lambda_{p,\ell} \xrightarrow{\text{a.s.}} \sigma^2 + \theta_\ell, \tag{7}$$

$$p\big(\lambda_{p,\ell} - \sigma^2 - \theta_\ell\big) \xrightarrow{\text{law}} W_\ell = \begin{cases} \displaystyle\sum_{j=1}^{s} \frac{\alpha_i \beta_j}{\alpha_i - \beta_j} |g_{ij}|^2, & \theta_\ell = \alpha_i, \\ \displaystyle\sum_{i=1}^{r} \frac{\alpha_i \beta_j}{\beta_j - \alpha_i} |g_{ij}|^2, & \theta_\ell = \beta_j. \end{cases} \tag{8}$$

where $|g_{ij}|^2 \overset{\text{i.i.d.}}{\sim} \chi_1^2$ in the real case, and $|g_{ij}|^2 \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1)$ in the complex case. Moreover, the convergence in law holds jointly for $\ell = 1, \dots, r+s$, with all W_ℓ defined using the same family (g_{ij}).

2.5 Known results for the classical spiked model

The three results of this subsection concern the classical spiked covariance model, in which the population covariance is deterministic. We state them as external inputs and we use them extensively to prove our results in Section 4.

Fix an integer $K \ge 1$, and let

$$\Lambda_p^{\mathrm{id}} = \mathrm{diag}(\sigma^2 + \theta_1, \dots, \sigma^2 + \theta_K, \sigma^2, \dots, \sigma^2),$$

where $\theta_1 > \cdots > \theta_K > 0$ are fixed. Let $y_{p,1}, \dots, y_{p,n}$ be i.i.d. $\mathcal{N}_{\mathbb{F}}(0, \Lambda_p^{\mathrm{id}})$, and let $\widehat{\lambda}_{p,1}^{\mathrm{id}} \ge \cdots \ge \widehat{\lambda}_{p,p}^{\mathrm{id}}$ be the ordered eigenvalues of the associated sample covariance matrix. Recall from Section 1.4 the functions

$$\phi_c(\theta) = (\sigma^2 + \theta)\Big(1 + \frac{c\,\sigma^2}{\theta}\Big), \qquad V_c(\theta) = (\sigma^2 + \theta)^2\Big(1 - \frac{c\,\sigma^4}{\theta^2}\Big). \tag{9}$$

A spike θ_ℓ is called supercritical when $\theta_\ell > \sigma^2\sqrt{c}$, and subcritical when $\theta_\ell \le \sigma^2\sqrt{c}$.

Theorem 2.4 (Limit of the spikes, BBP transition [6])**.** *Assume $p/n \to c \in (0, \infty)$. For each $\ell \le K$: if $\theta_\ell > \sigma^2\sqrt{c}$, then $\widehat{\lambda}_{p,\ell}^{\mathrm{id}} \to \phi_c(\theta_\ell)$ almost surely, and $\phi_c(\theta_\ell) > \sigma^2(1+\sqrt{c})^2$; if $0 < \theta_\ell \le \sigma^2\sqrt{c}$, then $\widehat{\lambda}_{p,\ell}^{\mathrm{id}} \to \sigma^2(1+\sqrt{c})^2$ almost surely.*

Theorem 2.5 (Gaussian fluctuations of a supercritical spike [2, 3])**.** *Assume $p/n \to c \in (0, \infty)$. Fix $\ell \le K$ and assume θ_ℓ is supercritical. Then*

$$\sqrt{n}\,\big(\widehat{\lambda}_{p,\ell}^{\mathrm{id}} - \phi_{p/n}(\theta_\ell)\big) \xrightarrow{\text{law}} \mathcal{N}\big(0, \kappa V_c(\theta_\ell)\big),$$

with κ given by (6). The fluctuations of distinct supercritical spikes are asymptotically independent.

Let TW denote the Tracy–Widom law of index 2 when $\mathbb{F} = \mathbb{C}$ and of index 1 when $\mathbb{F} = \mathbb{R}$, and define the deterministic centering and scaling

$$m_{p,n}^{+} = \frac{\sigma^2}{n}\big(\sqrt{n-1} + \sqrt{p}\big)^2, \qquad s_{p,n}^{+} = \frac{\sigma^2}{n}\big(\sqrt{n-1} + \sqrt{p}\big)\Big(\frac{1}{\sqrt{n-1}} + \frac{1}{\sqrt{p}}\Big)^{1/3} > 0. \tag{10}$$

Theorem 2.6 (Tracy–Widom fluctuations of the largest non-spiked eigenvalue [19, 17, 6, 10])**.** *Assume $p/n \to c \in (0,\infty)$ and that every spike is supercritical, that is, $\theta_K > \sigma^2\sqrt{c}$. Then*

$$\textit{(a)}\ \ \widehat{\lambda}_{p,K+1}^{\mathrm{id}} \to \sigma^2(1+\sqrt{c})^2 \ \ \textit{almost surely,} \qquad \textit{(b)}\ \ \frac{\widehat{\lambda}_{p,K+1}^{\mathrm{id}} - m_{p,n}^{+}}{s_{p,n}^{+}} \xrightarrow{\ \mathrm{law}\ } \mathrm{TW}.$$

The centering $m_{p,n}^{+}$ and scaling $s_{p,n}^{+}$ depend only on (n, p, σ^2), not on the spikes.

Part (a) follows from the limiting locations of the sample eigenvalues in the spiked model [6]. For part (b), eigenvalue sticking away from the BBP threshold transfers the edge fluctuations of the unspiked Wishart model to the largest non-outlier eigenvalue [10, Theorem 2.7 and Remark 8.7]. Since all spikes are fixed and supercritical, this gives the Tracy–Widom limit with the centering and scaling in (10).

2.6 Labeling of the eigenvalues and asymptotic regimes

The correspondence between sample and population eigenvalues announced at the end of Section 2.1 is made precise as follows. By Proposition 2.3, the population spike eigenvalues satisfy $\lambda_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_\ell$ for each $\ell \le r+s$, and the limits $\sigma^2+\theta_1 > \cdots > \sigma^2+\theta_{r+s}$ are distinct. Hence, on an event of probability tending to one, the population spikes are separated, and each of them is close to exactly one eigenvalue of $\widehat{\Sigma}_p$. We write $\widehat{\lambda}_{p,\ell}$ for the eigenvalue of $\widehat{\Sigma}_p$ associated in this way with θ_ℓ, and it is the object whose fluctuations we study.

We study four asymptotic regimes. The main object of the paper is the disproportional regime $p/n \to 0$, in which the ratio p^2/n separates three cases:

$$\text{the Haar-dominated regime} \quad \frac{p^2}{n} \to 0, \tag{11}$$

$$\text{the mixed regime} \quad \frac{p^2}{n} \to \tau \in (0,\infty), \tag{12}$$

$$\text{the sample-dominated regime} \quad \frac{p}{n} \to 0, \quad \frac{p^2}{n} \to \infty. \tag{13}$$

Theorem 3.1 covers the first two of them at once, that is the whole range $\tau \in [0,\infty)$, with a single proof. In addition, we also treat the proportional regime

$$\frac{p}{n} \to c \in (0,\infty). \tag{14}$$

The role of the ratio p^2/n, and the meaning of the two scales p and $\sqrt{n}$, were explained in Section 1.3.

All the objects needed to state our results are now defined: the model, the two sources of randomness, the classical results from RMT, and the four regimes. We now present our results.

3 Main results

In this section we state the fluctuation results announced in Section 1.4 as well as other relevant new results. The disproportional regime $p/n \to 0$ is the main object of the paper, so we treat

it first. Theorem 3.1 covers at once the Haar-dominated regime $p^2/n \to 0$ and the mixed regime $p^2/n \to \tau \in (0,\infty)$, both at the scale p; Theorem 3.4 covers the sample-dominated regime $p^2/n \to \infty$, at the scale $\sqrt{n}$. We then treat the proportional regime $p/n \to c > 0$ in Theorem 3.5, where the classical properties of the spiked model are recovered. Section 3.4 explains why each regime uses the scaling it uses, and completes Table 1. All the statements are given at a fixed spike θ_ℓ, and none of the limiting laws depends on the realization of O_p.

3.1 The $p^2/n \to \tau \in [0,\infty[$ regime theorem

We start with the regime $p^2/n \to \tau \in [0,\infty)$. It contains both the Haar-dominated case $\tau = 0$ and the mixed case $\tau \in (0,\infty)$, and the natural scaling in both is p. We give the proof of Theorem 3.1 in Section 4.2.

Theorem 3.1 (The $p^2/n \to \tau \in [0,\infty[$ regime theorem)**.** *Consider the model of Section 2. Assume that we are in the regime where*

$$\frac{p^2}{n} \to \tau \in [0,\infty).$$

For each fixed $\ell \in \{1,\dots,r+s\}$,

$$p\big(\widehat{\lambda}_{p,\ell} - \sigma^2 - \theta_\ell\big) \xrightarrow{\text{law}} W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}\, G_\ell + \tau\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}, \tag{15}$$

or equivalently,

$$p\big(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big) \xrightarrow{\text{law}} W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}G_\ell. \tag{16}$$

where $G_\ell \sim \mathcal{N}(0,1)$ is independent of W_ℓ, W_ℓ is given by (8)*, κ is given by* (6) *and $\phi_{p/n}(\cdot), V_0(\cdot)$ are given by* (9)*.*

Remark 3.2 (Theorem 3.1 covers both the Haar-regime and the mixed regime.). Theorem 3.1 covers the whole range $\tau \in [0,\infty)$ with a single proof, i.e. it gives the limit both in the Haar-dominated regime and the mixed regime as defined in Section 2.6. However, it produces a different limiting law in both regime, which is why we separate those regimes in Table 1. In the Haar regime $\tau = 0$, the Gaussian term $\sqrt{\kappa\tau}(\sigma^2+\theta_\ell)G_\ell$ and the deterministic shift $\tau\sigma^2(\sigma^2+\theta_\ell)/\theta_\ell$ both vanish, and (16) reduces to

$$p\big(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big) \xrightarrow{\text{law}} W_\ell.$$

The limit is then W_ℓ alone, that is, a linear combination of independent chi-square variables with one degree of freedom in the real case, and exponential variables in the complex case. It carries no Gaussian component.

Remark 3.3. The limit in (15) can be read as

$$p\big(\widehat{\lambda}_{p,\ell} - \sigma^2 - \theta_\ell\big) \xrightarrow{\text{law}} \underbrace{W_\ell}_{\text{Haar fluctuation}} + \underbrace{\sqrt{\kappa\tau}\,(\sigma^2+\theta_\ell)G_\ell}_{\text{sample fluctuation}} + \underbrace{\tau\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}}_{\text{bulk shift}}.$$

The variable W_ℓ comes from the Haar matrix. The Gaussian variable G_ℓ comes from the sample vectors. These two sources are independent before taking the limit, and this is why they are independent in the limit. The bulk shift is deterministic: it is not a third source of noise but a bias. It is the same finite-sample bias as in the classical spiked model: the sample spike does not estimate $\sigma^2+\theta_\ell$ but $\phi_{p/n}(\theta_\ell) = \sigma^2 + \theta_\ell + \frac{p}{n}\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}$, and the difference $\phi_{p/n}(\theta_\ell) - \sigma^2 - \theta_\ell$ is of order p/n. After scaling by p, it becomes $\frac{p^2}{n}\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}$, which converges to $\tau\,\sigma^2(\sigma^2+\theta_\ell)/\theta_\ell$.

3.2 The sample-dominated theorem

The formula of Theorem 3.1 suggests what happens when $\tau = \infty$. Heuristically, when replacing τ by p^2/n in (15), one expects $p(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell))$ to behave like

$$W_\ell + \sqrt{\kappa \frac{p^2}{n}}\,(\sigma^2 + \theta_\ell) G_\ell.$$

When $p^2/n \to \infty$, this quantity is not tight: the standard deviation of the Gaussian term grows like $\sqrt{p^2/n}$ while W_ℓ remains bounded. The right normalization is therefore obtained by dividing by $\sqrt{p^2/n}$, which amounts to scaling with $\sqrt{n}$ instead of p. This is what the next theorem does. We give its proof in Section 4.3.

Theorem 3.4 (Sample-dominated regime theorem)**.** *Consider the model of Section 2. Assume that we are in the sample-dominated regime, where*

$$\frac{p}{n} \to 0, \qquad \frac{p^2}{n} \to \infty.$$

Then, for each fixed $\ell \in \{1, \dots, r+s\}$,

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \sqrt{\kappa V_0(\theta_\ell)}\, G_\ell. \tag{17}$$

where $G_\ell \sim \mathcal{N}(0,1)$, κ is given by (6) *and $\phi_{p/n}(\cdot), V_0(\cdot)$ are given by* (9)*.*

As shown later in Proposition 3.6, the scaling by p has no limit here, so $\sqrt{n}$ is the only correct choice. In the sample-dominated regime and with the $\sqrt{n}$ scaling, only the Gaussian fluctuation remains. The Haar variable W_ℓ has disappeared from the limit, because it is of order $1/p$, which is negligible compared to the sample fluctuations of order $1/\sqrt{n}$ when $p^2/n \to \infty$.

Notice that in the sample-dominated regime, the centering is necessarily $\phi_{p/n}(\theta_\ell) = \sigma^2 + \theta_\ell + \frac{p}{n}\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}$ and it can't simply be $\sigma^2 + \theta_\ell$ as in Theorem 3.1. The deterministic bias $\frac{p}{n}\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}$ can't be left in the limit as it is of order p/n; after scaling by $\sqrt{n}$, it diverges since $\sqrt{n} \cdot p/n = p/\sqrt{n} \to \infty$. This deterministic bias must therefore be subtracted to be removed from the limit, exactly as in the classical Bai–Yao theorem.

3.3 The proportional regime theorem

We now treat the proportional regime $p/n \to c > 0$. Although point (ii) of Theorem 3.5 is similar to the content of Theorem 3.4, the proportional regime differs from the sample dominated in three ways. First, the proof techniques are different and rely on different tools. Second, the CLT presented in Theorem 3.5 only holds for supercritical spikes $\theta_\ell > \sigma^2\sqrt{c}$; this condition was automatic in the disproportional case. Third, in the proportional regime the bulk of the spectrum spreads over the interval $[\sigma^2(1-\sqrt{c})^2, \sigma^2(1+\sqrt{c})^2]$: the empirical spectral distribution converges to the Marchenko–Pastur law, and the largest non-spiked eigenvalue has Tracy–Widom fluctuations at the scale $n^{-2/3}$, as in the classical deterministic spiked model. The following theorem collects these facts and is proven in Section 4.4.

Theorem 3.5 (Limits and fluctuations in the proportional regime)**.** *Consider the model of Section 2. Assume that we are in the proportional regime, where*

$$p/n \to c \in (0, \infty).$$

Then:

(i) *The empirical spectral distribution of* $\widehat{\Sigma}_p$ *converges weakly, almost surely, to the Marchenko–Pastur law* μ_{c,σ^2}*, whose support has upper edge* $\sigma^2(1+\sqrt{c})^2$*.*

(ii) *For each* $\ell \in \{1,\dots,r+s\}$ *with* $\theta_\ell > \sigma^2\sqrt{c}$*,*

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \sqrt{\kappa V_c(\theta_\ell)}\, G_\ell, \qquad \text{and} \qquad \widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \phi_c(\theta_\ell).$$

(iii) *For each* $\ell \in \{1,\dots,r+s\}$ *with* $0 < \theta_\ell \le \sigma^2\sqrt{c}$*,* $\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2(1+\sqrt{c})^2$*.*

(iv) *If moreover every spike is supercritical, that is* $\theta_{r+s} > \sigma^2\sqrt{c}$*, then the largest bulk eigenvalue satisfies*

$$\frac{\widehat{\lambda}_{p,r+s+1} - m^+_{p,n}}{s^+_{p,n}} \xrightarrow{\text{law}} \mathrm{TW}, \qquad \text{and} \qquad \widehat{\lambda}_{p,r+s+1} = \sigma^2\big(1+\sqrt{p/n}\big)^2 + \mathrm{O}_{\mathbb{P}}\big(n^{-2/3}\big).$$

where $G_\ell \sim \mathcal{N}(0,1)$*,* κ *is given by* (6)*,* $\phi_{p/n}(\cdot), V_c(\cdot)$ *are given by* (9) *and* $m^+_{p,n}$*,* $s^+_{p,n}$ *are given by* (10)*. None of these limiting laws depends on* O_p*.*

The four parts describe the whole spectrum. Part (i) is the Marchenko–Pastur limit of the bulk; part (ii) gives the Gaussian fluctuations of the supercritical spikes, with the classical variance V_c; part (iii) shows that a subcritical spike does not produce an outlier in the sample spectrum: the corresponding sample eigenvalue converges to the upper edge $\sigma^2(1+\sqrt{c})^2$ of the bulk, regardless of the value of θ_ℓ; part (iv) locates the largest non-spiked eigenvalue and gives its Tracy–Widom fluctuations.

The final sentence stating that none of these laws depends on O_p, is the precise form of the statement that our model behaves like a classical spiked covariance model in the proportional regime: the Haar matrix is drawn once, but the Haar fluctuations are negligible compared to the sample fluctuations. Hence, the Haar matrix effect on the spikes is of lower order and does not enter any of the four limits.

3.4 Two possible scalings

Theorems 3.1, 3.4 and 3.5 use different scalings: the first uses p, the second and the third use $\sqrt{n}$. The next two propositions show that, outside the mixed regime, these choices are the only ones that give a limit: under the other scaling, the rescaled fluctuation diverges in probability, so that no subsequence converges in distribution. In the mixed regime $p^2/n \to \tau \in (0,\infty)$, both scalings give a limit, and they differ by the deterministic factor $\sqrt{n}/p \to \tau^{-1/2}$.

Proposition 3.6 (The other scaling has no limit)**.** *Fix* $\ell \in \{1,\dots,r+s\}$*.*

(i) *If* $p^2/n \to 0$*, then*

$$\sqrt{n}\,\big|\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big| \xrightarrow{\mathbb{P}} \infty.$$

(ii) *If* $p/n \to 0$ *and* $p^2/n \to \infty$*, then*

$$p\,\big|\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big| \xrightarrow{\mathbb{P}} \infty.$$

(iii) *If* $p/n \to c > 0$ *and* $\theta_\ell > \sigma^2\sqrt{c}$*, then*

$$p\,\big|\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\big| \xrightarrow{\mathbb{P}} \infty.$$

In each of the three cases the scaled sequence is not tight, and no subsequence of it converges in distribution to a real random variable.

In the mixed regime the scalings by p and $\sqrt{n}$ are equivalent and both yield a finite limit, since they differ by the bounded factor $\sqrt{n}/p \to \tau^{-1/2}$. For completeness we record the limit under the scaling by $\sqrt{n}$, which is the one used in the third row of Table 1.

Proposition 3.7 (In the mixed regime both scalings work)**.** *Assume $p^2/n \to \tau \in (0,\infty)$ and fix $\ell \in \{1,\dots,r+s\}$. Then*

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \tau^{-1/2} W_\ell + \sqrt{\kappa V_0(\theta_\ell)}\, G_\ell,$$

where $G_\ell \sim \mathcal{N}(0,1)$ is independent of W_ℓ.

The proofs of Propositions 3.6 and 3.7 are given in Section 4.5.

Remark 3.8 (Why we use the scaling by p when τ is finite). When $\tau \in (0,\infty)$, the two scaled sequences differ by the deterministic factor $\sqrt{n}/p$, which converges to $\tau^{-1/2} \in (0,\infty)$, and Proposition 3.7 shows that both converge. In this regime the choice is only a matter of presentation. We use p because it is the only one of the two which gives a limit on the whole range $\tau \in [0,\infty)$, since the scaling by $\sqrt{n}$ diverges when $\tau = 0$, by Proposition 3.6.

Before turning to the application, we illustrate the two theorems on the smallest instance of the model, where every quantity can be written in closed form.

3.5 An example: two rank-one perturbations

This example is the simplest instance of the model, with $r = s = 1$, and it can be computed in closed form. We first look at the population matrix, then at the sample eigenvalues.

The population spikes as a function of the angle between eigenvectors. We first drop the Haar assumption and compute the eigenvalues of any rank-two perturbation (unrelated to a Haar matrix). Let $a, b \in \mathbb{F}^p$ be two unit vectors, and let $t = |a^* b|^2 \in [0,1]$ be the squared overlap of the two directions, so that $t = 0$ when a and b are orthogonal, and $t = 1$ when a and b are colinear. The matrix $\alpha\, aa^* + \beta\, bb^*$ with $\alpha > \beta > 0$ has rank at most two, and its two nonzero eigenvalues are

$$\mu_\pm(t) = \frac{\alpha+\beta}{2} \pm \frac{1}{2}\sqrt{(\alpha-\beta)^2 + 4\alpha\beta\, t}. \tag{18}$$

When $t = 0$ the eigenvalues are α and β; when $t = 1$ they are $\alpha + \beta$ and 0 because the eigenvectors are colinear. The overlap t therefore measures how far the spikes are pushed away from α and β.

The Haar model corresponds to $t \to 0$. In our model $A_p = \operatorname{diag}(\alpha, 0, \dots, 0)$, $B_p = \operatorname{diag}(\beta, 0, \dots, 0)$, $a = e_1$ and $b = O_p e_1$, so the overlap is $t_p = |e_1^* O_p e_1|^2 = |o_{11}^{(p)}|^2$. By Proposition 2.2, $\sqrt{p}\, o_{11}^{(p)} \xrightarrow{\text{law}} g$, a standard Gaussian variable, so

$$t_p = |e_1^* O_p e_1|^2 = O_{\mathbb{P}}\left(\frac{1}{p}\right) \to 0.$$

Hence e_1 and $O_p e_1$ are asymptotically orthogonal. This is why the population spikes converge to α and β, and not to a mixture of them. A Taylor expansion of (18) at $t = 0$ gives

$$\mu_+(t) = \alpha + \frac{\alpha\beta}{\alpha-\beta} t + \mathrm{O}(t^2), \qquad \mu_-(t) = \beta - \frac{\alpha\beta}{\alpha-\beta} t + \mathrm{O}(t^2),$$

so, with $p\, t_p \xrightarrow{\text{law}} |g|^2$ and the continuous mapping theorem, and since $\lambda_{p,1} - \sigma^2$ and $\lambda_{p,2} - \sigma^2$ are the two nonzero eigenvalues $\mu_+(t_p)$ and $\mu_-(t_p)$ of $A_p + O_p B_p O_p^*$,

$$p(\lambda_{p,1} - \sigma^2 - \alpha) \xrightarrow{\text{law}} \frac{\alpha\beta}{\alpha-\beta}|g|^2 = W_1, \qquad p(\lambda_{p,2} - \sigma^2 - \beta) \xrightarrow{\text{law}} -\frac{\alpha\beta}{\alpha-\beta}|g|^2 = W_2.$$

The $O(t_p^2)$ remainders are of order $1/p^2$, so they vanish after scaling by p. This recovers Proposition 2.3 in the rank-one case, and shows that $W_1 = -W_2$.

Example 3.9 (Sample eigenvalues in this example)**.** *Keeping the notation above and adding the sample step, Theorem 3.1 gives, for the largest sample eigenvalue in the mixed regime $p^2/n \to \tau \in [0, \infty)$,*

$$p\big(\widehat{\lambda}_{p,1} - \sigma^2 - \alpha\big) \xrightarrow{\text{law}} \frac{\alpha\beta}{\alpha - \beta}|g|^2 + \sqrt{\kappa\tau}\,(\sigma^2 + \alpha)G + \tau\frac{\sigma^2(\sigma^2 + \alpha)}{\alpha}, \tag{19}$$

where G is a standard real Gaussian variable independent of g, with $|g|^2 \sim \chi_1^2$ in the real case and $|g|^2 \sim \mathrm{Exp}(1)$ in the complex case. The limit is a sum of two independent variables: the Haar variable $\frac{\alpha\beta}{\alpha-\beta}|g|^2$, which is positive, asymmetric and non-Gaussian, and the sample variable $\sqrt{\kappa\tau}\,(\sigma^2+\alpha)G$, which is centered Gaussian. When $\tau = 0$, only the term $W_1 = \frac{\alpha\beta}{\alpha-\beta}|g|^2$ survives; it is a positive random variable since $\alpha > \beta$, so $p(\widehat{\lambda}_{p,1} - \sigma^2 - \alpha)$ converges to a distribution supported on $[0, \infty)$, and the largest sample spike lies asymptotically above $\sigma^2 + \alpha$. As τ grows, the Gaussian term is centered with standard deviation of order $\sqrt{\tau}$, so it eventually dominates W_1, which does not depend on τ; the limiting law then becomes closer and closer to a Gaussian, and its asymmetry disappears.

3.6 Numerical experiments

The rank-one example above is also the simplest setting in which our results can be checked numerically [1]. We keep the parameters of this example, namely $r = s = 1$, and we set $\alpha = 10$, $\beta = 5$ and $\sigma^2 = 1$, so that the tracked spike is $\theta_1 = \alpha = 10$ and the Haar fluctuation is the explicit variable

$$W_1 = \frac{\alpha\beta}{\alpha - \beta}\,|g|^2 = 10\,|g|^2,$$

with $|g|^2 \sim \chi_1^2$ in the real case and $|g|^2 \sim \mathrm{Exp}(1)$ in the complex case. For each pair (p, n) we do 2000 independent simulations of the model: we sample a Haar matrix O_p, then n Gaussian vectors with covariance Σ_p conditionally on O_p, and we compute the largest eigenvalue $\widehat{\lambda}_{p,1}$ of the sample covariance matrix.

Let us emphasize that our results describe limits along sequences (p, n), while a simulation is run at a single pair (p, n), at which p^2/n is a finite number. We therefore do not compare the empirical fluctuation to a limiting law, but to the law obtained from Theorem 3.1 by replacing τ by p^2/n and V_c by $V_{p/n}$,

$$L_{p,n} := W_\ell + \sqrt{\kappa\,\frac{p^2}{n}\,V_{p/n}(\theta_\ell)}\;G_\ell, \tag{20}$$

where $G_\ell \sim \mathcal{N}(0, 1)$ is independent of W_ℓ. This is the reference at the scale p; at the scale $\sqrt{n}$ the two statistics differ by the factor $p/\sqrt{n}$, so the reference is

$$\frac{\sqrt{n}}{p}\,L_{p,n} = \sqrt{\frac{n}{p^2}}\;W_\ell + \sqrt{\kappa V_{p/n}(\theta_\ell)}\;G_\ell. \tag{21}$$

Formula (20) is a single expression covering the four regimes, so the same reference can be used in all the configurations we simulate. Along a sequence with $p^2/n \to \tau \in [0, \infty)$ it converges to the limit of Theorem 3.1. Along a sequence with $p^2/n \to \infty$ it does not converge, but (21) does: the coefficient $\sqrt{n/p^2}$ tends to zero and only the Gaussian $\sqrt{\kappa V_c(\theta_\ell)}\,G_\ell$ of Theorems 3.4 and 3.5(ii) remains in the limit law.

[1] The code to reproduce the figures of this section is available at `https://github.com/jeanlouvaleau/Internship-Random-Matrix-for-ML-/tree/main`.

We report two kinds of experiments: histograms of the fluctuation against the density of the reference law, and a measure of the distance between the two as p^2/n varies.

Histograms. Figures 1 and 2 plot the fluctuation at the scale $\sqrt{n}$ and at the scale p, over 2000 simulations, in the real and in the complex case, for four configurations of pairs (p, n) chosen so that p^2/n decreases over four orders of magnitude:

$$(2500, 2500), \qquad (500, 5000), \qquad (500, 50000), \qquad (100, 50000),$$

for which p^2/n equals 2500, 50, 5 and 0.2. The first pair is in the proportional regime $c = 1$, the second and the third have $p/n \to 0$, and the last one is the closest to the Haar-dominated regime that we can reach while keeping p large enough.

In the eight panels the histogram follows the reference law closely, in both fields and over the whole range of p^2/n, which is what the theorems predict.

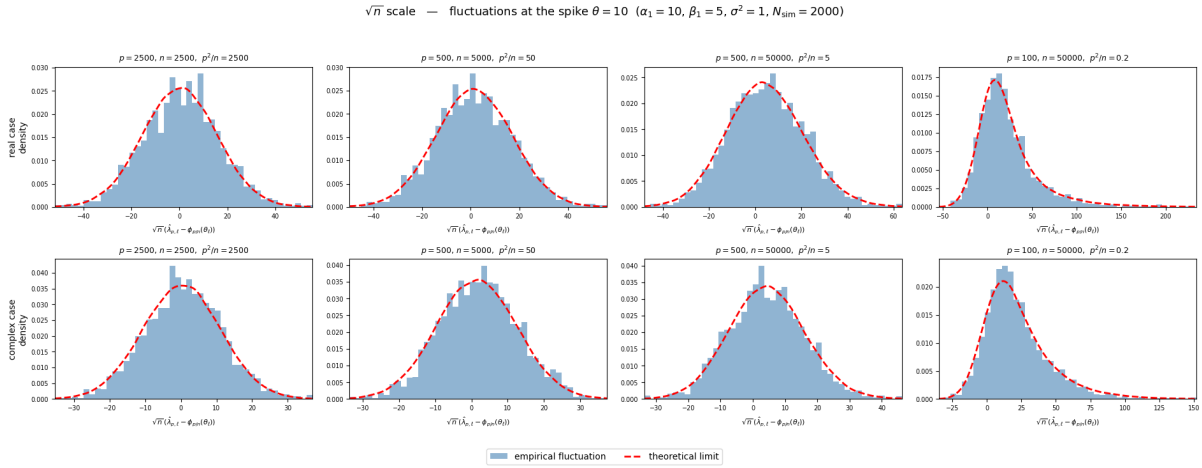


Figure 1: Fluctuations of $\sqrt{n}\,(\widehat{\lambda}_{p,1} - \phi_{p/n}(\theta_1))$ at the spike $\theta_1 = 10$, over 2000 simulations of the model, with $r = s = 1$, $\alpha = 10$, $\beta = 5$ and $\sigma^2 = 1$. Top row: real case. Bottom row: complex case. The four columns are the four pairs (p, n), with p^2/n decreasing from left to right. The red dashed curve is the reference law (21), that is $\sqrt{n/p^2}\,W_1 + \sqrt{\kappa V_{p/n}(\theta_1)}\,G_1$.

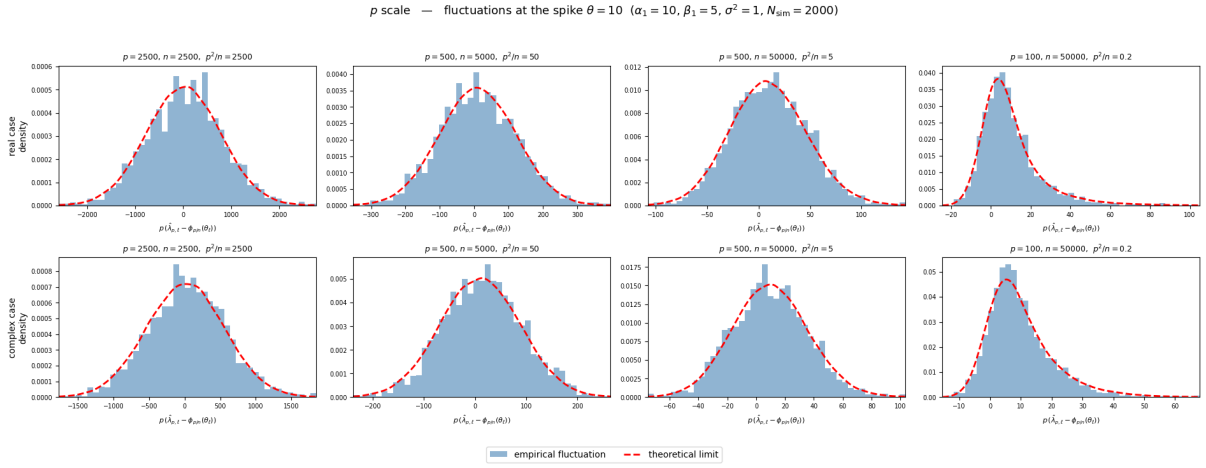


Figure 2: Fluctuations of $p\,(\widehat{\lambda}_{p,1} - \phi_{p/n}(\theta_1))$, for the same simulations and the same four pairs (p, n) as in Figure 1. The red dashed curve is the reference law (20), that is $W_1 + \sqrt{\kappa(p^2/n)V_{p/n}(\theta_1)}\,G_1$. Note the different horizontal scales across columns and between the two rows.

The two figures give complementary pictures. At the scale $\sqrt{n}$, the Gaussian part is essentially fixed across the four configurations, since $V_{p/n}$ varies little, and the Haar part grows as p^2/n decreases: the histogram widens and becomes skewed. The skewness is visible only in the last column, where the coefficient $\sqrt{n/p^2}$ in front of W_1 is 2.24; it is stronger in the real case, where $|g|^2$ is χ^2_1, than in the complex case, where it is Exp(1). At the scale p, it is the Haar part W_1 that stays fixed and the Gaussian part that grows as p^2/n increases. In the first two columns the law is then almost Gaussian, since W_1 is negligible next to it; the asymmetry only appears in the last column.

Only the second of these two figures shows the divergence of the fluctuations when the scaling is wrong. At the scale p, the Gaussian term of (20) has standard deviation $\sqrt{\kappa(p^2/n)V_{p/n}(\theta_\ell)}$, which is approximately 774, 110, and 7 in the real case for the first, second, and last configurations, respectively: this is why the fluctuations take huge values at scale p when $p^2/n \to \infty$. These standard deviations grow to ∞ as $p^2/n \to \infty$, as stated in Proposition 3.6(ii) and (iii). At the scale $\sqrt{n}$, the coefficient in front of W_ℓ is $\sqrt{n/p^2}$, which reaches only 2.24 in the last configuration, so the fluctuations do not take huge values and we don't observe a divergence. To observe the divergence of the fluctuations with a scaling in $\sqrt{n}$, one would need p^2/n smaller by several orders of magnitude, hence a value of p which is too small for the fluctuations to converge to their limit.

Distance of the fluctuation to the limit. Figure 3 measures how close the fluctuation is to its limit, on a grid of values of p at fixed $n = 20000$, so that p^2/n covers the different regimes. We take p from 10 to 1500, so that p^2/n ranges from 0.005 to 112.5, and we work in the real case. To do this, we plot the Wasserstein-1 distance between the normalised [2] fluctuation and three limiting laws: W_ℓ, the Gaussian, and $L_{p,n}$. These distances are computed over 2000 samples.

In the plots below, the dotted line is the Monte-Carlo noise level: even when the reference law is exactly right, the distance is not zero, because it is computed between two finite samples, whose empirical distributions differ by pure sampling error. We measure this non-zero noise level by drawing two independent samples of $L_{p,n}$, of the same sizes as the ones used for the

[2] The Wasserstein-1 distance has the same unit as the variables being compared, and the standard deviation of the reference law $L_{p,n}$ ranges from 14 to 166 over our grid. A raw distance would therefore grow from left to right for reasons of scale alone. We divide both the fluctuation and the reference law by that standard deviation, so that a given distance means the same thing at every point of the grid.

curves, and computing the same distance between them; the operation is repeated and averaged, and it is done at every point of the grid, since the level depends on (p, n) through the shape of $L_{p,n}$. When a curve reaches this level, the fluctuation and the candidate law are as close to each other as two samples of the same law would be, so the simulation cannot tell them apart. A curve staying above it reports a discrepancy that sampling error alone does not explain.

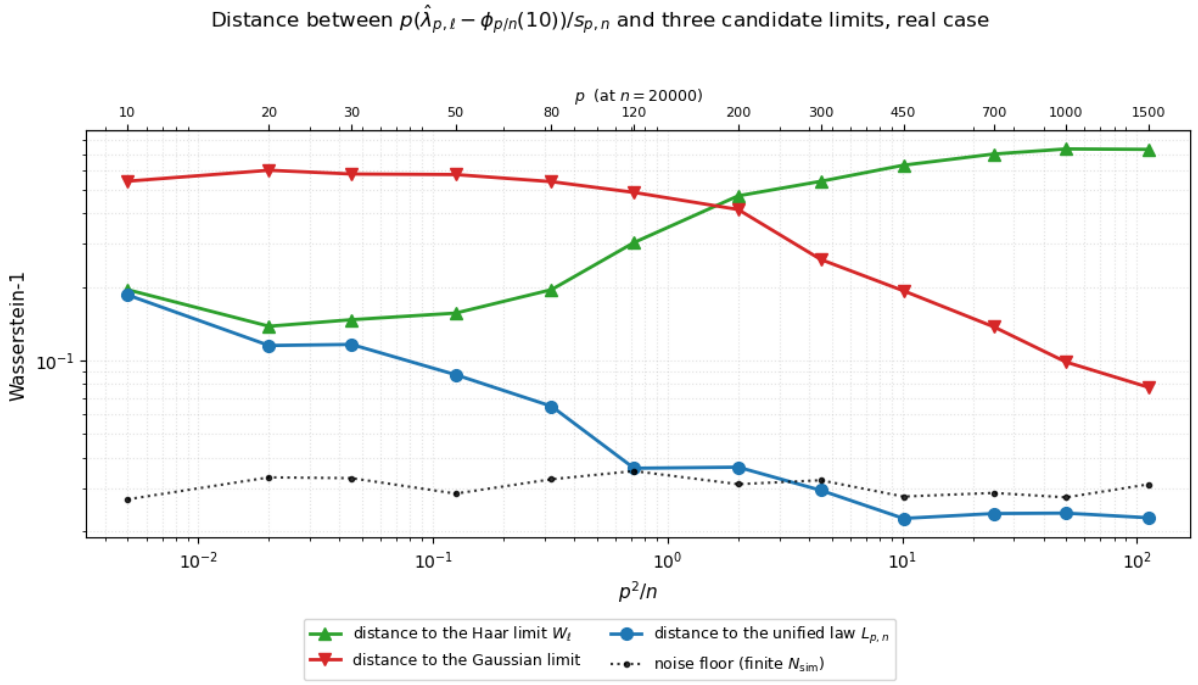


Figure 3: Wasserstein-1 distance between the fluctuation $p\,(\widehat{\lambda}_{p,1} - \phi_{p/n}(\theta_1))/s_{p,n}$ and three candidate limits, in the real case, with $n = 20000$ fixed and p from 10 to 1500, so that p^2/n ranges from 0.005 to 112.5. Both the fluctuation and each candidate are divided by $s_{p,n}$, the standard deviation of $L_{p,n}$, so that the distances are comparable across the grid. The dotted line is the Monte-Carlo noise level, computed from two independent samples of $L_{p,n}$ at the same sample sizes. The upper axis gives the corresponding values of p.

The figure separates the three laws clearly. The distance to $L_{p,n}$ decreases and reaches the noise level from p^2/n of order 2: past that point the simulation cannot distinguish the fluctuation from the reference law. The distance to W_ℓ alone is small only on the left and grows steadily, while the distance to the Gaussian alone does the opposite. The two curves cross around $p^2/n = 2$, which is the region where neither of the two terms can be dropped. Two things are worth noting. On the left of the grid, the distance to $L_{p,n}$ does not reach the noise level: there, p is between 10 and 80, which is too small for Proposition 2.3 to hold. On the right, the distance to the Gaussian alone decreases but is still above the noise level at $p^2/n = 112.5$: the Haar term is of relative size $\sqrt{n/p^2}$, which is still about 0.09, so a much larger p^2/n would be needed before the Gaussian alone becomes accurate.

3.7 Application and a spectral test

In Section 1.5 we introduced the linear mixed model with a continuous batch effect and explained why the Haar assumption makes the separation of the signal and the batch possible. Here we develop one application in detail, industrial quality control, we state precisely when the model is reasonable, and we build a test for one variance component. We only present the test in the regime $p^2/n \to \tau \in [0, \infty)$, which is where our test is new; when $p/n \to c > 0$ our model behaves

similarly to a classical spiked covariance model and some tests already exist: we discuss these at the end of the section.

3.7.1 An application: industrial quality control

A production line in a factory makes items one after another, and after each item is produced, we measure p physical characteristics of it (for instance a length, a mass, an electrical resistance...). Each item gives one measurement vector $x_{p,k} \in \mathbb{R}^p$, whose ℓ-th coordinate is the value of characteristic ℓ for item k, and over a production session we collect n such vectors. A modern line produces and measures a very large number of items, while the number p of measured characteristics is moderate, of order at most few hundreds. So n is large and p is moderate, hence $p/n \to 0$ and $p^2/n \to \tau < \infty$.

The three parts of the model $A_p + O_p B_p O_p^* + \sigma^2 I_p$ have a concrete interpretation. The signal covariance A_p defines the genuine product variation, that is the real differences in terms of quality between items produced in the factory; there are r main modes of variation, each a fixed direction $a_i \in \mathbb{R}^p$ with strength α_i. The batch covariance $O_p B_p O_p^*$ defines the continuous batch effect: when an item is measured, the instrument adds a distortion that moves several characteristics together along s directions $O_p e_1, \dots, O_p e_s$, with strengths $\beta_1, \dots, \beta_s$. Each item passes through the station on its own and is measured once, so its batch effect is drawn independently from the others, which is the continuous batch effect. The noise $\sigma^2 I_p$ is the small independent error made on each single characteristic.

The Haar assumption is reasonable here. The batch effect directions $O_p e_j$ come from the internal state of the measurement device, such as a thermal drift of one of its parts, a small misalignment of the item on the table, or a change of focus of a camera. We have no reason to expect the distortion to act along one combination of characteristics rather than another, so treating all directions as equally likely is the natural choice. It is also reasonable to assume the batch effect directions $O_p e_j$ to be unrelated to the product-variation directions a_i: the first come from the measurement device and the second from the production process, so the two have no reason to be aligned. The assumption fails when the measurement is done in a way that follows the quality of the product itself, so that the distortion and the genuine variation become correlated; in that case the spikes of Σ_p are no longer close to the union of the α_i and the β_j but a combination of both depending on the correlation between product quality and batch effect.

One consequence of the model matters for the test. By Theorem 3.1, the $r + s$ largest sample eigenvalues let us recover the sorted set of strengths $\{\alpha_i\} \cup \{\beta_j\}$, but the model does not say which strength is a product mode and which is a batch effect: differentiating the two requires external information. We come back to this after stating the test.

3.7.2 The hypothesis we test

Fix one spike θ_ℓ among the sorted signal and batch strengths $\{\alpha_i\}_{i=1}^r \cup \{\beta_j\}_{j=1}^s$. In practice, one wants to know whether this spike is strong enough to be considered significant, that is above a threshold $\gamma > 0$ chosen by the user. We test

$$H_0 : \theta_\ell = \gamma \qquad \text{against} \qquad H_1 : \theta_\ell > \gamma.$$

We calibrate the test at the boundary $\theta_\ell = \gamma$. For every fixed parameter configuration satisfying the assumptions with $\theta_\ell < \gamma$, the same argument gives $T_\ell \to -\infty$ in probability and a bounded critical value, so the rejection probability tends to zero. These are pointwise statements; uniform size over parameter sequences approaching the boundary is not claimed.

Remark 3.10. The limit law W_ℓ of Proposition 2.3 is not the same when θ_ℓ is a signal spike or a batch spike, so we must fix which of the two we test. In what follows, we assume for simplicity that θ_ℓ is a signal spike, that is $\theta_\ell = \alpha_i$ for some i. The construction is the same when θ_ℓ is a batch spike, the only thing that differs is that W_ℓ has to be replaced by the corresponding sum over the signal strengths (instead of batch strengths).

3.7.3 The test in the regime $p^2/n \to \tau \in [0, \infty)$

We recall the shape of the fluctuations, since this is what defines the test statistic. Write $c = p/n$ and recall the functions defined in (9), $\phi_c(y) = (\sigma^2 + y)(1 + c\sigma^2/y)$ and $V_c(y) = (\sigma^2 + y)^2(1 - c\sigma^4/y^2)$. By Theorem 3.1, under H_0 and when $p^2/n \to \tau \in [0, \infty)$,

$$p\big(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\gamma)\big) \xrightarrow{\text{law}} L(\gamma) := \sum_{j=1}^{s} \frac{\gamma\beta_j}{\gamma - \beta_j} Z_j + \sqrt{\kappa\tau}\,(\sigma^2 + \gamma)G, \tag{22}$$

where the Z_j are independent χ^2_1 variables in the real case or independent Exp(1) variables in the complex case, and $G \sim \mathcal{N}(0, 1)$ is independent of them.

To calibrate the test, we must estimate the noise level σ^2, the number of spikes, and the batch strengths $\beta_1, \dots, \beta_s$. We first estimate the noise level without knowing the number of spikes, then count the spikes and refine the noise estimate using the bulk. The centered spike eigenvalues are used to estimate the batch strengths. These estimates determine the null centering $\phi_{p/n}(\gamma)$ and the reference law. The Gaussian coefficient uses the observed ratio p^2/n directly, so no separate estimate of its limit τ is needed. The consistency of these estimates is proved in Section 4.6.

Once $\hat{\sigma}^2$, the $\hat{\beta}_j$ and $\widehat{r+s}$ are computed, the estimated reference law

$$\widehat{L} = \sum_{j=1}^{\widehat{r+s}-r} \frac{\gamma\,\hat{\beta}_j}{\gamma - \hat{\beta}_j} Z_j + \sqrt{\kappa\tfrac{p^2}{n}}\,(\hat{\sigma}^2 + \gamma)\,G$$

is an explicit function of the standard variables $Z_1, \dots, Z_{\hat{s}}, G$, where the Z_j are i.i.d. χ^2_1 if $\mathbb{F} = \mathbb{R}$ (resp. Exp(1) if $\mathbb{F} = \mathbb{C}$) and $G \sim \mathcal{N}(0, 1)$ is independent of them. All of these can be simulated by a standard random number generator, so we can draw from $\widehat{L}$ and approximate its $(1-a)$-quantile by the empirical quantile of the draws. Concretely, we draw M independent copies $c_1, \dots, c_M$ of $\widehat{L}$ and sort these copies in increasing order $c_{(1)} \le \cdots \le c_{(M)}$. The empirical $(1-a)$-quantile is then

$$\hat{q}_{1-a} = c_{(\lceil M(1-a)\rceil)},$$

We then reject H_0 when $T_\ell > \hat{q}_{1-a}$. As $M \to \infty$, $\hat{q}_{1-a}$ converges to the $(1-a)$-quantile of $\widehat{L}$ and we show in Section 4.6 that the latter converges to the quantile of the true reference law $L(\gamma)$.

The reference law W_ℓ is a sum over the batch strengths $\beta_1, \dots, \beta_s$, so to sample from it we must know which of the $\widehat{r+s}$ estimated spikes are signals and which are batch effects. We encode this by the ranks of the signal strengths: for each $i \in \{1, \dots, r\}$, let k_i be the position of α_i in the list of the $r+s$ spikes $\{\alpha_i\}_{i=1}^r \cup \{\beta_j\}_{j=1}^s$ sorted in decreasing order, so that $\theta_{k_i} = \alpha_i$. Knowing r and $\{k_1, \dots, k_r\}$ is what lets us remove the r signal-related centered eigenvalues and keep the $\hat{s}$ remaining ones as estimates $\hat{\beta}_1, \dots, \hat{\beta}_{\hat{s}}$ of the batch strengths.

For completeness, stop and do not reject if $\widehat{r+s} \notin \{r+1, \dots, p-1\}$, if a required signal rank is unavailable or duplicated, or if any retained batch estimate belongs to $\{0, \gamma\}$. The proof shows that these exceptional events have probability tending to zero, so this convention does not affect the asymptotic conclusions.

Algorithm 1 Spectral test in the regime $p^2/n \to \tau \in [0, \infty)$

Input: eigenvalues $\widehat{\lambda}_{p,1} \geq \cdots \geq \widehat{\lambda}_{p,p}$; threshold γ; tested rank ℓ; level a; margin ϵ_p; number of draws M; number of signals r and signal ranks $\{k_1, \ldots, k_r\}$.
$\tilde{\sigma}^2 \leftarrow \frac{1}{p} \sum_{k=1}^{p} \widehat{\lambda}_{p,k}$ ▷ preliminary noise estimate
$\widehat{r+s} \leftarrow \#\{k : \widehat{\lambda}_{p,k} > \tilde{\sigma}^2(1 + \sqrt{p/n})^2 + \epsilon_p\}$ ▷ number of spikes
$\hat{\sigma}^2 \leftarrow \frac{1}{p - \widehat{r+s}} \sum_{k > \widehat{r+s}} \widehat{\lambda}_{p,k}$ ▷ improved noise estimate
remove the r signal ranks; let $\hat{\beta}_1, \ldots, \hat{\beta}_{\hat{s}}$ be the remaining centered eigenvalues, $\hat{s} = \widehat{r+s} - r$
$T_\ell \leftarrow p\left(\widehat{\lambda}_{p,\ell} - (\hat{\sigma}^2 + \gamma)(1 + \frac{p}{n}\frac{\hat{\sigma}^2}{\gamma})\right)$
draw $c_1, \ldots, c_M$ i.i.d. from $\widehat{L} = \sum_{j=1}^{\hat{s}} \frac{\gamma\hat{\beta}_j}{\gamma - \hat{\beta}_j} Z_j + \sqrt{\kappa \frac{p^2}{n}}(\hat{\sigma}^2 + \gamma) G$
$\hat{q}_{1-a} \leftarrow c_{(\lceil M(1-a) \rceil)}$ ▷ empirical $(1-a)$-quantile
reject H_0 iff $T_\ell > \hat{q}_{1-a}$

Notice that the same procedure covers both $\tau = 0$ and $\tau > 0$. When $\tau > 0$ (the mixed regime), the reference law has both a Haar part and a Gaussian part. When $\tau = 0$ (the Haar-dominated regime), the plugged-in Gaussian scale $\sqrt{\kappa p^2/n}(\hat{\sigma}^2 + \gamma)$ tends to zero, so $\widehat{L}$ reduces to the weighted sum $\sum_j \frac{\gamma\hat{\beta}_j}{\gamma - \hat{\beta}_j} Z_j$ alone, and the test is calibrated against $W_\ell(\gamma)$ with no Gaussian part.

We collect the assumptions under which the test has the stated properties.

Assumption 3.11. *(A1) The data $x_{p,1}, \ldots, x_{p,n}$ follow the model of Section 2, that is $\Sigma_p = A_p + O_p B_p O_p^* + \sigma^2 I_p$ with O_p Haar, A_p, B_p diagonal as in Remark 2.1.*

(A2) The integers $r, s \geq 1$ are fixed, and the α_i and β_j are distinct and nonzero. Every spike other than the tested one is different from γ.

(A3) The margin satisfies $\epsilon_p > 0$, $\epsilon_p \to 0$, and $\sqrt{p/n} = \mathrm{o}(\epsilon_p)$.

(A4) The dimensions grow with $p^2/n \to \tau \in [0, \infty)$, so in particular $p/n \to 0$.

(A5) The tested spike θ_ℓ is a signal spike.[3]

(A6) The number of signals r and the ranks $\{k_1, \ldots, k_r\}$ of $\alpha_1, \ldots, \alpha_r$ in the sorted list of the $r + s$ spikes are known.

(A7) The number of Monte-Carlo draws is an integer sequence $M = M_p \to \infty$. Conditionally on the data, the draws are independent with law $\widehat{L}$, generated using auxiliary randomness independent of the data.

Theorem 3.12 (Level and consistency)**.** *Under assumptions (A1)–(A7), in the regime $p^2/n \to \tau \in [0, \infty)$, the test in Algorithm 1 is asymptotically of level a and consistent: under H_0, $\mathbb{P}(\text{reject}) \to a$; under H_1, $\mathbb{P}(\text{reject}) \to 1$.*

The proof is given in Section 4.6.1.

3.7.4 The regime $p^2/n \to \infty$, and a comparison

In the sample-dominated regime of Theorem 3.4, or in the proportional regime of Theorem 3.5, the Haar contribution is negligible at the sampling fluctuation scale. Under H_0, the limiting reference law is $\mathcal{N}(0, \kappa V_c(\gamma))$, provided that $\gamma > \sigma^2\sqrt{c}$ when $c > 0$. It depends only on γ, σ^2

[3] Assumed for simplicity; the test also holds when θ_ℓ is a batch spike, on replacing (A6) and the law W_ℓ accordingly.

and $c = \lim p/n$. With a noise estimator satisfying $\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{\mathbb{P}} 0$, the statistic $\sqrt{n}\big(\widehat{\lambda}_{p,\ell} - (\hat{\sigma}^2 + \gamma)(1 + \frac{p}{n}\frac{\hat{\sigma}^2}{\gamma})\big)$ can be compared with a Gaussian quantile of estimated variance $\kappa\hat{V}(\gamma)$, where $\hat{V}(\gamma) = (\hat{\sigma}^2+\gamma)^2(1 - \frac{p}{n}\frac{(\hat{\sigma}^2)^2}{\gamma^2})$. No Monte-Carlo step is needed for this Gaussian reference law. Related spectral procedures have been developed for the problem of detecting or counting signals, in the proportional subcase $p/n \to c > 0$: [30] estimate the number of signals from the sample eigenvalues and describe the detection threshold $\sigma^2\sqrt{p/n}$; [22] build a sequence of hypothesis tests in which each eigenvalue is compared to a Tracy–Widom quantile; and [9] compute the level and the power of a generalized likelihood ratio test for single-source detection.

It is interesting to compare the difficulty of the two regimes. In the classical spiked model setting without Haar fluctuation, the proportional regime $p/n \to c > 0$ is the most difficult setup: weak non-supercritical spikes can't be detected, and the sample covariance is not consistent in operator norm. The regime $p/n \to 0$ is easier in the classical spiked model because the sample covariance is consistent and every fixed spike is detectable.

In our model, calibration in the regime $p^2/n \to \tau < \infty$ requires information about the batch strengths because the Haar contribution is not negligible. After removing the deterministic sampling bias, the Haar fluctuation dominates when $\tau = 0$, whereas the Haar and sampling fluctuations have the same order when $\tau > 0$. The reference law therefore requires estimates of the batch strengths and information identifying the batch spikes. In the sample-dominated and proportional regimes considered above, the Gaussian reference law for a separated sample outlier does not involve these batch parameters.

3.7.5 Relaxing assumption (A6)

Assumption (A6) is the strongest of the list: it requires external knowledge of the signal ranks, that is, to know which of the $r + s$ spikes are signals and which are batch effects. We present three ways to weaken this assumption. The first is a special case where (A6) holds for free; the second replaces it by prior knowledge of the scale of the signals. Throughout we write, as in Algorithm 1, $\hat{\mu}_\ell = \widehat{\lambda}_{p,\ell} - \hat{\sigma}^2$ for the centered eigenvalue at rank ℓ, $\hat{\phi}(\gamma) = (\hat{\sigma}^2 + \gamma)(1 + \frac{p}{n}\frac{\hat{\sigma}^2}{\gamma})$ for the estimated centering, $\hat{v} = \sqrt{\kappa\, p^2/n}\,(\hat{\sigma}^2 + \gamma)$ for the estimated Gaussian scale, and $v_0 = \sqrt{\kappa\tau\, V_0(\gamma)} = \sqrt{\kappa\tau}\,(\sigma^2 + \gamma)$ for its limit.

A single signal dominating the batch effects. Suppose $r = 1$ and $\alpha_1 > \beta_j$ for every $j \in \{1, \dots, s\}$, that is, one signal stronger than every batch effect. Then α_1 is the largest spike, so its rank is $k_1 = 1$: the tested spike is $\theta_1 = \alpha_1$, and the batch effects are exactly the remaining spikes $\theta_2, \dots, \theta_{1+s}$. The signal ranks are therefore known from the ordering of the spikes alone, with no external information, and (A6) holds automatically. In practice this covers any setting where a single real effect clearly dominates every batch effect. The test itself is unchanged: the reference law still depends on the batch strengths $\beta_1, \dots, \beta_s$ through $W_1(\gamma) = \sum_{j=1}^{s} \frac{\gamma\beta_j}{\gamma - \beta_j}|g_{1j}|^2$, which are estimated from the remaining spikes, so the Monte-Carlo calibration is still needed.

Relaxing the ranks with prior knowledge of the scale of the signals. Suppose now that for each signal we have a prior approximate value α_i° of its strength α_i, available for instance from an earlier study or from the physical scale of the effect. Let $\mathcal{S} := \{\theta_1, \dots, \theta_{r+s}\}$ be the set of spikes, and recall that k_i denotes the rank of α_i, so that $\theta_{k_i} = \alpha_i$ and $\mathcal{K}_{\text{sig}} := \{k_1, \dots, k_r\}$ is the set of signal ranks. The idea is to assign to each prior value α_i° the closest centered eigenvalue.

Proposition 3.13 (Calibrated signal values)**.** *Assume (A1)–(A5), that the number of signals r is known (the first half of (A6)), (A7), and:*

(C1) for each $i \in \{1, \dots, r\}$, the prior value α_i° has α_i as its unique closest spike: $|\alpha_i^\circ - \alpha_i| < |\alpha_i^\circ - \theta|$ for every spike $\theta \in \mathcal{S}$ with $\theta \neq \alpha_i$.

Estimate each signal rank by $\hat{k}_i := \arg\min_{\ell \in \{1,\dots,\widehat{r+s}\}} |\hat{\mu}_\ell - \alpha_i^\circ|$. Then $\mathbb{P}\big(\{\hat{k}_1, \dots, \hat{k}_r\} = \mathcal{K}_{\mathrm{sig}}\big) \to 1$, and the rank-$\ell$ test of Algorithm 1 run with these estimated ranks is asymptotically of level a under H_0 and consistent under H_1.

The separation condition (C1) is what makes the nearest-spike assignment reliable; it holds, for example, whenever the prior values are closer to the truth than half the smallest gap between two spikes. A cleaner statement is available in the common case where the signals are known to lie above the batch effects on a fixed scale: then a single threshold separates the two groups, and one no longer needs a prior value for each signal.

Proposition 3.14 (Test under a known scale separation)**.** *Assume (A1)–(A5), that r is known, (A7), and:*

(S1) a threshold $\eta > 0$ is known with $\alpha_i > \eta > \beta_j$ for all i, j.

Estimate the set of signal ranks by $\widehat{\mathcal{K}}_{\mathrm{sig}} := \{\ell \in \{1, \dots, \widehat{r+s}\} : \hat{\mu}_\ell > \eta\}$, and run the rank-$\ell$ test for a signal rank $\ell \in \{1, \dots, r\}$ with batch ranks $\{1, \dots, \widehat{r+s}\} \setminus \widehat{\mathcal{K}}_{\mathrm{sig}}$. Then $\mathcal{K}_{\mathrm{sig}} = \{1, \dots, r\}$, $\mathbb{P}(\widehat{\mathcal{K}}_{\mathrm{sig}} = \mathcal{K}_{\mathrm{sig}}) \to 1$, and the test is asymptotically of level a under H_0 and consistent under H_1.

The proofs of both propositions are given in Section 4.6.2.

4 Proofs

All the proofs below start from the same common step. Conditionally on the Haar matrix O_p, the population covariance Σ_p is deterministic, so the eigenvalues of $\widehat{\Sigma}_p$ are those of a classical spiked model. This reduction is stated in Section 4.1. The remaining difficulty is that the conditional spikes $\lambda_{p,\ell} - \sigma^2$ are not fixed: they depend on p and converge to θ_ℓ only as $p \to \infty$, while the classical spiked covariance model results are stated for fixed spikes. Each regime handles this difficulty at its own scale and using specific tools:

- In the regime $p^2/n \to \tau$ (Section 4.2), the scale is p, and we expand the sample eigenvalue to second order using the perturbation Lemma 4.4.
- In the sample-dominated regime (Section 4.3), the scale is $\sqrt{n}$, at which a fixed-order expansion is no longer accurate, and we use instead a central limit theorem (Proposition 4.6) relatively similar to the Bai–Yao CLT; but applying specificaly to the case $p/n \to 0$.
- In the proportional regime (Section 4.4), the classical results of the deterministic spiked model are transferred to our model with non-fixed spikes using Lemma 4.8.

The results on the two scalings are proved in Section 4.5 and the properties of the spectral test are proven in Section 4.6.

4.1 Conditional reduction to a deterministic spiked model

This reduction is the structural core of the argument. The eigenvectors of the population covariance are random, but conditionally on the Haar matrix the eigenvalue distribution of the sample covariance depends only on the population eigenvalues.

Lemma 4.1 (Conditional diagonalization)**.** *Let $\Sigma_p = U_p \Lambda_p U_p^*$ be an $\mathcal{F}_p$-measurable spectral decomposition, with*

$$\Lambda_p = \operatorname{diag}(\lambda_{p,1}, \dots, \lambda_{p,p}).$$

If $y_{p,k} = U_p^* x_{p,k}$, then conditionally on $\mathcal{F}_p$, the vectors $y_{p,1}, \dots, y_{p,n}$ are independent centered Gaussian vectors with covariance Λ_p, and

$$U_p^* \widehat{\Sigma}_p U_p = \frac{1}{n} \sum_{k=1}^{n} y_{p,k} y_{p,k}^*.$$

In particular, the conditional eigenvalue distribution of $\widehat{\Sigma}_p$ depends only on Λ_p, that is, only on the population eigenvalues.

Proof. After conditioning on $\mathcal{F}_p$, the matrix U_p is deterministic. Orthogonal or unitary invariance of the Gaussian distribution implies that $U_p^* x_{p,k}$ is centered Gaussian with covariance

$$U_p^* \Sigma_p U_p = \Lambda_p.$$

Independence is preserved. The identity for $U_p^* \widehat{\Sigma}_p U_p$ follows by direct substitution. Orthogonal or unitary conjugation does not change eigenvalues. □

Corollary 4.2 (Representation by a standard Gaussian matrix)**.** *When discussing the conditional eigenvalue distribution, we may replace $\widehat{\Sigma}_p$ by*

$$\widetilde{\Sigma}_p = \Lambda_p^{1/2} \left(\frac{1}{n} Z_p Z_p^* \right) \Lambda_p^{1/2}, \tag{23}$$

where $Z_p = (z_{qk})_{1 \le q \le p,\, 1 \le k \le n}$ has independent standard real Gaussian entries in the real case and independent standard circular complex Gaussian entries in the complex case. The matrix Z_p is independent of $\mathcal{F}_p$.

Proof. The conditional law depends only on Λ_p, and the standard Gaussian variables used in the representation can be chosen independently of $\mathcal{F}_p$. □

Remark 4.3 (Randomness left after conditioning). After conditioning on $\mathcal{F}_p$, the Haar matrix O_p, the covariance matrix Σ_p, and all eigenvalues and eigenvectors of Σ_p are fixed deterministic objects. The only randomness left in the conditional model is the Gaussian randomness used to generate the independent sample vectors $x_{p,1}, \ldots, x_{p,n}$. There is no remaining Haar randomness inside the conditional law.

4.2 Proof of Theorem 3.1

The idea is to expand the sample eigenvalue to second order in the sample error matrix, conditionally on the Haar matrix, and then to add back the Haar fluctuation. The first-order term gives the Gaussian piece; the second-order term splits into a negligible spike part and the deterministic bulk shift; the third-order remainder vanishes after scaling by p; and the Haar fluctuation is added at the end using independence.

4.2.1 Second order perturbation lemma

Lemma 4.4 (Second-order eigenvalue perturbation)**.** *Let M be Hermitian, let λ be a simple eigenvalue of M, and let v be a corresponding unit eigenvector. Let $\{v_q\}$ be an orthonormal eigenbasis of M, with $Mv_q = \lambda_q v_q$, and let ℓ be the index such that $v_\ell = v$. Set*

$$\delta = \min_{\mu \in \mathrm{Spec}(M) \setminus \{\lambda\}} |\lambda - \mu| > 0.$$

If E is Hermitian and $\|E\|_{\mathrm{op}} \le \delta/4$, then the eigenvalue $\widehat{\lambda}$ of $M + E$ associated with λ satisfies

$$\widehat{\lambda} - \lambda = v^* E v + \sum_{q \neq \ell} \frac{|v_q^* E v|^2}{\lambda - \lambda_q} + R, \tag{24}$$

and there exists a universal constant $C > 0$ such that

$$|R| \le C\delta^{-2} \|E\|_{\mathrm{op}}^3. \tag{25}$$

The proof of Lemma 4.4 is given in Appendix A.7.

Remark 4.5 (Why a first-order expansion is not enough). Conditionally on the Haar matrix, the sample eigenvalue admits an expansion around the population eigenvalue $\lambda_{p,\ell}$,

$$\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell} = (\text{first order}) + (\text{second order}) + (\text{remainder}),$$

with the three terms of order $n^{-1/2}$, p/n and $(p/n)^{3/2}$. After multiplication by p, the first two terms are of order $\sqrt{p^2/n}$ and p^2/n, hence both of order one when $p^2/n \to \tau$, while the remainder is of order $(p^2/n)\sqrt{p/n}$ and tends to zero. A first-order expansion would keep only the first term and discard the second, but the second term contributes a deterministic bias of order τ that must be retained. This is why we expand to second order. In the proof, the second-order term is a sum over the other eigenvalue directions, which splits into a finite sum over the remaining spikes, negligible after scaling, and a sum over the $p - r - s$ bulk directions, which gives the deterministic shift.

4.2.2 Proof of Theorem 3.1

We fix one index $\ell \in \{1, \dots, r+s\}$ throughout the proof. All the perturbation identities below are stated on the good event Ω_p of Step 1, where the sample perturbation is small enough for Lemma 4.4 to hold. Since $\mathbb{P}(\Omega_p) \to 1$, what happens on Ω_p^c does not affect any limit.

Step 1: the good event.

Let $K = r + s$ and define

$$d_\ell = \frac{1}{4} \min \left\{ \theta_\ell, \min_{1 \le q \le K,\, q \ne \ell} |\theta_\ell - \theta_q| \right\} > 0.$$

By Proposition 2.3, for each $q \le K$,

$$\lambda_{p,q} - \sigma^2 \xrightarrow{\mathbb{P}} \theta_q.$$

For $q > K$, we have $\lambda_{p,q} = \sigma^2$ exactly. Hence the event

$$\Omega_p^{\mathrm{gap}} = \left\{ \min_{q \ne \ell} |\lambda_{p,\ell} - \lambda_{p,q}| \ge 2d_\ell, \quad \lambda_{p,\ell} - \sigma^2 \ge 2d_\ell \right\}$$

satisfies

$$\mathbb{P}(\Omega_p^{\mathrm{gap}}) \to 1. \tag{26}$$

The population eigenvalues are also uniformly bounded, because

$$\|\Sigma_p\|_{\mathrm{op}} \le \sigma^2 + \max_i \alpha_i + \max_j \beta_j.$$

After conditioning on the Haar matrix, let E_p be the sample error matrix in the population eigenbasis, defined in Step 2 below. Corollary A.3, applied conditionally on $\mathcal{F}_p$ as explained in Remark A.4, together with $p/n \to 0$, gives

$$\|E_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}\left(\sqrt{\frac{p}{n}}\right) = o_{\mathbb{P}}(1). \tag{27}$$

Set

$$\Omega_p = \Omega_p^{\mathrm{gap}} \cap \{\|E_p\|_{\mathrm{op}} \le d_\ell/2\}. \tag{28}$$

Then $\mathbb{P}(\Omega_p) \to 1$. On Ω_p, the eigenvalue of the sample covariance associated with $\lambda_{p,\ell}$ is unique. It is the eigenvalue denoted by $\widehat{\lambda}_{p,\ell}$ in Section 2.

All perturbation identities below are first proved on Ω_p. Since $\mathbb{P}(\Omega_p^c) \to 0$, replacing a random variable by any harmless value on Ω_p^c does not change its limit in probability or in distribution.

Step 2: the conditional diagonal model.

By Lemma 4.1 and Corollary 4.2, conditionally on $\mathcal{F}_p = \sigma(O_p)$ we may work with

$$\widetilde{\Sigma}_p = \Lambda_p^{1/2} \left(\frac{1}{n} Z_p Z_p^* \right) \Lambda_p^{1/2}, \qquad \Lambda_p = \mathrm{diag}(\lambda_{p,1}, \dots, \lambda_{p,p}),$$

where $Z_p = (z_{qk})$ has independent standard real or complex Gaussian entries and is independent of $\mathcal{F}_p$. This is not an extra assumption; it is a representation of the conditional law of the original sample covariance eigenvalues.

Put

$$E_p = \widetilde{\Sigma}_p - \Lambda_p.$$

If $z_q = (z_{q1}, \dots, z_{qn})$ is the q-th row of Z_p, then the entries of E_p are

$$(E_p)_{ab} = \sqrt{\lambda_{p,a}\lambda_{p,b}} \left(\frac{1}{n} \sum_{k=1}^{n} z_{ak} \overline{z_{bk}} - \delta_{ab} \right). \tag{29}$$

In the real case, the complex conjugate has no effect.

Step 3: second-order perturbation.

On Ω_p, Lemma 4.4 applies to $M = \Lambda_p$, $E = E_p$, and $\lambda = \lambda_{p,\ell}$. Therefore

$$\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell} = (E_p)_{\ell\ell} + \sum_{q \neq \ell} \frac{|(E_p)_{q\ell}|^2}{\lambda_{p,\ell} - \lambda_{p,q}} + R_{p,\ell}, \tag{30}$$

where

$$|R_{p,\ell}| \le C d_\ell^{-2} \|E_p\|_{\mathrm{op}}^3 \qquad \text{on } \Omega_p. \tag{31}$$

We split the second-order sum as

$$\sum_{q \neq \ell} = \sum_{\substack{1 \le q \le K \\ q \neq \ell}} + \sum_{q > K}.$$

The first part contains only finitely many spike directions. The second part contains the $p - K$ bulk directions.

Step 4: the first-order term.

Taking $a = b = \ell$ in (29) gives

$$(E_p)_{\ell\ell} = \frac{\lambda_{p,\ell}}{n} \sum_{k=1}^{n} \big(|z_{\ell k}|^2 - 1 \big). \tag{32}$$

Define

$$S_{p,\ell} = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \big(|z_{\ell k}|^2 - 1 \big).$$

The summands have mean zero and variance κ. The classical one-dimensional central limit theorem gives

$$S_{p,\ell} \xrightarrow{\text{law}} \sqrt{\kappa}\, G_\ell, \qquad G_\ell \sim \mathcal{N}(0,1). \tag{33}$$

Since Z_p is independent of $\mathcal{F}_p$, the random variable $S_{p,\ell}$ is independent of all Haar-measurable quantities.

Moreover,

$$p(E_p)_{\ell\ell} = \frac{p}{\sqrt{n}} \lambda_{p,\ell} S_{p,\ell}. \tag{34}$$

The coefficient satisfies

$$\frac{p}{\sqrt{n}} \lambda_{p,\ell} \xrightarrow{\mathbb{P}} \sqrt{\tau}(\sigma^2 + \theta_\ell). \tag{35}$$

Step 5: the other spike directions are negligible.

Let

$$\Pi_{p,\ell}^{\mathrm{sp}} = p \sum_{\substack{1 \le q \le K \\ q \neq \ell}} \frac{|(E_p)_{q\ell}|^2}{\lambda_{p,\ell} - \lambda_{p,q}}.$$

On Ω_p, all denominators in this finite sum have absolute value at least $2d_\ell$. Moreover $\Omega_p \subset \Omega_p^{\mathrm{gap}}$, and Ω_p^{gap} is $\mathcal{F}_p$-measurable, since it is defined by the population eigenvalues only. Hence

$$\mathbf{1}_{\Omega_p} \big| \Pi_{p,\ell}^{\mathrm{sp}} \big| \le \mathbf{1}_{\Omega_p^{\mathrm{gap}}} \frac{p}{2d_\ell} \sum_{\substack{1 \le q \le K \\ q \neq \ell}} |(E_p)_{q\ell}|^2,$$

where we used that the denominators are bounded below by $2d_\ell$ on Ω_p^{gap}. Taking conditional expectations, the indicator is now $\mathcal{F}_p$-measurable and can be taken out, so

$$\mathbb{E}\left[\mathbf{1}_{\Omega_p}|\Pi_{p,\ell}^{\mathrm{sp}}| \mid \mathcal{F}_p\right] \le \frac{p}{2d_\ell} \sum_{\substack{1\le q\le K\\ q\neq \ell}} \mathbb{E}\left[|(E_p)_{q\ell}|^2 \mid \mathcal{F}_p\right] \le C\frac{p}{n},$$

where the last inequality uses

$$\mathbb{E}\left[|(E_p)_{q\ell}|^2 \mid \mathcal{F}_p\right] = \frac{\lambda_{p,q}\lambda_{p,\ell}}{n}$$

together with the boundedness of the population eigenvalues.

Since $p/n \to 0$, Markov's inequality gives

$$\Pi_{p,\ell}^{\mathrm{sp}} = o_{\mathbb{P}}(1). \tag{36}$$

Step 6: the bulk directions give the deterministic shift.

For $q > K$, $\lambda_{p,q} = \sigma^2$. Put

$$L_q = \sum_{k=1}^{n} z_{qk}\overline{z_{\ell k}}.$$

Then

$$(E_p)_{q\ell} = \frac{\sqrt{\sigma^2\lambda_{p,\ell}}}{n} L_q. \tag{37}$$

Let $m_p = p - K$. Conditionally on $\mathcal{F}_p$ and on the row z_ℓ, the variables L_q, $q > K$, are independent centered Gaussian variables. In the real case their conditional variance is $\|z_\ell\|_2^2$; in the complex case the conditional second moment is also $\|z_\ell\|_2^2$. In both cases

$$\mathbb{E}\left[|L_q|^2 \mid \mathcal{F}_p, z_\ell\right] = \|z_\ell\|_2^2,$$

and

$$\mathrm{Var}\left(|L_q|^2 \mid \mathcal{F}_p, z_\ell\right) \le C\|z_\ell\|_2^4.$$

Therefore the conditional Chebyshev inequality gives

$$V_p := \frac{1}{m_p\|z_\ell\|_2^2} \sum_{q>K} |L_q|^2 \xrightarrow{\mathbb{P}} 1. \tag{38}$$

Also, by the law of large numbers,

$$U_p := \frac{\|z_\ell\|_2^2}{n} \xrightarrow{\mathbb{P}} 1. \tag{39}$$

On Ω_p, the bulk denominator is $\lambda_{p,\ell} - \sigma^2 > 0$. Hence the bulk part satisfies

$$\begin{aligned}\Pi_{p,\ell}^{\mathrm{bulk}} &:= p \sum_{q>K} \frac{|(E_p)_{q\ell}|^2}{\lambda_{p,\ell} - \sigma^2} \\ &= \frac{pm_p}{n} \frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2} U_p V_p. \end{aligned} \tag{40}$$

Since $m_p/p \to 1$, $p^2/n \to \tau$, and

$$\frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2} \xrightarrow{\mathbb{P}} \frac{\sigma^2(\sigma^2 + \theta_\ell)}{\theta_\ell},$$

Slutsky's theorem gives

$$\Pi_{p,\ell}^{\mathrm{bulk}} \xrightarrow{\mathbb{P}} \tau \frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}. \tag{41}$$

This is the deterministic second-order bulk shift.

Step 7: the third-order remainder vanishes.

From (31) and (27), on Ω_p,

$$\begin{aligned} p|R_{p,\ell}| &\le C d_\ell^{-2} p \|E_p\|_{\mathrm{op}}^3 \\ &= \mathrm{O}_{\mathbb{P}}\left(p\left(\frac{p}{n}\right)^{3/2}\right) \\ &= \mathrm{O}_{\mathbb{P}}\left(\frac{(p^2/n)^{3/2}}{\sqrt{p}}\right) = o_{\mathbb{P}}(1), \end{aligned}$$

so

$$pR_{p,\ell} = o_{\mathbb{P}}(1). \tag{42}$$

Step 8: the sample contribution.

Multiplying (30) by p and using Steps 4–7 gives

$$p\big(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell}\big) = \frac{p}{\sqrt{n}}\lambda_{p,\ell} S_{p,\ell} + \tau\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell} + o_{\mathbb{P}}(1). \tag{43}$$

More precisely, before replacing the bulk part by its limit, the second term is $\Pi_{p,\ell}^{\mathrm{bulk}}$ from (40); equation (41) justifies the replacement.

Step 9: adding the Haar fluctuation.

Write

$$\begin{aligned} p\big(\widehat{\lambda}_{p,\ell} - \sigma^2 - \theta_\ell\big) = p\big(\lambda_{p,\ell} - \sigma^2 - \theta_\ell\big) \\ + p\big(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell}\big). \end{aligned} \tag{44}$$

Set

$$H_{p,\ell} = p\big(\lambda_{p,\ell} - \sigma^2 - \theta_\ell\big), \qquad a_{p,\ell} = \frac{p}{\sqrt{n}}\lambda_{p,\ell}.$$

The pair $(H_{p,\ell}, a_{p,\ell})$ is $\mathcal{F}_p$-measurable. Proposition 2.3 and (35) imply

$$(H_{p,\ell}, a_{p,\ell}) \xrightarrow{\mathrm{law}} \big(W_\ell, \sqrt{\tau}(\sigma^2+\theta_\ell)\big).$$

The variable $S_{p,\ell}$ is a function of Z_p only and is independent of $\mathcal{F}_p$. Lemma A.16 therefore gives the joint convergence

$$(H_{p,\ell}, a_{p,\ell}, S_{p,\ell}) \xrightarrow{\mathrm{law}} \big(W_\ell, \sqrt{\tau}(\sigma^2+\theta_\ell), \sqrt{\kappa}G_\ell\big),$$

where G_ℓ is independent of W_ℓ. Combining this joint convergence with (43) and using Slutsky's theorem proves (15). Equation (16) follows by subtracting the deterministic center $\phi_{p/n}(\theta_\ell)$. □

4.3 Proof of Theorem 3.4

4.3.1 A specific CLT when $p/n \to 0$

We now state the deterministic spiked covariance input needed for the sample-dominated regime. This point deserves a separate proof. The standard Bai–Yao theorem is usually stated for a fixed limiting aspect ratio $p/n \to c \in (0,1)$. In the present paper the aspect ratio itself tends to zero, and the required statement is a triangular-array statement with $c_p = p/n \to 0$. Quoting the fixed-c theorem and then sending c to zero would leave a uniformity gap.

Proposition 4.6 (Low-aspect-ratio spike CLT for a deterministic covariance)**.** *Let $\sigma^2 > 0$, let K be fixed, and let*

$$\Lambda_p = \operatorname{diag}(\lambda_{p,1}, \dots, \lambda_{p,K}, \sigma^2, \dots, \sigma^2), \qquad \lambda_{p,j} \geq 0.$$

Fix $\ell \leq K$. Assume

$$\lambda_{p,\ell} \to \lambda_\ell^* > \sigma^2.$$

Assume also that there are constants $\delta_0 > 0$ and p_0 such that, for every $p \geq p_0$,

$$\lambda_{p,\ell} - \sigma^2 \geq \delta_0, \qquad |\lambda_{p,\ell} - \lambda_{p,j}| \geq \delta_0 \quad (j \neq \ell), \tag{45}$$

and

$$\sup_p \max_{1 \leq j \leq K} \lambda_{p,j} < \infty. \tag{46}$$

Let $y_{p,1}, \dots, y_{p,n}$ be independent centered real or complex Gaussian vectors with covariance Λ_p, and set

$$S_p = \frac{1}{n} \sum_{k=1}^{n} y_{p,k} y_{p,k}^*.$$

For a Hermitian matrix M, write its eigenvalues as

$$\mu_1(M) \geq \cdots \geq \mu_p(M).$$

For $p \geq p_0$, define

$$k_{p,\ell} = 1 + \#\{j \leq K : \lambda_{p,j} > \lambda_{p,\ell}\},$$

and define $k_{p,\ell}$ in any way for the finitely many values $p < p_0$. Put

$$\widehat{\lambda}_{p,\ell}^{\det} = \mu_{k_{p,\ell}}(S_p). \tag{47}$$

If $p, n \to \infty$ and $p/n \to 0$, then

$$\sqrt{n} \left(\widehat{\lambda}_{p,\ell}^{\det} - \lambda_{p,\ell} - \frac{p}{n} \frac{\sigma^2 \lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2} \right) \xrightarrow{\text{law}} N\big(0, \kappa(\lambda_\ell^*)^2\big). \tag{48}$$

The proof of Proposition 4.6 is provided in Appendix C

Remark 4.7 (Why the $p^2/n \to \tau \in [0, \infty[$ regime proof is not sufficient here). In the $p^2/n \to \tau \in [0, \infty[$ regime, expanding only up to second order is enough, because the third-order remainder is negligible after multiplying by p. In the regime $p^2/n \to \infty$, one works at the $\sqrt{n}$ scale, and a crude operator-norm bound for a fixed-order remainder is not uniform enough. Proposition 4.6 avoids this difficulty by using the Schur complement equation and the deterministic outlier location, which already resums the repeated interaction with the bulk.

4.3.2 Proof of Theorem 3.4

Proof. We condition on the Haar matrix, apply Proposition 4.6, and then show that the Haar fluctuation disappears at the $\sqrt{n}$ scale.

Step 1: Conditioning on the Haar matrix. Condition on $\mathcal{F}_p = \sigma(O_p)$. By Lemma 4.1, the conditional model is a deterministic spiked covariance model with population eigenvalues

$$\lambda_{p,1}, \dots, \lambda_{p,r+s}, \sigma^2, \dots, \sigma^2.$$

By Proposition 2.3,

$$\lambda_{p,q} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_q, \qquad q = 1, \dots, r+s.$$

Since the limiting spikes are distinct, the separation assumptions in Proposition 4.6 hold with probability tending to one. The boundedness assumption (46) holds for every p, because

$$\|A_p + O_p B_p O_p^*\|_{\mathrm{op}} \le \|A_p\|_{\mathrm{op}} + \|B_p\|_{\mathrm{op}}.$$

Let

$$X_{p,\ell} = \sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell} - \frac{p}{n}\frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2}\right),$$

and let

$$\phi_p(t) = \mathbb{E}\left[e^{\mathrm{i}tX_{p,\ell}} \mid \mathcal{F}_p\right].$$

We now justify the conditional limit without assuming a uniform version of Proposition 4.6. Take any subsequence. Since the finite vector $(\lambda_{p,1}, \ldots, \lambda_{p,r+s})$ converges in probability, this subsequence has a further subsequence on which the vector converges almost surely. Fix a sample point in this probability-one event. Along the further subsequence, the population eigenvalues form a deterministic triangular array that satisfies Proposition 4.6. Therefore, for every fixed t,

$$\phi_p(t) \to \exp\left(-\frac{1}{2}\kappa(\sigma^2 + \theta_\ell)^2 t^2\right)$$

almost surely along that further subsequence. Since this can be done for every subsequence, we conclude that

$$\phi_p(t) \xrightarrow{\mathbb{P}} \exp\left(-\frac{1}{2}\kappa(\sigma^2 + \theta_\ell)^2 t^2\right).$$

Lemma A.17 therefore gives the unconditional convergence

$$X_{p,\ell} \xrightarrow{\text{law}} \mathcal{N}(0, \kappa(\sigma^2 + \theta_\ell)^2). \tag{49}$$

Step 2: Replacing the random center by a deterministic center. Let

$$f(x) = \frac{\sigma^2 x}{x - \sigma^2}.$$

Set $I_\ell := [\sigma^2 + \theta_\ell/2,\ \sigma^2 + 2\theta_\ell]$. This interval does not contain σ^2, so f is C^1 on I_ℓ and there is a constant C_ℓ, depending only on σ^2 and θ_ℓ, with $|f'| \le C_\ell$ on I_ℓ. Proposition 2.3 implies

$$\lambda_{p,\ell} - \sigma^2 - \theta_\ell = \mathrm{O}_{\mathbb{P}}(1/p). \tag{50}$$

In particular $\lambda_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_\ell$, so $\mathbb{P}(\lambda_{p,\ell} \in I_\ell) \to 1$. On this event, both $\lambda_{p,\ell}$ and $\sigma^2 + \theta_\ell$ lie in I_ℓ, so the mean value theorem applies and gives

$$\begin{aligned}
&\sqrt{n}\frac{p}{n}\left|\frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell} - \sigma^2} - \frac{\sigma^2(\sigma^2 + \theta_\ell)}{\theta_\ell}\right| \\
&\quad \le C\sqrt{n}\frac{p}{n}|\lambda_{p,\ell} - \sigma^2 - \theta_\ell| = C\frac{1}{\sqrt{n}}\,p|\lambda_{p,\ell} - \sigma^2 - \theta_\ell| = o_{\mathbb{P}}(1).
\end{aligned} \tag{51}$$

We used $p|\lambda_{p,\ell} - \sigma^2 - \theta_\ell| = \mathrm{O}_{\mathbb{P}}(1)$ and $n \to \infty$. This holds on an event of probability tending to one, hence (51) holds as stated.

Step 3: The Haar population fluctuation vanishes at the $\sqrt{n}$ scale. Again by (50),

$$\sqrt{n}(\lambda_{p,\ell} - \sigma^2 - \theta_\ell) = \frac{\sqrt{n}}{p} \cdot p(\lambda_{p,\ell} - \sigma^2 - \theta_\ell) = o_{\mathbb{P}}(1), \tag{52}$$

because $p^2/n \to \infty$ is equivalent to $\sqrt{n}/p \to 0$.

Finally, decompose

$$\begin{aligned}
&\widehat{\lambda}_{p,\ell} - \sigma^2 - \theta_\ell - \frac{p}{n}\frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell} \\
&\quad = \left(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell} - \frac{p}{n}\frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell}-\sigma^2}\right) \\
&\qquad + (\lambda_{p,\ell} - \sigma^2 - \theta_\ell) \\
&\qquad + \frac{p}{n}\left(\frac{\sigma^2\lambda_{p,\ell}}{\lambda_{p,\ell}-\sigma^2} - \frac{\sigma^2(\sigma^2+\theta_\ell)}{\theta_\ell}\right).
\end{aligned}$$

Multiplying by $\sqrt{n}$, the first term converges to the Gaussian law in (49), while the second and third terms vanish by (52) and (51). Slutsky's theorem proves (17). □

4.4 Proof of Theorem 3.5

4.4.1 A lemma to deal with the fact that the $\lambda_{p,\ell}$ are not fixed

Lemma 4.8 (The CLT holds under a small perturbation of the spikes)**.** *Let $\vartheta_{p,1}, \dots, \vartheta_{p,K}$ be possibly random positive numbers, and let Z_p be as in Corollary 4.2. Put*

$$\Lambda_p = \mathrm{diag}(\sigma^2 + \vartheta_{p,1}, \dots, \sigma^2 + \vartheta_{p,K}, \sigma^2, \dots, \sigma^2),$$

and let Λ_p^{id} be as in Section 2.5. Consider the two sample covariance matrices built from the same Z_p,

$$\widehat{\Sigma}_p = \Lambda_p^{1/2}\Big(\frac{1}{n}Z_pZ_p^*\Big)\Lambda_p^{1/2}, \qquad \widehat{\Sigma}_p^{\mathrm{id}} = (\Lambda_p^{\mathrm{id}})^{1/2}\Big(\frac{1}{n}Z_pZ_p^*\Big)(\Lambda_p^{\mathrm{id}})^{1/2},$$

with ordered eigenvalues $\widehat{\lambda}_{p,\ell}$ and $\widehat{\lambda}_{p,\ell}^{\mathrm{id}}$. Assume $p/n \to c \in (0,\infty)$ and $\vartheta_{p,\ell} - \theta_\ell = \mathrm{O}_{\mathbb{P}}(1/p)$ for every $\ell \le K$. Then

$$\max_{\ell \le p} \sqrt{n}\,\big|\widehat{\lambda}_{p,\ell} - \widehat{\lambda}_{p,\ell}^{\mathrm{id}}\big| = o_{\mathbb{P}}(1).$$

Proof. **Step 1: the two population matrices are close.** By construction Λ_p and Λ_p^{id} differ only in their first K diagonal entries, and there the difference is $\vartheta_{p,\ell} - \theta_\ell$. Hence

$$\|\Lambda_p - \Lambda_p^{\mathrm{id}}\|_{\mathrm{op}} = \max_{\ell \le K} |\vartheta_{p,\ell} - \theta_\ell| = \mathrm{O}_{\mathbb{P}}(1/p).$$

In particular, on an event Ω_p with $\mathbb{P}(\Omega_p) \to 1$, all the diagonal entries of Λ_p and of Λ_p^{id} lie in the fixed compact set $[\sigma^2/2, \sigma^2 + \theta_1 + 1] \subset (0,\infty)$. The same bound then holds for the square roots: since $t \mapsto \sqrt{t}$ is Lipschitz on any interval bounded away from 0, there is a constant L with

$$\|\Lambda_p^{1/2} - (\Lambda_p^{\mathrm{id}})^{1/2}\|_{\mathrm{op}} \le L\,\|\Lambda_p - \Lambda_p^{\mathrm{id}}\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(1/p).$$

Step 2: the two sample matrices are close. Put $\Upsilon_p := \Lambda_p^{1/2} - (\Lambda_p^{\mathrm{id}})^{1/2}$. Adding and subtracting,

$$\widehat{\Sigma}_p - \widehat{\Sigma}_p^{\mathrm{id}} = \Upsilon_p\Big(\frac{1}{n}Z_pZ_p^*\Big)\Lambda_p^{1/2} + (\Lambda_p^{\mathrm{id}})^{1/2}\Big(\frac{1}{n}Z_pZ_p^*\Big)\Upsilon_p,$$

so, using $\|MN\|_{\mathrm{op}} \le \|M\|_{\mathrm{op}}\|N\|_{\mathrm{op}}$,

$$\|\widehat{\Sigma}_p - \widehat{\Sigma}_p^{\mathrm{id}}\|_{\mathrm{op}} \le \|\Upsilon_p\|_{\mathrm{op}}\Big\|\frac{1}{n}Z_pZ_p^*\Big\|_{\mathrm{op}}\big(\|\Lambda_p^{1/2}\|_{\mathrm{op}} + \|(\Lambda_p^{\mathrm{id}})^{1/2}\|_{\mathrm{op}}\big).$$

In the proportional regime $\|n^{-1}Z_pZ_p^*\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(1)$, since its largest eigenvalue converges almost surely to $(1+\sqrt{c})^2$ (Bai–Silverstein [1]), and the two square-root norms are $\mathrm{O}_{\mathbb{P}}(1)$ by Step 1. With $\|\Upsilon_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(1/p)$,

$$\|\widehat{\Sigma}_p - \widehat{\Sigma}_p^{\mathrm{id}}\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(1/p).$$

Step 3: from matrices to eigenvalues. By Weyl's inequality (Lemma A.10), for every ℓ,

$$|\widehat{\lambda}_{p,\ell} - \widehat{\lambda}^{\mathrm{id}}_{p,\ell}| \le \|\widehat{\Sigma}_p - \widehat{\Sigma}^{\mathrm{id}}_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(1/p).$$

Multiplying by $\sqrt{n}$ and using $\sqrt{n}/p = (1/\sqrt{n}) \cdot (n/p) \to 0$ in the proportional regime, since $n/p \to 1/c$ and $1/\sqrt{n} \to 0$, gives the claim. □

4.4.2 Proof of Theorem 3.5

Put $K = r + s$. Part (i) follows from a finite-rank comparison with a white Wishart matrix on the original probability space. For parts (ii)–(iv), we use the following coupling with a model whose spikes are fixed.

The ideal model. For parts (ii)–(iv), we work with the representation of Corollary 4.2: the eigenvalues of $\widehat{\Sigma}_p$ have the same law as those of $\Lambda_p^{1/2}(n^{-1}Z_pZ_p^*)\Lambda_p^{1/2}$, where Z_p is independent of $\mathcal{F}_p$. Using the same Z_p, we build the ideal model $\widehat{\Sigma}^{\mathrm{id}}_p$ of Section 4.4.1. Its population matrix Λ^{id}_p is deterministic and has the fixed spikes $\theta_1, \dots, \theta_K$. Thus Theorems 2.4, 2.5 and 2.6 apply directly to this ideal model under their respective hypotheses. All comparisons below are made in this coupling.

The comparison bound. By Proposition 2.3, $\vartheta_{p,\ell} := \lambda_{p,\ell} - \sigma^2$ satisfies $\vartheta_{p,\ell} - \theta_\ell = \mathrm{O}_{\mathbb{P}}(1/p)$ for every $\ell \le K$. The proof of Lemma 4.8 therefore gives

$$\max_{\ell \le p} \big|\widehat{\lambda}_{p,\ell} - \widehat{\lambda}^{\mathrm{id}}_{p,\ell}\big| = \mathrm{O}_{\mathbb{P}}(1/p) = o_{\mathbb{P}}(n^{-1/2}). \tag{53}$$

The last equality follows from $\sqrt{n}/p = (n/p)/\sqrt{n} \to 0$.

Proof of (i). Work on the original probability space and set

$$X_p = (x_{p,1}, \dots, x_{p,n}), \qquad Z_p^{(0)} = \Sigma_p^{-1/2} X_p.$$

Conditionally on O_p, the entries of $Z_p^{(0)}$ are independent standard real or circular complex Gaussian variables. This conditional law does not depend on O_p, so $Z_p^{(0)}$ is also a standard Gaussian matrix unconditionally. Put

$$M_p = \frac{1}{n} Z_p^{(0)} (Z_p^{(0)})^*, \qquad S_p^{(0)} = \sigma^2 M_p, \qquad D_p = \Sigma_p^{1/2} - \sigma I_p,$$

where σ is the positive square root of σ^2. Since $\Sigma_p - \sigma^2 I_p$ has rank at most K, the same is true of D_p. Moreover,

$$\widehat{\Sigma}_p - S_p^{(0)} = D_p M_p \Sigma_p^{1/2} + \sigma M_p D_p,$$

and hence $\operatorname{rank}(\widehat{\Sigma}_p - S_p^{(0)}) \le 2K$. For a Hermitian matrix M, let F^M denote its empirical spectral distribution function. The rank inequality [1] gives

$$\sup_{t \in \mathbb{R}} |F^{\widehat{\Sigma}_p}(t) - F^{S_p^{(0)}}(t)| \le \frac{\operatorname{rank}(\widehat{\Sigma}_p - S_p^{(0)})}{p} \le \frac{2K}{p} \longrightarrow 0.$$

By Theorem A.6, the empirical spectral distribution of $S_p^{(0)}$ converges weakly, almost surely, to μ_{c,σ^2}. The same conclusion therefore holds for $\widehat{\Sigma}_p$. □

Proof of (ii). Fix ℓ with $\theta_\ell > \sigma^2\sqrt{c}$; by assumption the spikes are distinct, so θ_ℓ is simple. We prove the central limit theorem; the convergence in probability then follows, since the central limit theorem gives $\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell) = \mathrm{O}_{\mathbb{P}}(n^{-1/2}) = o_{\mathbb{P}}(1)$, and $\phi_{p/n}(\theta_\ell) \to \phi_c(\theta_\ell)$ because $p/n \to c$; adding the two gives $\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \phi_c(\theta_\ell)$.

Step 1: the ideal model. The ideal model $\widehat{\Sigma}_p^{\mathrm{id}}$ of the preamble has fixed spikes $\theta_1, \dots, \theta_K$ and is built from the same Gaussian matrix Z_p. The spike θ_ℓ is supercritical and simple, so Theorem 2.5 applies to the ideal model and gives

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell}^{\mathrm{id}} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \mathcal{N}(0, \kappa V_c(\theta_\ell)). \tag{54}$$

Step 2: comparing the ideal model and our model. By Proposition 2.3, the true spikes satisfy $(\lambda_{p,\ell} - \sigma^2) - \theta_\ell = \mathrm{O}_{\mathbb{P}}(1/p)$, and both models are built from the same Z_p, so Lemma 4.8 applies with $\vartheta_{p,\ell} = \lambda_{p,\ell} - \sigma^2$ and gives, through (53),

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \widehat{\lambda}_{p,\ell}^{\mathrm{id}}\right) = o_{\mathbb{P}}(1).$$

Writing

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) = \sqrt{n}\left(\widehat{\lambda}_{p,\ell}^{\mathrm{id}} - \phi_{p/n}(\theta_\ell)\right) + \sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \widehat{\lambda}_{p,\ell}^{\mathrm{id}}\right),$$

and combining (54) with the $o_{\mathbb{P}}(1)$ term, Slutsky's theorem gives

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \mathcal{N}(0, \kappa V_c(\theta_\ell)).$$

Both inputs used here are unconditional. The ideal model $\widehat{\Sigma}_p^{\mathrm{id}}$ has deterministic spikes and is a function of Z_p alone, so Theorem 2.5 applies to it directly; and Lemma 4.8 is stated for possibly random spikes, its hypothesis $\vartheta_{p,\ell} - \theta_\ell = \mathrm{O}_{\mathbb{P}}(1/p)$ being supplied by Proposition 2.3. So the randomness of O_p is already absorbed by Lemma 4.8, and no conditioning on $\mathcal{F}_p$ is needed.

□

Proof of (iii). Fix ℓ with $0 < \theta_\ell \le \sigma^2\sqrt{c}$. Theorem 2.4, applied to the ideal model with fixed spikes, gives

$$\widehat{\lambda}_{p,\ell}^{\mathrm{id}} \xrightarrow{\text{a.s.}} \sigma^2(1+\sqrt{c})^2.$$

By (53),

$$\widehat{\lambda}_{p,\ell} - \widehat{\lambda}_{p,\ell}^{\mathrm{id}} = \mathrm{O}_{\mathbb{P}}(1/p) = o_{\mathbb{P}}(1).$$

Combining these statements yields $\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2(1+\sqrt{c})^2$. □

Proof of (iv). Assume $\theta_K > \sigma^2\sqrt{c}$. From (10) and $p/n \to c \in (0,\infty)$, we obtain

$$s_{p,n}^+ = \sigma^2(1+\sqrt{c})^{4/3} c^{-1/6} n^{-2/3}(1+o(1)).$$

Also, the centering satisfies the exact identity

$$m_{p,n}^+ - \sigma^2\big(1+\sqrt{p/n}\big)^2 = -\frac{\sigma^2}{n}\left(1 + \frac{2\sqrt{p}}{\sqrt{n}+\sqrt{n-1}}\right) = \mathrm{O}(n^{-1}).$$

All spikes of the ideal model are supercritical, so Theorem 2.6 gives

$$\frac{\widehat{\lambda}_{p,K+1}^{\mathrm{id}} - m_{p,n}^+}{s_{p,n}^+} \xrightarrow{\text{law}} \mathrm{TW}.$$

On the other hand, (53) implies

$$\frac{\widehat{\lambda}_{p,K+1} - \widehat{\lambda}_{p,K+1}^{\mathrm{id}}}{s_{p,n}^+} = \mathrm{O}_{\mathbb{P}}\left(\frac{n^{2/3}}{p}\right) = o_{\mathbb{P}}(1).$$

Slutsky's theorem therefore yields

$$\frac{\widehat{\lambda}_{p,K+1} - m_{p,n}^+}{s_{p,n}^+} \xrightarrow{\text{law}} \mathrm{TW}.$$

Convergence in distribution implies tightness, so $\widehat{\lambda}_{p,K+1} - m_{p,n}^{+} = \mathrm{O}_{\mathbb{P}}(n^{-2/3})$. Combining this with the centering identity and $n^{-1} = o(n^{-2/3})$ proves

$$\widehat{\lambda}_{p,K+1} = \sigma^2\big(1 + \sqrt{p/n}\big)^2 + \mathrm{O}_{\mathbb{P}}(n^{-2/3}).$$

□

4.5 Proofs of the divergence of the fluctuations with a wrong scaling

Proof of Proposition 3.6. We apply Lemma A.18(ii) three times, and we check its two hypotheses in each case: convergence in distribution under one scaling, with a limit having no atom at zero, and divergence of the ratio of the two scalings.

(i) The assumption $p^2/n \to 0$ is the case $\tau = 0$ of Theorem 3.1, so

$$p\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} W_\ell,$$

and Corollary A.21 with $\tau = 0$ gives $\mathbb{P}(W_\ell = 0) = 0$. Moreover $(p/\sqrt{n})^2 = p^2/n \to 0$ and $p/\sqrt{n} > 0$, so $\sqrt{n}/p \to \infty$. Lemma A.18(ii) with $u_p = p$ and $v_p = \sqrt{n}$ gives the claim.

(ii) By Theorem 3.4,

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \mathcal{N}\big(0, \kappa V_0(\theta_\ell)\big).$$

The variance $\kappa V_0(\theta_\ell) = \kappa(\sigma^2+\theta_\ell)^2$ is positive, so this law has no atom at zero by Corollary A.21. Moreover $p/\sqrt{n} = \sqrt{p^2/n} \to \infty$. Lemma A.18(ii) with $u_p = \sqrt{n}$ and $v_p = p$ gives the claim.

(iii) By Theorem 3.5(ii),

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \mathcal{N}\big(0, \kappa V_c(\theta_\ell)\big).$$

The variance is positive: the assumption $\theta_\ell > \sigma^2\sqrt{c}$ gives $\theta_\ell^2 > c\sigma^4$, hence $1 - c\sigma^4/\theta_\ell^2 > 0$ and $V_c(\theta_\ell) > 0$. So this law has no atom at zero by Corollary A.21. Moreover $p/\sqrt{n} = (p/n)\sqrt{n} \to \infty$, because $p/n \to c > 0$ and $n \to \infty$. Lemma A.18(ii) with $u_p = \sqrt{n}$ and $v_p = p$ gives the claim.

The last assertion follows from Lemma A.19 in the three cases. □

Proof of Proposition 3.7. Theorem 3.1 gives

$$p\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}\, G_\ell.$$

The assumption $p^2/n \to \tau$ means $(p/\sqrt{n})^2 \to \tau$; since $p/\sqrt{n} > 0$ and the square root is continuous on $[0,\infty)$, this gives $p/\sqrt{n} \to \sqrt{\tau}$, and $\tau > 0$ gives $\sqrt{n}/p \to \tau^{-1/2} \in (0,\infty)$. Lemma A.18(i) with $u_p = p$, $v_p = \sqrt{n}$ and $\gamma = \tau^{-1/2}$ therefore gives

$$\sqrt{n}\left(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)\right) \xrightarrow{\text{law}} \tau^{-1/2}\Big(W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}\, G_\ell\Big) = \tau^{-1/2} W_\ell + \sqrt{\kappa V_0(\theta_\ell)}\, G_\ell,$$

because $\tau^{-1/2}\sqrt{\kappa\tau V_0(\theta_\ell)} = \sqrt{\kappa V_0(\theta_\ell)}$. □

4.6 Proofs of the properties of the spectral test

In section 4.6.1, we prove the consistency and level of our test under assumptions (A1)-(A7). In section 4.6.2, we prove the properties of the test when relaxing assumption (A6).

4.6.1 Proof of consistency and level of the test

We first isolate the continuity property needed to replace the unknown reference-law parameters by consistent estimates. The statement also covers a vanishing Gaussian component and weights of either sign.

Lemma 4.9 (Continuity of the reference law and its quantile)**.** *Fix $\gamma > 0$, $a \in (0,1)$, and an integer $m \geq 1$. Let $Z_1, \dots, Z_m$ be independent χ^2_1 variables in the real case or independent* Exp(1) *variables in the complex case, and let $G \sim \mathcal{N}(0,1)$ be independent of them. For $b_j \notin \{0, \gamma\}$ and $v \geq 0$, define*

$$L(b,v) = \sum_{j=1}^{m} \frac{\gamma b_j}{\gamma - b_j} Z_j + vG, \qquad F_{b,v}(x) = \mathbb{P}(L(b,v) \leq x),$$

and $q(b,v) = \inf\{x : F_{b,v}(x) \geq 1-a\}$. The law of $L(b,v)$ is absolutely continuous, and its cdf is strictly increasing on the interior of its support. Its quantile lies in that interior and satisfies $F_{b,v}(q(b,v)) = 1-a$. If $(b,v) \to (\beta, v_0)$, where $\beta_j \notin \{0,\gamma\}$ and $v_0 \geq 0$, then

$$F_{b,v}(x) \to F_{\beta,v_0}(x) \quad \text{for every } x \in \mathbb{R}, \qquad q(b,v) \to q(\beta, v_0).$$

Proof. (A) Regularity. All coefficients are nonzero, so Lemma A.20 gives absolute continuity. The support is $\mathbb{R}$ if $v > 0$ or if the coefficients have both signs; otherwise it is one of the two closed half-lines with endpoint zero. Every nonempty open interval in the interior of this support has positive probability, since the Z_j have positive densities on $(0,\infty)$ and G has a positive density on $\mathbb{R}$. Thus the cdf is continuous and strictly increasing on that interior. Since $1-a \in (0,1)$, its quantile is an interior point and its cdf value there is $1-a$.

(B) Continuity. Using the same auxiliary variables for all parameters gives $L(b,v) \to L(\beta, v_0)$ almost surely. Absolute continuity of the limiting law implies convergence of the cdfs at every fixed x. Write $q = q(\beta, v_0)$. For every sufficiently small $\varepsilon > 0$,

$$F_{\beta,v_0}(q-\varepsilon) < 1-a < F_{\beta,v_0}(q+\varepsilon).$$

The same inequalities hold with $F_{b,v}$ for all parameters sufficiently close to (β, v_0), and imply $q-\varepsilon < q(b,v) \leq q+\varepsilon$. This proves quantile continuity. □

We now prove Theorem 3.12. Steps 1–3 establish consistency of the nuisance-parameter estimates. Steps 4–5 prove convergence of the plug-in and Monte-Carlo critical values, and Steps 6–7 establish level and consistency.

Notation and tools. We write $\lambda_{p,k} := \lambda_k(\Sigma_p)$ for the k-th largest eigenvalue of the population covariance. Since $A_p + O_p B_p O_p^*$ has rank at most $r+s$, we have $\lambda_{p,k} = \sigma^2$ for $k > r+s$, and for $k \leq r+s$ the number $\lambda_{p,k} - \sigma^2$ is the k-th largest eigenvalue of $A_p + O_p B_p O_p^*$, which converges to θ_k. We use three facts repeatedly.

(T1) Weyl's inequality (Lemma A.10): $\max_k |\widehat{\lambda}_{p,k} - \lambda_{p,k}| \leq \|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}}$.

(T2) Operator-norm bound: by Corollary A.3 and Remark A.4, since $\|\Sigma_p\|_{\mathrm{op}}$ is bounded by a deterministic constant, and since $p/n \to 0$ in this regime, $\|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$.

(T3) Eigenvalue limit: by Theorem 3.1, $\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_\ell$.

Step 1: $\tilde{\sigma}^2 \xrightarrow{\mathbb{P}} \sigma^2$ and $\hat{\sigma}^2 - \sigma^2 = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn})$. Split $\tilde{\sigma}^2 - \sigma^2 = \frac{1}{p}(\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p) + \frac{1}{p}(\operatorname{tr}\Sigma_p - p\sigma^2)$. The deterministic part equals $\frac{1}{p}\sum_{\ell \leq r+s}(\lambda_{p,\ell} - \sigma^2) = \mathrm{O}(1/p)$, a sum of $r+s$ bounded terms. For the random part, using $\operatorname{tr}\widehat{\Sigma}_p = \frac{1}{n}\sum_{k=1}^{n} \|x_{p,k}\|_2^2$, this is, conditionally on $\mathcal{F}_p$, an average of

n i.i.d. terms of common mean $\mathbb{E}[\|x_{p,1}\|_2^2 \mid \mathcal{F}_p] = \operatorname{tr}\Sigma_p$. Writing $x_{p,1} = \Sigma_p^{1/2} z$ with $z \sim \mathcal{N}_{\mathbb{F}}(0, I_p)$ and diagonalising, $\|x_{p,1}\|_2^2 = \sum_k \lambda_{p,k}|\tilde{z}_k|^2$ with $|\tilde{z}_k|^2$ i.i.d. of variance κ, so $\operatorname{Var}(\|x_{p,1}\|_2^2 \mid \mathcal{F}_p) = \kappa \operatorname{tr}(\Sigma_p^2)$ and $\operatorname{Var}(\operatorname{tr}\widehat{\Sigma}_p \mid \mathcal{F}_p) = \frac{\kappa}{n}\operatorname{tr}(\Sigma_p^2)$. Since $\operatorname{tr}(\Sigma_p^2) = (p-r-s)\sigma^4 + \sum_{k\le r+s}\lambda_{p,k}^2 = \mathrm{O}(p)$ with a constant depending only on σ^2 and the θ_k, the conditional variance is bounded uniformly in $\mathcal{F}_p$:

$$\operatorname{Var}\big(\operatorname{tr}\widehat{\Sigma}_p \mid \mathcal{F}_p\big) = \frac{\kappa}{n}\operatorname{tr}(\Sigma_p^2) \le C\,\frac{p}{n},$$

for a constant C not depending on the realisation of O_p. The conditional Chebyshev inequality gives, for every $t > 0$,

$$\mathbb{P}\big(|\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p| \ge t \mid \mathcal{F}_p\big) \le \frac{\operatorname{Var}(\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p \mid \mathcal{F}_p)}{t^2} = \frac{\operatorname{Var}(\operatorname{tr}\widehat{\Sigma}_p \mid \mathcal{F}_p)}{t^2} \le \frac{C\,p/n}{t^2},$$

and the right-hand side is deterministic; taking expectations over $\mathcal{F}_p$ (the tower property) yields $\mathbb{P}(|\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p| \ge t) \le C\,(p/n)/t^2$. With $t = K\sqrt{p/n}$ this is $\le C/K^2$, uniformly in p, so $\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$. Dividing by p,

$$\tfrac{1}{p}\big(\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p\big) = \mathrm{O}_{\mathbb{P}}\big(\tfrac{1}{p}\sqrt{p/n}\big) = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}).$$

Together with the deterministic part $\frac{1}{p}(\operatorname{tr}\Sigma_p - p\sigma^2) = \mathrm{O}(1/p)$, this gives $\tilde{\sigma}^2 - \sigma^2 = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}) + \mathrm{O}(1/p) \to 0$.

For $\hat{\sigma}^2$, we work on $\{\widehat{r+s} = r+s\}$, whose probability tends to 1 by Step 2. On this event, $\hat{\sigma}^2 = \frac{1}{p-r-s}\sum_{k>r+s}\widehat{\lambda}_{p,k}$, and since $\lambda_{p,k} = \sigma^2$ for $k > r+s$, subtracting gives $(p-r-s)(\hat{\sigma}^2 - \sigma^2) = \sum_{k>r+s}(\widehat{\lambda}_{p,k} - \lambda_{p,k})$. Writing $\sum_{k>r+s} = \sum_{k=1}^{p} - \sum_{k\le r+s}$,

$$(p-r-s)(\hat{\sigma}^2 - \sigma^2) = \underbrace{(\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p)}_{=:S_1} - \underbrace{\sum_{k\le r+s}(\widehat{\lambda}_{p,k} - \lambda_{p,k})}_{=:S_2}.$$

For S_1, the bound above gives $S_1 = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$. For S_2, each of the $r+s$ terms is $\le \|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}}$ by (T1), so $|S_2| \le (r+s)\|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$ by (T2). Dividing by $p-r-s$, which is of order p,

$$|\hat{\sigma}^2 - \sigma^2| \le \frac{|S_1| + |S_2|}{p-r-s} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n}/p) = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}).$$

This holds on $\{\widehat{r+s} = r+s\}$, hence unconditionally. We record the consequence used below, at scale p: $p(\hat{\sigma}^2 - \sigma^2) = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n}) \to 0$.

Step 2: $\mathbb{P}(\widehat{r+s} = r+s) \to 1$**.** Write $\tilde{\theta}_p := \tilde{\sigma}^2(1+\sqrt{p/n})^2 + \epsilon_p$. The count equals $r+s$ when the $r+s$ spike eigenvalues exceed $\tilde{\theta}_p$ and the $p-r-s$ bulk eigenvalues fall below it. From Step 1, we can write

$$\tilde{\sigma}^2 - \sigma^2 = R_p + D_p, \qquad R_p := \tfrac{1}{p}\big(\operatorname{tr}\widehat{\Sigma}_p - \operatorname{tr}\Sigma_p\big), \quad D_p := \tfrac{1}{p}\operatorname{tr}\Sigma_p - \sigma^2 = \tfrac{1}{p}\textstyle\sum_{k\le r+s}(\lambda_{p,k} - \sigma^2).$$

with $D_p = \mathrm{O}(1/p)$, $R_p = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn})$, so

$$\tilde{\theta}_p - \sigma^2 = (D_p + R_p)(1+\sqrt{p/n})^2 + \sigma^2(2\sqrt{p/n} + p/n) + \epsilon_p. \tag{55}$$

For every bulk index $k > r+s$, $\lambda_{p,k} = \sigma^2$, so by (T1)–(T2) $\widehat{\lambda}_{p,k} - \sigma^2 \le \|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$ uniformly in k. Dropping the nonnegative terms in (55), $\tilde{\theta}_p - \sigma^2 \ge R_p(1+\sqrt{p/n})^2 + \epsilon_p$, hence

$$\widehat{\lambda}_{p,k} - \tilde{\theta}_p = \widehat{\lambda}_{p,k} - \sigma^2 + \sigma^2 - \tilde{\theta}_p \le -\epsilon_p + \mathrm{O}_{\mathbb{P}}(\sqrt{p/n}) + \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}).$$

Both error terms are $o_{\mathbb{P}}(\epsilon_p)$, since $\sqrt{p/n} = o(\epsilon_p)$ by (A3) and $1/\sqrt{pn} \le \sqrt{p/n}$. So with probability tending to one every bulk eigenvalue is below $\tilde{\theta}_p$.

For $k \le r+s$, $\widehat{\lambda}_{p,k} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_k$ by Theorem 3.1. On the other hand (55) gives $\tilde{\theta}_p \xrightarrow{\mathbb{P}} \sigma^2$, since $D_p \to 0$, $R_p \xrightarrow{\mathbb{P}} 0$, $p/n \to 0$ and $\epsilon_p \to 0$. Hence $\widehat{\lambda}_{p,k} - \tilde{\theta}_p \xrightarrow{\mathbb{P}} \theta_k > 0$, so every spike exceeds the threshold with probability tending to one.

So $\widehat{r+s} = r+s$ with probability tending to one, and, r being known, $\hat{s} = \widehat{r+s} - r = s$ on the same event.

Step 3: the centered eigenvalues converge. On $\{\widehat{r+s} = r+s\}$, for each $\ell \in [r+s]$,

$$\hat{\mu}_\ell - \theta_\ell = \underbrace{(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell})}_{\text{(i)}} + \underbrace{\big((\lambda_{p,\ell} - \sigma^2) - \theta_\ell\big)}_{\text{(ii)}} + \underbrace{(\sigma^2 - \hat{\sigma}^2)}_{\text{(iii)}},$$

where $\hat{\mu}_\ell = \widehat{\lambda}_{p,\ell} - \hat{\sigma}^2$. Term (i) is $O_{\mathbb{P}}(\sqrt{p/n})$ by (T1)–(T2); term (ii) is the gap between the ℓ-th eigenvalue of $A_p + O_p B_p O_p^*$ and its limit θ_ℓ, which is $O_{\mathbb{P}}(1/p)$ by Proposition 2.3; term (iii) is $O_{\mathbb{P}}(1/\sqrt{pn})$ by Step 1. All three tend to 0, so $\hat{\mu}_\ell \xrightarrow{\mathbb{P}} \theta_\ell$. Applying this to the ranks labelled as batch effects gives $\hat{\beta}_j \xrightarrow{\mathbb{P}} \beta_j$ for every $j \in [s]$.

Step 4: convergence of the plug-in reference law. Relabel the batch strengths in decreasing order, and order their estimates in the same way. This does not change the reference law. By Steps 1–3 and (A6), with probability tending to one, $\hat{s} = s$ and the selected batch estimates satisfy

$$\max_{1 \le j \le s} |\hat{\beta}_j - \beta_j| \xrightarrow{\mathbb{P}} 0, \qquad \hat{v} := \sqrt{\kappa p^2/n}\,(\hat{\sigma}^2 + \gamma) \xrightarrow{\mathbb{P}} v_0 := \sqrt{\kappa\tau}\,(\sigma^2 + \gamma).$$

Assumption (A2) gives $\min_j \min\{|\beta_j|, |\beta_j - \gamma|\} > 0$. Hence there is an event E_p, with $\mathbb{P}(E_p) \to 1$, on which $\hat{s} = s$, all required ranks are available, and $\hat{\beta}_j \notin \{0, \gamma\}$ for every j. For the convergence statements below, extend the fitted parameters by (β, v_0) on E_p^c; this is only a proof convention.

Put $F_0 = F_{\beta, v_0}$, $q = q(\beta, v_0)$, and $\widehat{F}_p = F_{\hat{\beta}, \hat{v}}$. Lemma 4.9 and the continuous mapping theorem give

$$q(\hat{\beta}, \hat{v}) \xrightarrow{\mathbb{P}} q, \qquad \widehat{F}_p(x) \xrightarrow{\mathbb{P}} F_0(x) \quad \text{for every fixed } x \in \mathbb{R}.$$

These conclusions hold under both hypotheses. Under H_0, F_0 is the cdf of the limiting test statistic before the negligible noise-estimation correction.

Step 5: convergence of the Monte-Carlo critical value. Write $u = 1 - a$, and let $\mathcal{D}_p$ be the sigma-field generated by the data. Conditionally on $\mathcal{D}_p$, generate the $M = M_p$ draws independently from $\widehat{L}$ using fresh auxiliary randomness. Their empirical cdf

$$F_{p,M}(x) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}_{\{c_m \le x\}}$$

satisfies, on E_p,

$$\mathbb{E}[F_{p,M}(x) \mid \mathcal{D}_p] = \widehat{F}_p(x), \qquad \mathrm{Var}(F_{p,M}(x) \mid \mathcal{D}_p) \le \frac{1}{4M}.$$

Fix a sufficiently small $\varepsilon > 0$. By Lemma 4.9, $F_0(q - \varepsilon) < u < F_0(q + \varepsilon)$. Choose $\delta > 0$ smaller than both of these cdf gaps. Step 4 implies that

$$H_p = E_p \cap \{\widehat{F}_p(q - \varepsilon) \le u - \delta,\ \widehat{F}_p(q + \varepsilon) \ge u + \delta\}$$

has probability tending to one. Conditional Chebyshev inequalities at $q-\varepsilon$ and $q+\varepsilon$, followed by integration over the data, give

$$\mathbb{P}(|\hat{q}_{1-a} - q| > \varepsilon) \le \mathbb{P}(H_p^c) + \frac{1}{2M\delta^2}.$$

Indeed, if the empirical cdf is below u at $q-\varepsilon$ and at least u at $q+\varepsilon$, then $c_{(\lceil Mu \rceil)} \in (q-\varepsilon, q+\varepsilon]$. Since $M_p \to \infty$, this proves $\hat{q}_{1-a} \xrightarrow{\mathbb{P}} q$. We henceforth write $q_{1-a} = q$. No rate condition relating M_p to p or n is required.

Step 6: asymptotic level. Under H_0 ($\theta_\ell = \gamma$),

$$T_\ell = \underbrace{p(\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\gamma))}_{\text{(A)}} + \underbrace{p(\phi_{p/n}(\gamma) - \hat{\phi}(\gamma))}_{\text{(B)}}.$$

Since $\theta_\ell = \gamma$, Theorem 3.1 applies at rank ℓ. By (A5), $\theta_\ell = \alpha_i$ for some signal index i, and

$$W_\ell(\gamma) = \sum_{j=1}^{s} \frac{\gamma\beta_j}{\gamma - \beta_j} |g_{ij}|^2.$$

Thus

$$\text{(A)} \xrightarrow{\text{law}} L(\gamma) = W_\ell(\gamma) + v_0 G.$$

For (B), with $\phi_c(y) = \sigma^2 + y + c\,\sigma^2(\sigma^2+y)/y$, $c = p/n$, and $\hat{\phi}(y) = \hat{\sigma}^2 + y + \frac{p\,\hat{\sigma}^2(\hat{\sigma}^2+y)}{ny}$,

$$\text{(B)} = p(\sigma^2 - \hat{\sigma}^2) + \frac{p^2}{n} \cdot \frac{1}{\gamma}\big(\sigma^2(\sigma^2+\gamma) - \hat{\sigma}^2(\hat{\sigma}^2+\gamma)\big).$$

By Step 1, $\hat{\sigma}^2 - \sigma^2 = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn})$, so $p(\sigma^2 - \hat{\sigma}^2) = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n}) \xrightarrow{\mathbb{P}} 0$ since $p/n \to 0$. For the second term, the identity $a(a+\gamma) - b(b+\gamma) = (a-b)(a+b+\gamma)$ gives $\hat{\sigma}^2(\hat{\sigma}^2+\gamma) - \sigma^2(\sigma^2+\gamma) = (\hat{\sigma}^2 - \sigma^2)(\hat{\sigma}^2 + \sigma^2 + \gamma) = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}) = \mathrm{o}_{\mathbb{P}}(1)$, and $p^2/n \to \tau < \infty$ is $\mathrm{O}(1)$, so the second term is $\mathrm{O}(1) \cdot \mathrm{o}_{\mathbb{P}}(1) \xrightarrow{\mathbb{P}} 0$. Thus (B) $\xrightarrow{\mathbb{P}} 0$, and by Slutsky $T_\ell \xrightarrow{\text{law}} L(\gamma)$.

By Step 5, $\hat{q}_{1-a} \xrightarrow{\mathbb{P}} q_{1-a}$, a constant, so Slutsky gives $T_\ell - \hat{q}_{1-a} \xrightarrow{\text{law}} L(\gamma) - q_{1-a}$. Now $\mathbb{P}(T_\ell > \hat{q}_{1-a}) = \mathbb{E}[h(T_\ell - \hat{q}_{1-a})]$ with $h(x) = \mathbf{1}_{\{x>0\}}$, which is bounded and continuous except at 0; and the limit $L(\gamma) - q_{1-a}$ has no atom at 0 because $L(\gamma)$ has a density (part (A) of the lemma). The Portmanteau theorem then gives

$$\mathbb{P}(T_\ell > \hat{q}_{1-a}) \to \mathbb{P}(L(\gamma) > q_{1-a}) = a,$$

by the definition of q_{1-a} and the continuity of $F_{L(\gamma)}$.

Step 7: consistency. Under H_1, $\theta_\ell > \gamma$ is a fixed positive gap and $p/n \to 0$. By Theorem 3.1, $\widehat{\lambda}_{p,\ell} \xrightarrow{\mathbb{P}} \sigma^2 + \theta_\ell$, and $\hat{\phi}(\gamma) \xrightarrow{\mathbb{P}} \sigma^2 + \gamma$, so $\widehat{\lambda}_{p,\ell} - \hat{\phi}(\gamma) \xrightarrow{\mathbb{P}} \theta_\ell - \gamma > 0$, and multiplying by $p \to \infty$ gives $T_\ell \xrightarrow{\mathbb{P}} +\infty$. Moreover $\hat{q}_{1-a} = \mathrm{O}_{\mathbb{P}}(1)$, since Step 5 gives $\hat{q}_{1-a} \xrightarrow{\mathbb{P}} q_{1-a}$ using only $\hat{\beta}_j \to \beta_j$, $\widehat{r+s} = r+s$ and $\hat{\sigma}^2 \to \sigma^2$, which hold under H_1 as well. For any fixed $\xi > 0$, $\mathbb{P}(T_\ell > \hat{q}_{1-a}) \ge \mathbb{P}(T_\ell > \xi) - \mathbb{P}(\hat{q}_{1-a} > \xi)$; the first term tends to 1 for every ξ, and the second can be made arbitrarily small by taking ξ large; hence $\mathbb{P}(T_\ell > \hat{q}_{1-a}) \to 1$.

□

4.6.2 Proofs of the properties of the test when relaxing assumption (A6)

All constructions rely on a single fact: the centered eigenvalues are consistent estimators of the spikes. We isolate it as a lemma.

Lemma 4.10 (Consistency of the centered eigenvalues)**.** *Under assumptions (A1)–(A4), the event* $\{\widehat{r+s} = r+s\}$ *has probability tending to* 1*, and on this event*

$$\hat{\mu}_\ell = \widehat{\lambda}_{p,\ell} - \hat{\sigma}^2 \ \xrightarrow{\mathbb{P}}\ \theta_\ell \qquad \text{for every } \ell \in \{1, \dots, r+s\}.$$

Proof. That $\mathbb{P}(\widehat{r+s} = r+s) \to 1$ is Step 2 of the proof of Theorem 3.12. We work on this event, so that $\hat{\sigma}^2$ is the average of the $p-(r+s)$ smallest eigenvalues.

Fix $\ell \in \{1, \dots, r+s\}$ and let $\lambda_{p,\ell}$ be the ℓ-th largest eigenvalue of the population covariance Σ_p. We decompose the error into three parts by adding and subtracting $\lambda_{p,\ell}$ and σ^2:

$$\hat{\mu}_\ell - \theta_\ell = \underbrace{\big(\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell}\big)}_{\text{(i)}} + \underbrace{\big((\lambda_{p,\ell} - \sigma^2) - \theta_\ell\big)}_{\text{(ii)}} + \underbrace{\big(\sigma^2 - \hat{\sigma}^2\big)}_{\text{(iii)}}.$$

We bound each term separately.

Term (i): By Weyl's inequality (Lemma A.10), $|\widehat{\lambda}_{p,\ell} - \lambda_{p,\ell}| \le \|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}}$. By the operator-norm bound (Corollary A.3) and since $p/n \to 0$ in this regime, $\|\widehat{\Sigma}_p - \Sigma_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$. Hence term (i) is $\mathrm{O}_{\mathbb{P}}(\sqrt{p/n}) \to 0$.

Term (ii): The number $\lambda_{p,\ell} - \sigma^2$ is the ℓ-th largest eigenvalue of $A_p + O_p B_p O_p^*$, and by Proposition 2.3, $p\big((\lambda_{p,\ell} - \sigma^2) - \theta_\ell\big)$ converges in law to W_ℓ. A sequence that converges in law after multiplication by p is $\mathrm{O}_{\mathbb{P}}(1/p)$, so term (ii) is $\mathrm{O}_{\mathbb{P}}(1/p) \to 0$.

Term (iii): By Step 1 of the proof of Theorem 3.12, $\hat{\sigma}^2 - \sigma^2 = \mathrm{O}_{\mathbb{P}}(1/\sqrt{pn})$, so term (iii) is $\mathrm{O}_{\mathbb{P}}(1/\sqrt{pn}) \to 0$.

The three terms tend to 0 in probability, hence so does their sum, and $\hat{\mu}_\ell \xrightarrow{\mathbb{P}} \theta_\ell$. □

We now use this lemma to prove the properties of our test with prior knowledge of the scale of the signals and under known scale separation.

Proof of Proposition 3.13. The proof has two parts: first we show that the estimated ranks are correct with probability tending to 1; then we transfer the guarantees of Theorem 3.12 by a coupling with an oracle that knows the true ranks.

Part 1: the estimated ranks are correct. Fix a signal index $i \in \{1, \dots, r\}$. Assumption (C1) says that α_i is strictly closer to α_i° than any other spike, so the separation margin

$$\rho_i := \min_{\theta \in \mathcal{S},\ \theta \neq \alpha_i} |\alpha_i^\circ - \theta| - |\alpha_i^\circ - \alpha_i|$$

is a minimum over finitely many terms, each strictly larger than $|\alpha_i^\circ - \alpha_i|$, hence $\rho_i > 0$. By definition of ρ_i, every spike $\theta_\ell \neq \alpha_i$ (that is, every $\ell \neq k_i$) satisfies

$$|\alpha_i^\circ - \theta_\ell| \ \ge\ |\alpha_i^\circ - \alpha_i| + \rho_i. \tag{56}$$

Work on $\{\widehat{r+s} = r+s\}$, where there are exactly $r+s$ centered eigenvalues. For each ℓ set

$$\varepsilon_\ell := |\hat{\mu}_\ell - \alpha_i^\circ| - |\theta_\ell - \alpha_i^\circ|.$$

By the reverse triangle inequality $\big||u| - |v|\big| \le |u - v|$, applied with $u = \hat{\mu}_\ell - \alpha_i^\circ$ and $v = \theta_\ell - \alpha_i^\circ$, we get $|\varepsilon_\ell| \le |\hat{\mu}_\ell - \theta_\ell|$, which tends to 0 in probability by Lemma 4.10. As there are $r+s$ indices (a fixed finite number), $\max_\ell |\varepsilon_\ell| \xrightarrow{\mathbb{P}} 0$.

Now compare, for any $\ell \neq k_i$, the distance of $\hat{\mu}_\ell$ to α_i° with that of $\hat{\mu}_{k_i}$. Writing each distance as its population value plus ε, and using $\theta_{k_i} = \alpha_i$,

$$|\hat{\mu}_\ell - \alpha_i^\circ| - |\hat{\mu}_{k_i} - \alpha_i^\circ| = \big(|\theta_\ell - \alpha_i^\circ| - |\alpha_i - \alpha_i^\circ|\big) + (\varepsilon_\ell - \varepsilon_{k_i}) \overset{(56)}{\geq} \rho_i - |\varepsilon_\ell| - |\varepsilon_{k_i}|.$$

Consider the event $E_p^{(i)} := \big\{\max_\ell |\varepsilon_\ell| < \rho_i/2\big\}$. On $E_p^{(i)}$ the right-hand side is $> \rho_i - \rho_i/2 - \rho_i/2 = 0$, so $\hat{\mu}_{k_i}$ is the strict unique minimizer of $\ell \mapsto |\hat{\mu}_\ell - \alpha_i^\circ|$, that is $\hat{k}_i = k_i$. Since $\max_\ell |\varepsilon_\ell| \xrightarrow{\mathbb{P}} 0$, we have $\mathbb{P}(E_p^{(i)}) \to 1$.

Intersecting over the r signals, the event $\{\widehat{r+s} = r+s\} \cap \bigcap_{i=1}^r E_p^{(i)}$ is a finite intersection of events each of probability tending to 1, hence has probability tending to 1. On it, $\hat{k}_i = k_i$ for every i. The true ranks $k_1, \dots, k_r$ are distinct, because the spikes are distinct by (A2); therefore $\{\hat{k}_1, \dots, \hat{k}_r\} = \{k_1, \dots, k_r\} = \mathcal{K}_{\text{sig}}$, which proves Part 1.

Part 2: coupling with the oracle. Consider two decision rules run on the same data and the same Monte-Carlo draws:

- ψ_p, our test, which uses the estimated signal ranks $\{\hat{k}_1, \dots, \hat{k}_r\}$ to decide which centered eigenvalues become the batch estimates $\hat{\beta}_1, \dots, \hat{\beta}_{\hat{s}}$;
- $\psi_p^\star$, an oracle test, identical in every respect but using the true signal ranks $\{k_1, \dots, k_r\}$.

The oracle $\psi_p^\star$ is exactly the test of Theorem 3.12, which assumes the ranks known; so under H_0, $\mathbb{P}(\psi_p^\star = 1) \to a$, and under H_1, $\mathbb{P}(\psi_p^\star = 1) \to 1$.

Let $R_p := \{\hat{k}_i = k_i \ \forall i\} \cap \{\widehat{r+s} = r+s\}$. On R_p, both rules keep the same centered eigenvalues as batch estimates, so they build the same $\hat{\beta}_j$, the same estimated law $\widehat{L}$, draw from it the same M copies (the draws are shared), obtain the same quantile $\hat{q}_{1-a}$, and compare it to the same statistic T_ℓ; hence $\psi_p = \psi_p^\star$ on R_p. Therefore $\{\psi_p \neq \psi_p^\star\} \subseteq R_p^c$, and

$$\big|\mathbb{P}(\psi_p = 1) - \mathbb{P}(\psi_p^\star = 1)\big| \leq \mathbb{P}(\psi_p \neq \psi_p^\star) \leq \mathbb{P}(R_p^c) \ \longrightarrow \ 0$$

by Part 1. Combining with the limits of $\mathbb{P}(\psi_p^\star = 1)$ gives $\mathbb{P}(\psi_p = 1) \to a$ under H_0 and $\mathbb{P}(\psi_p = 1) \to 1$ under H_1. □

The proof technique is similar when we assume a known scale separation between batch spikes and signal spikes.

Proof of Proposition 3.14. Under (S1) every signal exceeds η and every batch effect is below η, so the r signals are the r largest spikes. Since the spikes are sorted decreasingly, their ranks are $1, \dots, r$, that is $\mathcal{K}_{\text{sig}} = \{1, \dots, r\}$.

Let

$$\delta := \min\Big(\min_{i \in \{1,\dots,r\}} (\alpha_i - \eta), \ \min_{j \in \{1,\dots,s\}} (\eta - \beta_j)\Big),$$

a minimum of finitely many strictly positive numbers by (S1), so $\delta > 0$. Then $\theta_\ell - \eta \geq \delta$ for every signal rank $\ell \leq r$, and $\eta - \theta_\ell \geq \delta$ for every batch rank $\ell > r$.

By Lemma 4.10, $\hat{\mu}_\ell - \theta_\ell \xrightarrow{\mathbb{P}} 0$ for each ℓ. Hence, for a signal rank $\ell \leq r$,

$$\mathbb{P}(\hat{\mu}_\ell \leq \eta) = \mathbb{P}\big(\hat{\mu}_\ell - \theta_\ell \leq \eta - \theta_\ell\big) \leq \mathbb{P}\big(\hat{\mu}_\ell - \theta_\ell \leq -\delta\big) \leq \mathbb{P}\big(|\hat{\mu}_\ell - \theta_\ell| \geq \delta\big) \ \longrightarrow \ 0,$$

and symmetrically, for a batch rank $\ell > r$, $\mathbb{P}(\hat{\mu}_\ell > \eta) \leq \mathbb{P}(|\hat{\mu}_\ell - \theta_\ell| \geq \delta) \to 0$.

Consider the event

$$R_p := \{\widehat{r+s} = r+s\} \cap \bigcap_{\ell \leq r} \{\hat{\mu}_\ell > \eta\} \cap \bigcap_{\ell > r} \{\hat{\mu}_\ell \leq \eta\}.$$

By a union bound over the $r+s$ complementary events, each of probability tending to 0, and since $\mathbb{P}(\widehat{r+s} = r+s) \to 1$, we get $\mathbb{P}(R_p) \to 1$. On R_p, the centered eigenvalues above η are exactly those of the signals, so $\widehat{\mathcal{K}}_{\mathrm{sig}} = \{1, \dots, r\} = \mathcal{K}_{\mathrm{sig}}$.

For level and consistency we argue by coupling with an oracle, as in the proof of Proposition 3.13. Let ψ_p be the present test, which uses the estimated signal ranks $\widehat{\mathcal{K}}_{\mathrm{sig}}$, and let $\psi_p^\star$ be the oracle test of Theorem 3.12, identical but using the true ranks $\mathcal{K}_{\mathrm{sig}}$, run on the same data and Monte-Carlo draws. On the event $R_p := \{\widehat{\mathcal{K}}_{\mathrm{sig}} = \mathcal{K}_{\mathrm{sig}}\} \cap \{\widehat{r+s} = r+s\}$ the two rules select the same batch estimates, hence build the same $\widehat{L}$, the same quantile and the same statistic, so $\psi_p = \psi_p^\star$. Therefore $\{\psi_p \neq \psi_p^\star\} \subseteq R_p^c$, and

$$\big|\mathbb{P}(\psi_p = 1) - \mathbb{P}(\psi_p^\star = 1)\big| \leq \mathbb{P}(\psi_p \neq \psi_p^\star) \leq \mathbb{P}(R_p^c) \;\longrightarrow\; 0,$$

since $\mathbb{P}(R_p) \to 1$ by the first part of the proof. As $\mathbb{P}(\psi_p^\star = 1) \to a$ under H_0 and $\to 1$ under H_1 by Theorem 3.12, the same limits hold for ψ_p. □

5 Discussion

5.1 What the transition means for estimation

Table 1 can be viewed as a statement on the estimation of a spike. The sample eigenvalue does not estimate $\sigma^2 + \theta_\ell$ directly: as in the classical spiked model, it is centered at $\phi_{p/n}(\theta_\ell)$, which differs from $\sigma^2 + \theta_\ell$ by a deterministic bias of order p/n. Once this bias is substracted, two errors remain: the sampling error, of order $n^{-1/2}$, and the Haar fluctuation, of order $1/p$. The size of the total error $\widehat{\lambda}_{p,\ell} - \phi_{p/n}(\theta_\ell)$ is that of the larger of the two, and comparing $n^{-1/2}$ with $1/p$ amounts to comparing n with p^2. This is why the correct normalization is $\sqrt{n}$ when $p^2/n \to \tau \in (0, \infty]$, as in the classical spiked model, and p when $p^2/n \to 0$. The last case has no counterpart in the classical spiked model, where the population matrix is fixed and where the sampling error is the only one.

The shape of the limiting law is also important. In the two regimes where the Haar fluctuation is not negligible compared to the sampling fluctuation, that is $p^2/n \to \tau \in [0, \infty)$, the limit contains the variable W_ℓ, which is neither Gaussian nor centered. Hence, a confidence interval for the value of a spike θ_ℓ built from a Gaussian approximation would then be misspecified. Our results give the law to be used instead, and Algorithm 1 shows how to simulate it.

5.2 Limits of the model and open questions

We list the assumptions that we would like to remove.

Non-Gaussian samples. The results of Bai and Yao [2, 3] cover non-Gaussian spiked models under suitable moment and independence assumptions. Applying such results to the present random population model, however, requires additional work. Lemma 4.1 uses Gaussian rotational invariance, which generally fails when the independent Gaussian entries are replaced by non-Gaussian entries in a fixed coordinate system. Moreover, the sampling fluctuation variance may depend on fourth cumulants and on the population eigenvectors; in general it is not obtained by simply replacing κ in the Gaussian formula. The population fluctuation W_ℓ is unaffected as long as the population model is unchanged. We leave these extensions to future work.

Fixed ranks. The ranks r and s are fixed, and our results do not hold anymore when letting them grow with p. Proposition 2.3 treats the $r \times s$ block of overlaps $a_i^* O_p b_j$ as a small perturbation, and this block is small only because it has a fixed number of entries, each of size $p^{-1/2}$: if r and s grow too fast, it is no longer small and Proposition 2.3 does not hold anymore.

Another difficulty is that the strengths $\alpha_1, \dots, \alpha_r$ and $\beta_1, \dots, \beta_s$ are assumed to be distinct, while the weights $\alpha_i\beta_j/(\alpha_i - \beta_j)$ appearing in W_ℓ grow to infinity when two of them become close: a growing number of strengths in a bounded range would have to be accompanied by an assumption on how they separate. We do not know which limiting law to expect in that setting.

Beyond Haar rotations. Exact Haar invariance can be weakened by imposing conditions directly on the overlaps. Let Q_p be an orthogonal or unitary random matrix, and let $a_1, \dots, a_r$ and $b_1, \dots, b_s$ be orthonormal eigenvectors associated with the nonzero eigenvalues of A_p and B_p. Set

$$H_p = (a_i^* Q_p b_j)_{1 \le i \le r,\, 1 \le j \le s}.$$

Suppose that $H_p \to 0$ almost surely and that $\sqrt{p}\, H_p$ converges jointly in distribution to a matrix of independent standard real or circular complex Gaussian entries. Then the same finite-rank perturbation calculation gives the population limits in Proposition 2.3, with the same random variables W_ℓ. If the conditional Gaussian sampling model is retained, with the standard Gaussian noise independent of Q_p, Lemma 4.1 and the sampling arguments remain valid. Thus the asymptotic conclusions of Theorems 3.1, 3.4 and 3.5 are unchanged. For the population fluctuation limit alone, the joint weak convergence of $\sqrt{p}\, H_p$ is sufficient; the almost-sure condition is used to retain the almost-sure population convergence. Different overlap scales or dependence in their limiting law may change the population fluctuation and its competition with sampling.

Eigenvectors. We only study eigenvalues. The sample eigenvector associated with $\widehat{\lambda}_{p,\ell}$ carries the same two sources of the randomness, coming from the sampling and the Haar matrix, so one may ask whether its behaviour is also governed by p^2/n, as for sample eigenvalues. We did not treat this question and we keep it for further work.

The test. Algorithm 1 has two limitations. The first one is assumption (A6): the reference law W_ℓ is a sum over the batch strengths only, so one has to know which of the estimated spikes are signals and which are batch effects. Section 3.7.5 weakens this assumption but does not remove it, and we do not know how to decide, from the sample eigenvalues alone, whether a given spike comes from A_p or from $O_p B_p O_p^*$. The second one is that the test assumes we are in the regime where $p^2/n \to \tau < \infty$. In reality, a practitioner often only has access to n, p and not always to the rate at which p^2 grows compared to n. A procedure which does not require this knowledge, and which would adapt to the regime (whether $p^2/n \to \infty$ or $p^2/n \to \tau < \infty$), would be more useful in practice.

A Appendix: frequently used facts

This appendix collects standard facts that are used several times in the paper. They are stated in the form needed in the main text.

A.1 A fixed interval for a separated spike

Lemma A.1 (A fixed interval around a separated spike)**.** *Let K be fixed. Suppose that*

$$\lambda_p \to \lambda_* > \sigma^2,$$

and that, for some $\delta_0 > 0$ and all large p,

$$\lambda_p - \sigma^2 \geq \delta_0, \qquad |\lambda_p - \lambda_{p,j}| \geq \delta_0 \quad (2 \leq j \leq K).$$

Set

$$\nu = \frac{1}{16}\min\{\delta_0, \lambda_* - \sigma^2\},$$
$$J = [\lambda_* - \nu, \lambda_* + \nu], \qquad I = [\lambda_* - 2\nu, \lambda_* + 2\nu].$$

Then, for all large p,

$$\lambda_p \in J \subset \operatorname{int} I, \qquad \operatorname{dist}(\lambda_p, \partial I) \geq \nu,$$
$$\operatorname{dist}\big(I, \{\sigma^2\} \cup \{\lambda_{p,j} : 2 \leq j \leq K\}\big) \geq 4\nu.$$

Moreover, there is a fixed $\varepsilon > 0$ such that

$$I \subset (\sigma^2 + \varepsilon, \infty).$$

Proof. Since $\lambda_p \to \lambda_*$, for all large p, $|\lambda_p - \lambda_*| \leq \nu$. This gives the first line. For $2 \leq j \leq K$,

$$|\lambda_* - \lambda_{p,j}| \geq |\lambda_p - \lambda_{p,j}| - |\lambda_p - \lambda_*| \geq \delta_0 - \nu.$$

Every point of I is at distance at most 2ν from λ_*. Hence

$$\operatorname{dist}(I, \lambda_{p,j}) \geq \delta_0 - 3\nu \geq 4\nu.$$

The same argument with σ^2 gives $\operatorname{dist}(I, \sigma^2) \geq 4\nu$. Finally, if $a = \lambda_* - \sigma^2$, then $2\nu \leq a/8$. Thus the left end of I is at least $\sigma^2 + 7a/8$, so one may take, for example, $\varepsilon = a/2$. □

A.2 Gaussian sample covariance bounds

Proposition A.2 (Koltchinskii–Lounici bound in the Gaussian case)**.** *Let $X_1, \dots, X_n$ be independent centered Gaussian vectors in $\mathbb{R}^m$, with covariance matrix Γ. Put*

$$\widehat{\Gamma} = \frac{1}{n}\sum_{k=1}^{n} X_k X_k^*.$$

There is a universal constant $C > 0$ such that, for every $u \geq 1$, with probability at least $1 - e^{-u}$,

$$\|\widehat{\Gamma} - \Gamma\|_{\mathrm{op}} \leq C\|\Gamma\|_{\mathrm{op}}\left(\sqrt{\frac{r(\Gamma)}{n}} + \frac{r(\Gamma)}{n} + \sqrt{\frac{u}{n}} + \frac{u}{n}\right),$$

where the effective rank is

$$r(\Gamma) = \frac{\operatorname{Tr}\Gamma}{\|\Gamma\|_{\mathrm{op}}}$$

when $\Gamma \neq 0$, and $r(0) = 0$. The same conclusion holds in the complex circular Gaussian case, up to a change of the universal constant.

Proof. This is the Gaussian finite-dimensional form of Koltchinskii–Lounici [21, Theorem 4 and Corollary 2]. In the complex case, one applies the real result to the realification of the complex vector; this only changes absolute constants. □

Corollary A.3 (Operator-norm estimates used in the paper)**.** *Let Z_p be a $p \times n$ standard real or complex Gaussian matrix. If $p/n \to 0$, then*

$$\left\| \frac{1}{n} Z_p Z_p^* - I_p \right\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}\left(\sqrt{\frac{p}{n}} + \frac{p}{n} \right) = \mathrm{O}_{\mathbb{P}}\left(\sqrt{\frac{p}{n}} \right).$$

More generally, if D_p is deterministic positive semidefinite, $\sup_p \|D_p\|_{\mathrm{op}} < \infty$, and $\operatorname{rank} D_p \le Cp$, then

$$\left\| D_p^{1/2} \left(\frac{1}{n} Z_p Z_p^* \right) D_p^{1/2} - D_p \right\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}\left(\sqrt{\frac{p}{n}} + \frac{p}{n} \right).$$

In particular, for $E_p = \Lambda_p^{1/2}(n^{-1} Z_p Z_p^*)\Lambda_p^{1/2} - \Lambda_p$ with $\sup_p \|\Lambda_p\|_{\mathrm{op}} < \infty$,

$$\|E_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}\left(\sqrt{\frac{p}{n}} \right).$$

Proof. Apply Proposition A.2. For D_p, the covariance is D_p, and its effective rank is bounded by a constant times p under the stated assumptions. □

Remark A.4 (The case of a random Λ_p). In the paper, Corollary A.3 is used with Λ_p random and $\mathcal{F}_p$-measurable, while Z_p is independent of $\mathcal{F}_p$. The conclusion still holds, for the following reason. In our model

$$\|\Lambda_p\|_{\mathrm{op}} = \|\Sigma_p\|_{\mathrm{op}} \le M_0 := \sigma^2 + \max_{i \le r} \alpha_i + \max_{j \le s} \beta_j,$$

for every p and every realization of O_p, so the bound is deterministic. Conditionally on $\mathcal{F}_p$, the matrix Λ_p is deterministic and Z_p is still a standard Gaussian matrix, so Corollary A.3 applies conditionally. The constant it produces depends only on M_0 and on the universal constant of Proposition A.2, and not on the realization of O_p. Hence, for every $\varepsilon > 0$ there is a constant C_ε, not depending on p nor on the realization, such that for all large p

$$\mathbb{P}\Big(\|E_p\|_{\mathrm{op}} > C_\varepsilon \sqrt{p/n} \;\Big|\; \mathcal{F}_p \Big) \le \varepsilon.$$

The right-hand side is a constant, so taking expectations gives the same bound unconditionally, that is $\|E_p\|_{\mathrm{op}} = \mathrm{O}_{\mathbb{P}}(\sqrt{p/n})$.

Lemma A.5 (A chi-square tail consequence)**.** *Let $x_1, \dots, x_n$ be independent $N_{\mathbb{F}}(0, \sigma^2 I_m)$ vectors, real or complex. If $m/n \to 0$, then*

$$\max_{k \le n} \frac{\|x_k\|^2}{n} = o_{\mathbb{P}}(1).$$

Proof. In the real case, $\|x_k\|^2/\sigma^2 \sim \chi^2_m$. Laurent–Massart's bound [25, Lemma 1] gives

$$\mathbb{P}(\chi^2_m \ge m + 2\sqrt{mu} + 2u) \le e^{-u}.$$

Taking $u = 2 \log n$ and using a union bound over $k \le n$ proves the result, because $(m + \log n)/n \to 0$. The complex case is the same after writing a complex standard Gaussian vector as two real Gaussian vectors with variance $1/2$. □

A.3 Marchenko–Pastur facts

For $c > 0$, let

$$a_c = \sigma^2(1 - \sqrt{c})^2, \qquad b_c = \sigma^2(1 + \sqrt{c})^2.$$

The Marchenko–Pastur distribution with aspect ratio c and variance level σ^2 is

$$\mu_{c,\sigma^2} = \max\left\{1 - \frac{1}{c}, 0\right\}\delta_0 + \frac{\sqrt{(b_c - x)(x - a_c)}}{2\pi c\sigma^2 x}\mathbf{1}_{[a_c,b_c]}(x)\,dx.$$

When $c = 0$, we use the convention $\mu_{0,\sigma^2} = \delta_{\sigma^2}$. In the main proof we only use $c \to 0$, so eventually $0 < c < 1$, and there is no atom at zero.

Theorem A.6 (Marchenko–Pastur empirical law)**.** *Let Z be an $m \times n$ standard real or complex Gaussian matrix and let*

$$C_{m,n} = \frac{\sigma^2}{n} ZZ^*.$$

If $m/n \to c \in [0, \infty)$, then the empirical eigenvalue distribution of $C_{m,n}$ converges weakly, in probability and almost surely, to μ_{c,σ^2}.

Proof. This is the Marchenko–Pastur theorem [29]; see also Bai–Silverstein [1, Chapter 3] or Yao–Zheng–Bai [37, Chapter 2]. □

For $0 < c < 1$ and $t > b_c$, define the positive-resolvent transforms

$$s_c(t) = c\int \frac{1}{t - x}\,\mu_{c,\sigma^2}(dx), \qquad h_c(t) = c\int \frac{x}{t - x}\,\mu_{c,\sigma^2}(dx). \tag{57}$$

For $c = 0$, set $s_0(t) = h_0(t) = 0$. Since $x/(t - x) = t/(t - x) - 1$,

$$h_c(t) = ts_c(t) - c. \tag{58}$$

Proposition A.7 (Marchenko–Pastur self-consistent equation)**.** *Let $0 < c < 1$ and $t > b_c$. Then $s = s_c(t)$ satisfies*

$$G_t(s) := \sigma^2 ts^2 - \{t - \sigma^2(1 - c)\}s + c = 0. \tag{59}$$

Equivalently, for $s \neq 1/\sigma^2$,

$$F_t(s) := ts - c - \frac{\sigma^2 s}{1 - \sigma^2 s} = 0 \tag{60}$$

if and only if $G_t(s) = 0$. More precisely,

$$(1 - \sigma^2 s)F_t(s) = -G_t(s).$$

Proof. Let $m_{c,\sigma^2}(z) = \int (x - z)^{-1}\mu_{c,\sigma^2}(dx)$ be the usual Stieltjes transform. It satisfies

$$c\sigma^2 z m_{c,\sigma^2}(z)^2 + \{z - \sigma^2(1 - c)\}m_{c,\sigma^2}(z) + 1 = 0$$

for $z \in \mathbb{C} \setminus [a_c, b_c]$; see Bai–Silverstein [1, Chapter 3]. For real $t > b_c$, $s_c(t) = -cm_{c,\sigma^2}(t)$. Substitution gives (59). The identity relating F_t and G_t follows by direct algebra. □

Proposition A.8 (Root identification)**.** *Let $0 < c < 1$ and $t > b_c$. The equation $G_t(s) = 0$ has two distinct real roots*

$$0 < s_-(t) < s_+(t) < \frac{1}{\sigma^2}.$$

Moreover,

$$s_c(t) = s_-(t).$$

Equivalently, among the roots of $F_t(s) = 0$ in $(0, 1/\sigma^2)$, the Marchenko–Pastur value $s_c(t)$ is the one for which $F_t'(s) > 0$.

Proof. The discriminant of G_t is

$$\{t - \sigma^2(1-c)\}^2 - 4\sigma^2 ct = (t - a_c)(t - b_c) > 0.$$

Thus the two roots are real and distinct. Their sum and product are positive, so both roots are positive. Also

$$G_t(1/\sigma^2) = 1 > 0,$$

and the midpoint of the two roots is smaller than $1/(2\sigma^2)$. Hence both roots lie in $(0, 1/\sigma^2)$.

On $(-\infty, 1/\sigma^2)$,

$$F_t''(s) = -\frac{2\sigma^4}{(1 - \sigma^2 s)^3} < 0.$$

So F_t is strictly concave. Since $F_t(0) = -c < 0$ and the two roots are $s_-(t)$ and $s_+(t)$, strict concavity implies

$$F_t'(s_-(t)) > 0, \qquad F_t'(s_+(t)) < 0.$$

Finally, $t \mapsto s_c(t)$ is continuous on (b_c, ∞), and it must equal one of these two continuous roots. For large t,

$$s_c(t) \le \frac{c}{t - b_c} \to 0,$$

whereas $s_+(t)$ stays bounded away from zero. Therefore $s_c(t) = s_-(t)$ for large t, and by continuity it holds for every $t > b_c$. □

A.4 Gaussian quadratic forms

Lemma A.9 (Gaussian quadratic form identities)**.** *Let $z = (z_1, \dots, z_n)$ be a standard Gaussian row vector, real or complex circular. Let $A = A^* \in M_n(\mathbb{F})$ be deterministic, or independent of z. Then, conditionally on A,*

$$\mathbb{E}_z[zAz^*] = \operatorname{Tr} A,$$

and

$$\operatorname{Var}_z(zAz^*) = \kappa \operatorname{Tr}(A^2), \qquad \kappa = \begin{cases} 2, & \mathbb{F} = \mathbb{R}, \\ 1, & \mathbb{F} = \mathbb{C}. \end{cases}$$

Consequently,

$$\operatorname{Var}_z(zAz^*) \le \kappa \operatorname{rank}(A) \|A\|_{\mathrm{op}}^2.$$

If $x \sim N_{\mathbb{F}}(0, \sigma^2 I_n)$, the same formulas hold with the right-hand sides multiplied by σ^2 for the mean and σ^4 for the variance.

Proof. Diagonalize $A = U \operatorname{diag}(a_1, \dots, a_n) U^*$. By orthogonal or unitary invariance of the standard Gaussian vector, zU has the same distribution as z. Thus

$$zAz^* \overset{d}{=} \sum_{j=1}^{n} a_j |z_j|^2.$$

The mean is $\sum_j a_j = \operatorname{Tr} A$. In the real case, $|z_j|^2 \sim \chi^2_1$, whose variance is 2. In the complex circular case, $|z_j|^2 \sim \mathrm{Exp}(1)$, whose variance is 1. This proves the variance formula. The final bound follows from $\operatorname{Tr}(A^2) = \sum_j a_j^2 \le \operatorname{rank}(A)\|A\|_{\mathrm{op}}^2$. □

A.5 Linear algebra facts

We use the following elementary facts without further comment.

Lemma A.10 (Weyl's eigenvalue inequality)**.** *Let A and B be Hermitian $d \times d$ matrices, and write their eigenvalues in decreasing order. Then*

$$\max_{1 \le j \le d} |\mu_j(A) - \mu_j(B)| \le \|A - B\|_{\mathrm{op}}.$$

Lemma A.11 (Schur complement eigenvalue equation)**.** *Let*

$$S = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix},$$

where a is a scalar and C is Hermitian. If $tI - C$ is invertible, then

$$\det(tI - S) = \det(tI - C)\{t - a - b^*(tI - C)^{-1}b\}.$$

In particular, if $t \notin \mathrm{Spec}(C)$, then t is an eigenvalue of S if and only if

$$t = a + b^*(tI - C)^{-1}b.$$

Proof. Multiply $tI - S$ on the left by

$$\begin{pmatrix} 1 & b^*(tI - C)^{-1} \\ 0 & I \end{pmatrix}.$$

Its determinant is one, and the product is block lower triangular with diagonal blocks $t - a - b^*(tI - C)^{-1}b$ and $tI - C$. Taking determinants gives the formula. □

Lemma A.12 (Sherman–Morrison formula)**.** *Let M be invertible and let w be a vector. If $1 - w^*M^{-1}w \neq 0$, then*

$$(M - ww^*)^{-1} = M^{-1} + \frac{M^{-1}ww^*M^{-1}}{1 - w^*M^{-1}w}.$$

Lemma A.13 (Resolvent and finite-rank trace bounds)**.** *If A and B are Hermitian and $tI - A$, $tI - B$ are invertible, then*

$$(tI - A)^{-1} - (tI - B)^{-1} = (tI - A)^{-1}(A - B)(tI - B)^{-1}.$$

Also, if M has rank at most r, then

$$|\operatorname{Tr} M| \le r\|M\|_{\mathrm{op}}.$$

Lemma A.14 (Finite-rank resolvent trace comparison)**.** *Let A and B be Hermitian matrices of the same size. Let $I \subset \mathbb{R}$ be compact, and assume that, for some $d > 0$,*

$$\operatorname{dist}(I, \mathrm{Spec}(A)) \ge d, \qquad \operatorname{dist}(I, \mathrm{Spec}(B)) \ge d.$$

If

$$\operatorname{rank}(A - B) \le r, \qquad \|A - B\|_{\mathrm{op}} \le M,$$

then

$$\sup_{t \in I} \left|\operatorname{Tr}\{(tI - A)^{-1} - (tI - B)^{-1}\}\right| \le \frac{rM}{d^2}.$$

Proof. The resolvent identity gives

$$(tI - A)^{-1} - (tI - B)^{-1} = (tI - A)^{-1}(A - B)(tI - B)^{-1}.$$

The matrix on the right has rank at most r and operator norm at most M/d^2. The result follows from $|\operatorname{Tr} H| \le \operatorname{rank}(H)\|H\|_{\mathrm{op}}$. □

A.6 Convergence lemmas

Lemma A.15 (Localization)**.** *Let $\mathbb{P}(\Omega_p) \to 1$ and let $u_p > 0$. If*

$$\mathbb{E}\left[\mathbf{1}_{\Omega_p}|X_p|\right] = o(u_p),$$

then $X_p = o_{\mathbb{P}}(u_p)$.

Proof. For every $\delta > 0$, Markov's inequality gives

$$\mathbb{P}(|X_p| > \delta u_p) \le \mathbb{P}(\Omega_p^c) + \frac{\mathbb{E}[\mathbf{1}_{\Omega_p}|X_p|]}{\delta u_p}.$$

The right-hand side tends to zero. □

Lemma A.16 (Joint convergence for independent arrays)**.** *Let X_p be $\mathcal{F}_p$-measurable and let Y_p be independent of $\mathcal{F}_p$. If $X_p \xrightarrow{\text{law}} X$ and $Y_p \xrightarrow{\text{law}} Y$, then*

$$(X_p, Y_p) \xrightarrow{\text{law}} (X, Y^\circ),$$

where Y° has the same distribution as Y and is independent of X.

Proof. For real t, s, independence gives

$$\mathbb{E}\exp\bigl(\mathrm{i}tX_p + \mathrm{i}sY_p\bigr) = \mathbb{E}\exp(\mathrm{i}tX_p)\,\mathbb{E}\exp(\mathrm{i}sY_p).$$

The right-hand side converges to $\mathbb{E}e^{\mathrm{i}tX}\,\mathbb{E}e^{\mathrm{i}sY}$, which is the characteristic function of an independent pair. The result follows from Lévy's continuity theorem. □

Lemma A.17 (Deconditioning by characteristic functions)**.** *Let X_p be real-valued random variables and let $\mathcal{F}_p$ be sigma-fields. Suppose that, for every $t \in \mathbb{R}$,*

$$\mathbb{E}\left[\exp(\mathrm{i}tX_p) \mid \mathcal{F}_p\right] \xrightarrow{\mathbb{P}} \phi(t),$$

where ϕ is the characteristic function of a probability distribution. Then X_p converges in distribution to that probability distribution.

Proof. The conditional characteristic functions have modulus at most one. Convergence in probability together with boundedness implies convergence of expectations. Hence

$$\mathbb{E}\exp(\mathrm{i}tX_p) = \mathbb{E}\left[\mathbb{E}\left[\exp(\mathrm{i}tX_p) \mid \mathcal{F}_p\right]\right] \longrightarrow \phi(t)$$

for every t. Lévy's continuity theorem proves the assertion. □

Lemma A.18 (Change of scaling)**.** *Let X_p be real random variables, and let $u_p > 0$ and $v_p > 0$ be deterministic. Assume that $u_p X_p \xrightarrow{\text{law}} L$ for some real random variable L.*

(i) *If $v_p/u_p \to \gamma \in [0, \infty)$, then $v_p X_p \xrightarrow{\text{law}} \gamma L$.*

(ii) *If* $v_p/u_p \to \infty$ *and* $\mathbb{P}(L=0)=0$*, then* $|v_pX_p| \xrightarrow{\mathbb{P}} \infty$.

Proof. (i) Write $v_pX_p = (v_p/u_p)(u_pX_p)$. The first factor is a deterministic sequence converging to γ, and the second one converges in distribution to L. Slutsky's theorem gives the claim.

(ii) Fix $M > 0$ and $\varepsilon > 0$. Since $v_p/u_p \to \infty$, we have $u_p/v_p \to 0$, so there is p_0 with $Mu_p/v_p \le \varepsilon$ for every $p \ge p_0$. For such p, since $u_p, v_p > 0$,

$$\big\{|v_pX_p| \le M\big\} = \Big\{|u_pX_p| \le M\,\frac{u_p}{v_p}\Big\} \subseteq \big\{u_pX_p \in [-\varepsilon,\varepsilon]\big\}.$$

The set $[-\varepsilon,\varepsilon]$ is closed, so the portmanteau theorem gives

$$\limsup_{p\to\infty} \mathbb{P}\big(|v_pX_p| \le M\big) \le \limsup_{p\to\infty} \mathbb{P}\big(u_pX_p \in [-\varepsilon,\varepsilon]\big) \le \mathbb{P}\big(|L| \le \varepsilon\big).$$

This holds for every $\varepsilon > 0$. The events $\{|L| \le 1/k\}$ decrease to $\{L=0\}$ as $k \to \infty$, so, by continuity from above, $\mathbb{P}(|L| \le 1/k) \to \mathbb{P}(L=0) = 0$. Taking $\varepsilon = 1/k$ and letting $k \to \infty$ gives $\limsup_p \mathbb{P}(|v_pX_p| \le M) = 0$. □

Lemma A.19 (Divergence in probability excludes weak limits)**.** *Let* Y_p *be real random variables with* $|Y_p| \xrightarrow{\mathbb{P}} \infty$*. Then no subsequence of* (Y_p) *converges in distribution to a real random variable. In particular,* (Y_p) *is not tight.*

Proof. Suppose that $Y_{p_k} \xrightarrow{\text{law}} Y$ along a subsequence (p_k), for some real random variable Y. The events $\{|Y| < M\}$, $M \in \mathbb{N}$, increase to the whole space, so there is $M > 0$ with $\mathbb{P}(|Y| < M) \ge 1/2$. The set $(-M, M)$ is open, so the portmanteau theorem gives

$$\liminf_{k\to\infty} \mathbb{P}\big(Y_{p_k} \in (-M,M)\big) \ge \mathbb{P}\big(Y \in (-M,M)\big) \ge \frac{1}{2}.$$

On the other hand $\mathbb{P}(Y_{p_k} \in (-M,M)) \le \mathbb{P}(|Y_{p_k}| \le M) \to 0$ by assumption, a contradiction. Since every tight sequence has a subsequence converging in distribution, by Prokhorov's theorem, (Y_p) is not tight. □

Lemma A.20 (Convolution with an absolutely continuous law)**.** *Let* ξ *and* η *be independent real random variables, and assume that the law of* ξ *is absolutely continuous with respect to the Lebesgue measure, with density* f*. Then the law of* $\xi+\eta$ *is absolutely continuous, with density*

$$h(x) = \int_{\mathbb{R}} f(x-y)\,\mathbb{P}_\eta(\mathrm{d}y),$$

where $\mathbb{P}_\eta$ *is the law of* η*. In particular* $\mathbb{P}(\xi+\eta = x) = 0$ *for every* $x \in \mathbb{R}$*.*

Proof. Let $A \subseteq \mathbb{R}$ be a Borel set. By independence and by Tonelli's theorem, all the integrands being non-negative and measurable,

$$\mathbb{P}(\xi+\eta \in A) = \int_{\mathbb{R}} \mathbb{P}(\xi \in A-y)\,\mathbb{P}_\eta(\mathrm{d}y) = \int_{\mathbb{R}} \Big(\int_A f(x-y)\,\mathrm{d}x\Big)\mathbb{P}_\eta(\mathrm{d}y) = \int_A h(x)\,\mathrm{d}x.$$

So h is a density of $\xi+\eta$. A single point has Lebesgue measure zero, which gives the last assertion. □

Corollary A.21 (The limit laws have no atom at zero)**.** *Fix* $\ell \in \{1,\dots,r+s\}$*, let* W_ℓ *be given by* (8)*, and let* $G_\ell \sim \mathcal{N}(0,1)$ *be independent of* W_ℓ*. For every* $\tau \in [0,\infty)$*, the law of*

$$W_\ell + \sqrt{\kappa\tau V_0(\theta_\ell)}\,G_\ell$$

is absolutely continuous with respect to the Lebesgue measure; in particular it puts no mass at zero. The same holds for $\mathcal{N}(0,v)$ *for every* $v > 0$*.*

Proof. We first treat W_ℓ, and we assume $\theta_\ell = \alpha_i$ for some i; the case $\theta_\ell = \beta_j$ is identical, with the roles of r and s exchanged. Write

$$W_\ell = \sum_{j=1}^{s} c_j |g_{ij}|^2, \qquad c_j = \frac{\alpha_i \beta_j}{\alpha_i - \beta_j}.$$

The numbers $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_s$ are positive and all distinct, so every c_j is well defined and nonzero. The variable $|g_{i1}|^2$ is absolutely continuous: it follows the chi-square law with one degree of freedom in the real case, and the exponential law of parameter one in the complex case. Multiplying by the nonzero constant c_1 preserves absolute continuity. Put

$$\xi = c_1 |g_{i1}|^2, \qquad \eta = \sum_{j=2}^{s} c_j |g_{ij}|^2,$$

with $\eta = 0$ if $s = 1$, which is allowed in Lemma A.20. The variables ξ and η are independent because the g_{ij} are, so Lemma A.20 shows that W_ℓ is absolutely continuous.

If $\tau = 0$ the claim is proved. If $\tau > 0$, then $\sqrt{\kappa\tau V_0(\theta_\ell)} G_\ell$ is a Gaussian variable with positive variance, since $\kappa \geq 1$, $\tau > 0$ and $V_0(\theta_\ell) = (\sigma^2 + \theta_\ell)^2 > 0$. It is therefore absolutely continuous, and independent of W_ℓ, so Lemma A.20 applies again. The last assertion is the absolute continuity of a Gaussian law with positive variance. □

A.7 Second order eigenvalue perturbation lemma

We recall Lemma 4.4 before proving it.

Lemma A.22 (Second-order eigenvalue perturbation)**.** *Let M be Hermitian, let λ be a simple eigenvalue of M, and let v be a corresponding unit eigenvector. Let $\{v_q\}$ be an orthonormal eigenbasis of M, with $Mv_q = \lambda_q v_q$, and let ℓ be the index such that $v_\ell = v$. Set*

$$\delta = \min_{\mu \in \mathrm{Spec}(M) \setminus \{\lambda\}} |\lambda - \mu| > 0.$$

If E is Hermitian and $\|E\|_{\mathrm{op}} \leq \delta/4$, then the eigenvalue $\widehat{\lambda}$ of $M + E$ associated with λ satisfies

$$\widehat{\lambda} - \lambda = v^* E v + \sum_{q \neq \ell} \frac{|v_q^* E v|^2}{\lambda - \lambda_q} + R, \tag{61}$$

and there exists a universal constant $C > 0$ such that

$$|R| \leq C \delta^{-2} \|E\|_{\mathrm{op}}^3. \tag{62}$$

Proof. Let $P = vv^*$ and $Q = I - P$. With respect to the decomposition $\mathrm{Ran}(P) \oplus \mathrm{Ran}(Q)$,

$$M + E = \begin{pmatrix} \lambda + v^* E v & v^* E Q \\ Q E v & Q(M + E)Q \end{pmatrix}.$$

Weyl's inequality gives $|\widehat{\lambda} - \lambda| \leq \|E\|_{\mathrm{op}}$. Since the spectrum of QMQ is at distance at least δ from λ and $\|E\|_{\mathrm{op}} \leq \delta/4$,

$$\mathrm{dist}\big(\widehat{\lambda}, \mathrm{Spec}(Q(M + E)Q)\big) \geq \delta - 2\|E\|_{\mathrm{op}} \geq \frac{\delta}{2}.$$

Thus

$$\Big\| \big(\widehat{\lambda} I - Q(M + E)Q\big)^{-1} \Big\|_{\mathrm{op}} \leq \frac{2}{\delta}.$$

The Schur complement equation gives

$$\widehat{\lambda} - \lambda = v^* E v + v^* E Q \big(\widehat{\lambda} I - Q(M+E)Q\big)^{-1} Q E v. \tag{63}$$

Let

$$R_0 = (\lambda I - QMQ)^{-1}.$$

Then $\|R_0\|_{\mathrm{op}} \le 1/\delta$. The resolvent identity gives

$$\begin{aligned} &\big(\widehat{\lambda} I - Q(M+E)Q\big)^{-1} - R_0 \\ &\quad = \big(\widehat{\lambda} I - Q(M+E)Q\big)^{-1} \big((\lambda - \widehat{\lambda}) I + QEQ\big) R_0. \end{aligned}$$

Therefore

$$\Big\| \big(\widehat{\lambda} I - Q(M+E)Q\big)^{-1} - R_0 \Big\|_{\mathrm{op}} \le \frac{4}{\delta^2} \|E\|_{\mathrm{op}}.$$

Replacing the resolvent in (63) by R_0 produces an error bounded by $4\delta^{-2}\|E\|_{\mathrm{op}}^3$. Finally,

$$v^* E Q R_0 Q E v = \sum_{q \neq \ell} \frac{|v_q^* E v|^2}{\lambda - \lambda_q}.$$

This proves the claim. □

B Proof of Proposition 2.3

B.1 Notations

Throughout this section, O_p is Haar distributed on $U(p)$ or $O(p)$, and $\hat{O}^{(p)} = (o_{ij}^{(p)})_{i\in[r],\, j\in[s]}$ is its rectangular truncation. Recall from Section 2.1 the diagonal matrices A_p, B_p with nonzero eigenvalues $\alpha_1, \dots, \alpha_r$ and $\beta_1, \dots, \beta_s$, and set $\hat{A} = \mathrm{diag}(\alpha_1, \dots, \alpha_r)$, $\hat{B} = \mathrm{diag}(\beta_1, \dots, \beta_s)$. Let $P(x, y)$ be a noncommutative polynomial in two variables with $P(0, 0) = 0$. Define

$$P_1(x) := P(x, 0), \qquad Q_1(y) := P(0, y),$$

and decompose

$$P(x, y) = P_1(x) + Q_1(y) + P_2(x, y) + Q_2(x, y) + P_3(x, y) + Q_3(x, y) + R(x, y),$$

where

$$P_2(x, y) = \sum_{k,\ell\geq 1} a_{k,\ell}\, x^k y^\ell, \qquad Q_2(x, y) = \sum_{k,\ell\geq 1} b_{k,\ell}\, y^\ell x^k,$$

$$P_3(x, y) = \sum_{k,\ell,m\geq 1} c_{k,\ell,m}\, x^k y^\ell x^m, \qquad Q_3(x, y) = \sum_{k,\ell,m\geq 1} d_{k,\ell,m}\, y^\ell x^k y^m,$$

and R is a linear combination of monomials of length greater than three (the length counts each x^k and each y^ℓ as one). We use the conventions $a_{k,0}$ = coefficient of x^k in P_1, $b_{0,\ell}$ = coefficient of y^ℓ in Q_1, $c_{k,\ell,0} := a_{k,\ell}$ and $d_{k,\ell,0} := b_{k,\ell}$ for $k, \ell \geq 1$. For a matrix X we write $\|X\| := \sqrt{\mathrm{Tr}(X^*X)}$ for the Hilbert–Schmidt norm.

B.2 The proposition in the general case

Proposition B.1 (Fluctuations of the eigenvalues of a polynomial in A_p and $O_pB_pO_p^*$; extension of Theorem 1.3 of [12])**.** *With the notation above, the matrix $P(A_p, O_pB_pO_p^*)$ has $p-r-s$ trivial zero eigenvalues. The remaining $r+s$ eigenvalues, called the non-trivial eigenvalues, converge almost surely to $\{P_1(\alpha_i)\}_{i\in[r]} \cup \{Q_1(\beta_j)\}_{j\in[s]}$ as $p \to \infty$. If these $r+s$ limiting values are pairwise distinct, then the non-trivial eigenvalues are of the form*

$$P_1(\alpha_i) + \frac{1}{p}\sum_{j\in[s]} p_{i,j}\big|\sqrt{p}\, o_{ij}^{(p)}\big|^2 + \frac{\xi_i^{(p)}}{p^{3/2}}, \qquad i \in [r], \tag{64}$$

$$Q_1(\beta_j) + \frac{1}{p}\sum_{i\in[r]} q_{i,j}\big|\sqrt{p}\, o_{ij}^{(p)}\big|^2 + \frac{\zeta_j^{(p)}}{p^{3/2}}, \qquad j \in [s], \tag{65}$$

where

$$p_{i,j} := P_2(\alpha_i, \beta_j) + P_3(\alpha_i, \beta_j) + \frac{\big[P_1(\alpha_i) + P_2(\alpha_i, \beta_j)\big]\big[Q_1(\beta_j) + Q_2(\alpha_i, \beta_j)\big]}{P_1(\alpha_i) - Q_1(\beta_j)}, \tag{66}$$

$$q_{i,j} := Q_2(\alpha_i, \beta_j) + Q_3(\alpha_i, \beta_j) - \frac{\big[P_1(\alpha_i) + P_2(\alpha_i, \beta_j)\big]\big[Q_1(\beta_j) + Q_2(\alpha_i, \beta_j)\big]}{P_1(\alpha_i) - Q_1(\beta_j)}. \tag{67}$$

The constants $p_{i,j}, q_{i,j}$ are the same for the two groups, and $\xi_i^{(p)}, \zeta_j^{(p)}$ converge in law to $\mathbb{C}$-valued random variables. The family $\big\{|\sqrt{p}\, o_{ij}^{(p)}|^2 : i \in [r],\, j \in [s]\big\}$ converges in law to a family of i.i.d. random variables, whose common law is χ_1^2 in the orthogonal case and $\mathrm{Exp}(1) = \frac{1}{2}\chi_2^2$ in the unitary case.

Proposition 2.3 is the special case $P(x,y) = x + y$, that is, $\Sigma_p - \sigma^2 I_p = A_p + O_p B_p O_p^*$. Then $P_1(x) = x$, $Q_1(y) = y$, and P_2, P_3, Q_2, Q_3 all vanish, so $P_1(\alpha_i) = \alpha_i$, $Q_1(\beta_j) = \beta_j$, and (66)–(67) reduce to

$$p_{i,j} = \frac{\alpha_i \beta_j}{\alpha_i - \beta_j}, \qquad q_{i,j} = \frac{\alpha_i \beta_j}{\beta_j - \alpha_i}.$$

By Proposition 2.2 and the continuous mapping theorem,

$$\big(|\sqrt{p}\, o_{ij}^{(p)}|^2\big)_{i\in[r],\, j\in[s]} \xrightarrow{\text{law}} \big(|g_{ij}|^2\big)_{i\in[r],\, j\in[s]}.$$

Since r, s are fixed and $\xi_i^{(p)}, \zeta_j^{(p)}$ are tight, all the remainders $\xi_i^{(p)}/\sqrt{p}$ and $\zeta_j^{(p)}/\sqrt{p}$ converge jointly to zero in probability. After centering and multiplying (64)–(65) by p, the continuous mapping theorem and Slutsky's theorem therefore give

$$\big(p(\lambda_{p,\ell} - \sigma^2 - \theta_\ell)\big)_{\ell=1}^{r+s} \xrightarrow{\text{law}} (W_\ell)_{\ell=1}^{r+s}.$$

The limiting spike values are distinct, so the eigenvalue branches agree with the decreasing ordering for all sufficiently large p, almost surely. All the variables W_ℓ use the same family (g_{ij}) in (8). Together with the almost sure convergence in Proposition B.1, this proves Proposition 2.3.

B.3 Auxiliary lemmas for the population fluctuations

Lemma B.2 (Density of the rectangular truncation)**.** *Let $r, s \geq 1$ with $r + s \leq p$, and let O_p be Haar distributed on $U(p)$ or $O(p)$. Let $\hat{O}^{(p)} = (o_{ij}^{(p)})_{i\in[r],\, j\in[s]}$ be the rectangular truncation of O_p.*

1. ***Orthogonal case ($O(p)$).*** *Assume without loss of generality that $r \leq s$ (otherwise swap the roles using $\det(I_r - \hat{O}\hat{O}^\top) = \det(I_s - \hat{O}^\top \hat{O})$). Then $\hat{O}^{(p)}$ has probability density*

$$c_{p,r,s}\, \det\big(I_r - \hat{O}\hat{O}^\top\big)^{(p-r-s-1)/2}\, \mathbf{1}_{\{\|\hat{O}\|_{\mathrm{op}} \leq 1\}}$$

on $M_{r\times s}(\mathbb{R})$, with respect to the Lebesgue measure.

2. ***Unitary case ($U(p)$).*** *The truncation $\hat{O}^{(p)}$ has probability density*

$$c'_{p,r,s}\, \det\big(I_r - \hat{O}\hat{O}^*\big)^{p-r-s}\, \mathbf{1}_{\{\|\hat{O}\|_{\mathrm{op}} \leq 1\}}$$

on $M_{r\times s}(\mathbb{C})$, with respect to the Lebesgue measure $\prod_{i=1}^r \prod_{j=1}^s dx_{ij}\, dy_{ij}$, where $x_{ij} = \Re(o_{ij})$ and $y_{ij} = \Im(o_{ij})$.

In particular, in both cases $\hat{O}^{(p)}$ admits a density with respect to the Lebesgue measure. The asymptotic behavior of $\sqrt{p}\,\hat{O}^{(p)}$ is given by Proposition 2.2.

Proof. The results come from [16, Lemma 2.5] for the orthogonal case and [12, Lemma 2.1] for the unitary case. □

Lemma B.3 (Haar projection)**.** *Let $O_p \in U(p)$ be Haar distributed, and let o_j be its j-th column. Let $v \in \mathbb{F}^p$ be a fixed deterministic unit vector ($\|v\|_2 = 1$). Then:*

1. *The vector o_j is uniformly distributed on the unit sphere $\mathbb{S}(\mathbb{F}^p) := \{x \in \mathbb{F}^p : \|x\|_2 = 1\}$.*
2. *The inner product $\langle v, o_j\rangle$ satisfies $\langle v, o_j\rangle = \mathrm{O}_{\mathbb{P}}(p^{-1/2})$.*
3. *Let $(u_1, \ldots, u_k)$ be a fixed orthonormal family in $\mathbb{F}^p$ and set $v_i := O_p u_i$ for $i \in \{1, \ldots, k\}$. Then for all fixed deterministic unit vectors $w_1, \ldots, w_k \in \mathbb{F}^p$ (in particular for $w_i = u_i$), and all i, j, $\langle w_i, v_j\rangle = \mathrm{O}_{\mathbb{P}}(p^{-1/2})$.*

Proof. **Part 1: Uniform distribution on the sphere.** By the definition of the Haar measure on $U(p)$, the distribution of O_p is invariant under left-multiplication by any fixed deterministic $R \in U(p)$: that is, $RO_p \overset{d}{=} O_p$.

The j-th column of O_p is $o_j = O_p e_j$, where e_j is the j-th vector of the canonical basis. Applying the deterministic rotation R to this column yields

$$Ro_j = R(O_p e_j) = (RO_p)e_j \overset{d}{=} O_p e_j = o_j.$$

So the distribution of o_j is invariant under any $R \in U(p)$. The group $U(p)$ acts transitively on the unit sphere $\mathbb{S}(\mathbb{F}^p)$ (for $\mathbb{F} = \mathbb{R}$ this is the action of $O(p)$ on S^{p-1}; for $\mathbb{F} = \mathbb{C}$ it is the action of $U(p)$ on the complex unit sphere), and o_j has unit norm because $O_p \in U(p)$ has orthonormal columns. A probability measure on the sphere that is invariant under a transitive group of isometries is the uniform measure. Hence o_j is uniform on $\mathbb{S}(\mathbb{F}^p)$.

Part 2: Asymptotic order of the projection. Since v is a deterministic unit vector, there exists a fixed $R_v \in U(p)$ such that $R_v v = e_1$, where $e_1 = (1, 0, \ldots, 0)^\top$ (such an R_v exists by transitivity from Part 1). By the invariance of the inner product under R_v (which holds because $R_v^* R_v = I$),

$$\langle v, o_j \rangle = \langle R_v v, R_v o_j \rangle = \langle e_1, R_v o_j \rangle.$$

Since o_j is uniform on $\mathbb{S}(\mathbb{F}^p)$, the rotated vector $u \equiv R_v o_j$ is also uniform on $\mathbb{S}(\mathbb{F}^p)$. The inner product is then the first coordinate of this uniform random vector:

$$\langle v, o_j \rangle \overset{d}{=} \langle e_1, u \rangle = \bar{u}_1.$$

We compute the second moment of u_1. Since u lies on the unit sphere, $\sum_{k=1}^p |u_k|^2 = 1$ almost surely, so $\sum_{k=1}^p \mathbb{E}[|u_k|^2] = 1$. By the permutation symmetry of the uniform distribution on the sphere, all coordinates have the same second moment, so $p\,\mathbb{E}[|u_1|^2] = 1$, which gives

$$\mathbb{E}\big[|\langle v, o_j \rangle|^2\big] = \mathbb{E}[|u_1|^2] = \frac{1}{p}.$$

By Markov's inequality applied to $|\langle v, o_j \rangle|^2$: for any $M > 0$,

$$\mathbb{P}\big(\sqrt{p}\,|\langle v, o_j \rangle| \geq M\big) = \mathbb{P}\Big(|\langle v, o_j \rangle|^2 \geq \tfrac{M^2}{p}\Big) \leq \frac{p\,\mathbb{E}[|\langle v, o_j \rangle|^2]}{M^2} = \frac{1}{M^2}.$$

This bound is uniform in p and tends to 0 as $M \to \infty$, so $\sqrt{p}\,|\langle v, o_j \rangle|$ is bounded in probability, i.e. $\langle v, o_j \rangle = \mathrm{O}_{\mathbb{P}}(p^{-1/2})$.

Part 3. Choose a fixed $R_i \in U(p)$ with $R_i u_j = e_1$; then

$$\langle w_i, v_j \rangle = \langle w_i, O_p u_j \rangle = \langle w_i, O_p R_i^* e_1 \rangle = \langle (O_p R_i^*)^* w_i, e_1 \rangle = \overline{\langle e_1, (O_p R_i^*)^* w_i \rangle}.$$

Now $O_p R_i^* \overset{d}{=} O_p$ by right-invariance of the Haar measure. Hence

$$(O_p R_j^*)^* w_i \overset{d}{=} O_p^* w_i.$$

Since O_p^* is also Haar distributed, $O_p^* w_i$ is uniformly distributed on $\mathbb{S}(\mathbb{F}^p)$ by Part 1. Therefore, writing $z := (O_p R_j^*)^* w_i$, we get

$$\langle w_i, v_j \rangle \overset{d}{=} \overline{\langle e_1, z \rangle} = \overline{z_1}.$$

By the computation of Part 2, $\mathbb{E}[|z_1|^2] = 1/p$, so by Markov's inequality $\langle w_i, v_j \rangle = \mathrm{O}_{\mathbb{P}}(p^{-1/2})$. Taking $w_i = u_i$ gives $\langle u_i, v_j \rangle = \mathrm{O}_{\mathbb{P}}(p^{-1/2})$. Notice that this means that the fixed family (u_i) and its random rotation $(v_j) = (O_p u_j)$ are asymptotically orthogonal. □

Lemma B.4 (Almost sure decay of the truncated blocks)**.** *For O_p Haar distributed on $U(p)$, we have $o_{ij}^{(p)} \to 0$ almost surely as $p \to \infty$, for every fixed pair (i, j).*

Proof. The first column of O_p is uniformly distributed on the unit sphere of $\mathbb{F}^p$ (this is Part 1 of Lemma B.3, valid for the two groups). We compute the fourth moment of one entry.

Orthogonal case ($O(p)$). The first column is uniform on the real sphere $S^{p-1} \subset \mathbb{R}^p$, so $(o_{11}^{(p)})^2$ is Beta$(1/2, (p-1)/2)$-distributed and

$$\mathbb{E}[(o_{ij}^{(p)})^4] = \frac{3}{p(p+2)} = \mathrm{O}(p^{-2}).$$

Unitary case ($U(p)$). The first column is uniform on the complex sphere of $\mathbb{C}^p$. The fourth moment is

$$\mathbb{E}[|o_{ij}^{(p)}|^4] = \frac{2}{p(p+1)} = \mathrm{O}(p^{-2}),$$

which is the formula used in [12] (proof of Lemma 2.3).

In both cases $\mathbb{E}[|o_{ij}^{(p)}|^4] = \mathrm{O}(p^{-2})$, so

$$\mathbb{E}\Big[\sum_{p=1}^{\infty} |o_{ij}^{(p)}|^4\Big] = \sum_{p=1}^{\infty} \mathbb{E}\big[|o_{ij}^{(p)}|^4\big] \le C \sum_{p=1}^{\infty} \frac{1}{p^2} < \infty.$$

So the random series $\sum_p |o_{ij}^{(p)}|^4$ is finite almost surely, which forces $|o_{ij}^{(p)}| \to 0$ almost surely. □

Lemma B.5 (Perturbation series for simple eigenvalues)**.** *Let $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_d) \in M_d(\mathbb{C})$ with $\lambda_1, \dots, \lambda_d$ pairwise distinct. There exist homogeneous polynomials $\Pi_{q,k}(X)$, $q \in [d]$, $k \ge 1$, of degree k in the d^2 entries of $X = \{x_{ij}\}_{i,j\in[d]}$, and a constant $C > 0$, such that for every $X \in M_d(\mathbb{C})$ with $\|X\| < C$ the eigenvalues of $\Lambda + X$ are*

$$\mu_q(X) = \lambda_q + \sum_{k=1}^{\infty} \Pi_{q,k}(X), \qquad q \in [d],$$

with absolutely convergent series. The first two terms are

$$\Pi_{q,1}(X) = x_{qq}, \qquad \Pi_{q,2}(X) = \sum_{i \ne q} \frac{x_{iq}\, x_{qi}}{\lambda_q - \lambda_i}.$$

Proof. See Collins et al. [12], where this is Lemma 2.5. □

B.4 Proof of Proposition 2.3

Step 1: matrix representation in an adapted basis. Let o_j denote the j-th column of O_p. The family

$$\mathcal{B} = (e_1, \dots, e_r,\ o_1, \dots, o_s,\ e_{r+1}, \dots, e_{p-s})$$

is a basis of $\mathbb{F}^p$ with probability one. Indeed, the change-of-basis matrix has determinant equal, up to sign, to the determinant of the $s \times s$ block of O_p made of rows $\{p-s+1, \dots, p\}$ and columns $\{1, \dots, s\}$; by Lemma B.2 this block admits a density with respect to Lebesgue measure (on $M_s(\mathbb{R})$ in the orthogonal case, on $M_s(\mathbb{C})$ in the unitary case), so its determinant is non-zero a.s. Throughout the proof we work on this event.

A direct calculation, using $Ae_i = \alpha_i e_i$ for $i \in [r]$, $Ae_i = 0$ for $i > r$, the relation $O_p^* o_j = e_j$ (which holds because $O_p^* O_p = I$, valid for both groups), $Be_j = \beta_j e_j$ for $j \in [s]$ and $Be_j = 0$ for $j > s$, gives

$$\tilde{A} := [A]_{\mathcal{B}} = \begin{pmatrix} \hat{A} & \hat{A}\hat{O} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \tilde{B} := [O_p B O_p^*]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \\ \hat{B}\hat{O}^* & \hat{B} & * \\ 0 & 0 & 0 \end{pmatrix},$$

where $\hat{O}^* = (\overline{o_{ij}})_{j\in[s],i\in[r]}$ is the $s \times r$ adjoint of $\hat{O}$, and the $*$ in $\tilde{B}$ denotes a block depending on entries of O_p outside the truncation $\hat{O}$ (it will play no role).

The only place where the field matters here is the adjoint: in the orthogonal case $O_p^* = O_p^\top$ so $\hat{O}^* = \hat{O}^\top$ has the same real entries as $\hat{O}$ with indices swapped; in the unitary case $\hat{O}^*$ has the conjugated entries $\overline{o_{ij}}$. The block shapes are identical.

Consequently $P(A, O_p B O_p^*)$ has the same eigenvalues as $P(\tilde{A}, \tilde{B})$, and in particular $p - r - s$ trivial zero eigenvalues coming from the last block of rows/columns (which is zero in both $\tilde{A}$ and $\tilde{B}$, so it annihilates every monomial of degree ≥ 1, and $P(0,0) = 0$ removes the constant term). The remaining $r + s$ non-trivial eigenvalues coincide with those of the upper-left $(r+s)\times(r+s)$ corner of $P(\tilde{A}, \tilde{B})$.

Step 2: products in the algebra generated by $\tilde{A}, \tilde{B}$. By induction on $k, \ell \geq 1$,

$$\tilde{A}^k = \begin{pmatrix} \hat{A}^k & \hat{A}^k\hat{O} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \tilde{B}^\ell = \begin{pmatrix} 0 & 0 & 0 \\ \hat{B}^\ell\hat{O}^* & \hat{B}^\ell & * \\ 0 & 0 & 0 \end{pmatrix}.$$

For $k, \ell, m \geq 1$, a direct block multiplication yields

$$\tilde{A}^k\tilde{B}^\ell = \begin{pmatrix} \hat{A}^k\hat{O}\hat{B}^\ell\hat{O}^* & \hat{A}^k\hat{O}\hat{B}^\ell & * \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\tilde{B}^\ell\tilde{A}^k = \begin{pmatrix} 0 & 0 & 0 \\ \hat{B}^\ell\hat{O}^*\hat{A}^k & \hat{B}^\ell\hat{O}^*\hat{A}^k\hat{O} & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\tilde{A}^k\tilde{B}^\ell\tilde{A}^m = \begin{pmatrix} \hat{A}^k\hat{O}\hat{B}^\ell\hat{O}^*\hat{A}^m & \hat{A}^k\hat{O}\hat{B}^\ell\hat{O}^*\hat{A}^m\hat{O} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\tilde{B}^\ell\tilde{A}^k\tilde{B}^m = \begin{pmatrix} 0 & 0 & 0 \\ \hat{B}^\ell\hat{O}^*\hat{A}^k\hat{O}\hat{B}^m\hat{O}^* & \hat{B}^\ell\hat{O}^*\hat{A}^k\hat{O}\hat{B}^m & * \\ 0 & 0 & 0 \end{pmatrix}.$$

These computations are the same in the two groups: they use only the block shapes from Step 1 and the associativity of matrix multiplication. The only difference is again that $\hat{O}^*$ stands for $\hat{O}^\top$ (orthogonal) or the conjugate transpose (unitary).

Any monomial of length > 3 in $\tilde{A}, \tilde{B}$ has all of its blocks of order $\mathrm{O}(\|\hat{O}\|^3)$ or zero. Hence, using the conventions for $c_{k,\ell,0}, d_{k,\ell,0}$,

$$P(\tilde{A}, \tilde{B}) = \begin{pmatrix} P_1(\hat{A}) & 0 & 0 \\ 0 & Q_1(\hat{B}) & * \\ 0 & 0 & 0 \end{pmatrix} + \sum_{k\geq 1,\, \ell \geq 0} \begin{pmatrix} 0 & a_{k,\ell}\hat{A}^k\hat{O}\hat{B}^\ell & * \\ 0 & 0 & * \\ 0 & 0 & 0 \end{pmatrix}$$
$$+ \sum_{k\geq 0,\, \ell \geq 1} \begin{pmatrix} 0 & 0 & * \\ b_{k,\ell}\hat{B}^\ell\hat{O}^*\hat{A}^k & 0 & * \\ 0 & 0 & 0 \end{pmatrix} + \sum_{\substack{k,\ell\geq 1 \\ m\geq 0}} \begin{pmatrix} c_{k,\ell,m}\hat{A}^k\hat{O}\hat{B}^\ell\hat{O}^*\hat{A}^m & 0 & 0 \\ 0 & d_{k,\ell,m}\hat{B}^\ell\hat{O}^*\hat{A}^k\hat{O}\hat{B}^m & * \\ 0 & 0 & 0 \end{pmatrix}$$

$$+\begin{pmatrix} \mathrm{O}(\|\hat{O}\|^3) & \mathrm{O}(\|\hat{O}\|^2) & * \\ \mathrm{O}(\|\hat{O}\|^2) & \mathrm{O}(\|\hat{O}\|^3) & * \\ 0 & 0 & 0 \end{pmatrix}.$$

In $M := \begin{pmatrix} P_1(\hat{A}) & 0 \\ 0 & Q_1(\hat{B}) \end{pmatrix}$, the polynomials P_1, Q_1 are applied entry-wise to $\hat{A}$ and $\hat{B}$.

Step 3: a.s. convergence of non-trivial eigenvalues. The upper-left $(r+s)\times(r+s)$ corner of $P(\tilde{A}, \tilde{B})$ equals $M + V_p$ where

$$V_p \to 0 \text{ a.s. (by Lemma B.4).}$$

This holds for the two groups, since Lemma B.4 gives $\hat{O}^{(p)} \to 0$ a.s. in both of them, and every entry of V_p is a polynomial in the entries of $\hat{O}, \hat{O}^*$ with no constant term. By the argument principle applied to the characteristic polynomials (see Lemma 4.1 from [12]), the eigenvalues of $M + V_p$ converge a.s. to those of M, namely $\{P_1(\alpha_i)\}_{i\in[r]} \cup \{Q_1(\beta_j)\}_{j\in[s]}$. This proves the a.s. convergence stated in Proposition B.1.

Step 4: perturbation expansion under the simplicity assumption. Assume from now on that the $r+s$ limits $\{P_1(\alpha_i)\} \cup \{Q_1(\beta_j)\}$ are pairwise distinct. Then Lemma B.5 applies to $M + V_p$ (for p large enough so that $\|V_p\| < C$, which happens a.s.). Label the non-trivial eigenvalues so that $\mu_i^{(p)} \to P_1(\alpha_i)$ $(i \in [r])$ and $\mu_{r+j}^{(p)} \to Q_1(\beta_j)$ $(j \in [s])$. Lemma B.5 gives, up to order two,

$$\mu_i^{(p)} = P_1(\alpha_i) + (V_p)_{i,i} + \sum_{\substack{q\in[r+s] \\ q\neq i}} \frac{(V_p)_{q,i}(V_p)_{i,q}}{P_1(\alpha_i) - \lambda_q} + \mathrm{O}(\|V_p\|^3), \tag{68}$$

where $\lambda_q \in \{P_1(\alpha_{i'})\} \cup \{Q_1(\beta_j)\}$ is the eigenvalue corresponding to index q. This expansion is the same in the two groups (Lemma B.5 is deterministic and the entries of V_p are complex numbers in all cases).

From the block decomposition of $P(\tilde{A}, \tilde{B})$ in Step 2:

- The diagonal entry $(V_p)_{i,i}$ for $i \in [r]$ is $\sum_{k,\ell\geq 1,\, m\geq 0} c_{k,\ell,m} \big(\hat{A}^k \hat{O} \hat{B}^\ell \hat{O}^* \hat{A}^m\big)_{i,i} + \mathrm{O}(\|\hat{O}\|^3)$.
- For $q = i' \in [r] \setminus \{i\}$, $(V_p)_{i',i}$ and $(V_p)_{i,i'}$ are $\mathrm{O}(\|\hat{O}\|^2)$, so their product is $\mathrm{O}(\|\hat{O}\|^4)$ and contributes to the $\mathrm{O}(\|\hat{O}\|^3)$ remainder.
- For $q = r + j$ $(j \in [s])$, the entries $(V_p)_{r+j,i}$ and $(V_p)_{i,r+j}$ are, up to $\mathrm{O}(\|\hat{O}\|^3)$,

$$\sum_{k\geq 0,\, \ell\geq 1} b_{k,\ell} \big(\hat{B}^\ell \hat{O}^* \hat{A}^k\big)_{j,i}, \qquad \sum_{k\geq 1,\, \ell\geq 0} a_{k,\ell} \big(\hat{A}^k \hat{O} \hat{B}^\ell\big)_{i,j}.$$

Step 5: entry-wise evaluation. Since $\hat{A}, \hat{B}$ are diagonal, a direct calculation gives, for any k, ℓ, m, k', ℓ' (using $(\hat{O}^*)_{j,i} = \overline{o_{ij}}$):

$$\big(\hat{A}^k \hat{O} \hat{B}^\ell \hat{O}^* \hat{A}^m\big)_{i,i} = \alpha_i^{k+m} \sum_{j\in[s]} \beta_j^\ell \, o_{ij} \, \overline{o_{ij}} = \alpha_i^{k+m} \sum_{j\in[s]} \beta_j^\ell \, |o_{ij}|^2, \tag{69}$$

$$\big(\hat{A}^k \hat{O} \hat{B}^\ell\big)_{i,j} = \alpha_i^k \, \beta_j^\ell \, o_{ij}, \tag{70}$$

$$\big(\hat{B}^{\ell'} \hat{O}^* \hat{A}^{k'}\big)_{j,i} = \alpha_i^{k'} \, \beta_j^{\ell'} \, \overline{o_{ij}}. \tag{71}$$

The product of (70) and (71) equals

$$\alpha_i^{k+k'} \beta_j^{\ell+\ell'} \, o_{ij} \, \overline{o_{ij}} = \alpha_i^{k+k'} \beta_j^{\ell+\ell'} \, |o_{ij}|^2.$$

The orthogonal and unitary cases differ, and it is the reason why the building block is o_{ij}^2 in one case and $|o_{ij}|^2$ in the other. In the orthogonal case ($\mathbb{F} = \mathbb{R}$) the entries are real, $\overline{o_{ij}} = o_{ij}$, so $o_{ij}\,\overline{o_{ij}} = o_{ij}^2 = |o_{ij}|^2$. In the unitary case ($\mathbb{F} = \mathbb{C}$) the adjoint $\hat{O}^*$ carries the complex conjugate, so the product is $o_{ij}\,\overline{o_{ij}} = |o_{ij}|^2$. In both cases the result is the same non-negative real quantity $|o_{ij}|^2$.

Substituting (69)–(71) into (68) and factoring out $|o_{ij}|^2$:

$$\begin{aligned}\mu_i^{(p)} = P_1(\alpha_i) &+ \sum_{j\in[s]} \Big[\sum_{\substack{k,\ell\geq 1\\ m\geq 0}} c_{k,\ell,m}\,\alpha_i^{k+m}\beta_j^{\ell}\Big]|o_{ij}|^2 \\ &+ \sum_{j\in[s]} \frac{1}{P_1(\alpha_i)-Q_1(\beta_j)} \Big[\sum_{\substack{k\geq 1\\ \ell\geq 0}} a_{k,\ell}\,\alpha_i^k\beta_j^{\ell}\Big]\Big[\sum_{\substack{k'\geq 0\\ \ell'\geq 1}} b_{k',\ell'}\,\alpha_i^{k'}\beta_j^{\ell'}\Big]|o_{ij}|^2 \\ &+ \mathrm{O}(\|\hat{O}\|^3).\end{aligned}$$

Using the conventions $c_{k,\ell,0} = a_{k,\ell}$ and $a_{k,0} = $ (coeff. of x^k in P_1), we recognize:

$$\begin{aligned}\sum_{k,\ell\geq 1,\, m\geq 0} c_{k,\ell,m}\,\alpha_i^{k+m}\beta_j^{\ell} &= P_2(\alpha_i,\beta_j) + P_3(\alpha_i,\beta_j),\\ \sum_{k\geq 1,\,\ell\geq 0} a_{k,\ell}\,\alpha_i^k\beta_j^{\ell} &= P_1(\alpha_i) + P_2(\alpha_i,\beta_j),\\ \sum_{k\geq 0,\,\ell\geq 1} b_{k,\ell}\,\alpha_i^k\beta_j^{\ell} &= Q_1(\beta_j) + Q_2(\alpha_i,\beta_j).\end{aligned}$$

Hence

$$\mu_i^{(p)} = P_1(\alpha_i) + \sum_{j\in[s]} p_{i,j}\,|o_{ij}|^2 + \mathrm{O}(\|\hat{O}\|^3), \tag{72}$$

The analogous calculation for $j \in [s]$ yields

$$\mu_{r+j}^{(p)} = Q_1(\beta_j) + \sum_{i\in[r]} q_{i,j}\,|o_{ij}|^2 + \mathrm{O}(\|\hat{O}\|^3),$$

where $p_{i,j}$ and $q_{i,j}$ are given by:

$$\begin{aligned}p_{i,j} &:= P_2(\alpha_i,\beta_j) + P_3(\alpha_i,\beta_j) + \frac{\big[P_1(\alpha_i)+P_2(\alpha_i,\beta_j)\big]\big[Q_1(\beta_j)+Q_2(\alpha_i,\beta_j)\big]}{P_1(\alpha_i)-Q_1(\beta_j)},\\ q_{i,j} &:= Q_2(\alpha_i,\beta_j) + Q_3(\alpha_i,\beta_j) - \frac{\big[P_1(\alpha_i)+P_2(\alpha_i,\beta_j)\big]\big[Q_1(\beta_j)+Q_2(\alpha_i,\beta_j)\big]}{P_1(\alpha_i)-Q_1(\beta_j)}.\end{aligned}$$

The sign change in the second term of $q_{i,j}$ comes from $Q_1(\beta_j) - P_1(\alpha_i) = -(P_1(\alpha_i) - Q_1(\beta_j))$ in Lemma B.5. The constants $p_{i,j}, q_{i,j}$ do not depend on the group: only the building block $|o_{ij}|^2$ does (it is o_{ij}^2 for $O(p)$ and $|o_{ij}|^2$ for $U(p)$).

Step 6: rescaling and limiting law.

Power series structure. By Lemma B.5, the eigenvalue $\mu_i^{(p)}$ admits an absolutely convergent expansion

$$\mu_i^{(p)} - P_1(\alpha_i) = \sum_{k=1}^{\infty} \Pi_{i,k}(V_p),$$

true a.s. for p large enough so that $\|V_p\| < C$, which holds by Lemma B.4. Each $\Pi_{i,k}$ is a homogeneous polynomial of degree k in the entries of V_p. Since the matrix V_p is a polynomial in the entries $\{o_{ij}\}_{i\in[r],\,j\in[s]}$ of $\hat{O}$ and their conjugates $\{\overline{o_{ij}}\}$, with no constant term (as shown

in Step 2), $\sum_{k=1}^{\infty} \Pi_{i,k}(V_p)$ is a polynomial in the entries of $\hat{O}$ and their conjugates. Substituting these expressions and regrouping by total degree in the real coordinates of the entries $\{o_{ij}\}$, we obtain

$$\mu_i^{(p)} - P_1(\alpha_i) = \sum_{n=2}^{\infty} \Phi_{i,n}(\hat{O}), \tag{73}$$

where each $\Phi_{i,n}$ is a homogeneous polynomial of degree n in the real coordinates of the entries of $\hat{O}$ (rs real coordinates in the orthogonal case, $2rs$ in the unitary case, since each complex entry has a real and an imaginary part), and the series converges absolutely for $\|\hat{O}\|$ small enough. The sum starts at $n = 2$ because: the diagonal entries $(V_p)_{q,q}$ are $\mathrm{O}(\|\hat{O}\|^2)$ (Step 4), and the off-diagonal entries $(V_p)_{i,r+j}$, $(V_p)_{r+j,i}$ are $\mathrm{O}(\|\hat{O}\|)$ but appear in $\Pi_{i,2}$ only through products, which are $\mathrm{O}(\|\hat{O}\|^2)$. Hence no degree-1 term remains: $\Phi_{i,1} \equiv 0$.

The degree-2 term $\Phi_{i,2}$ is exactly the explicit $\mathrm{O}(\|\hat{O}\|^2)$ contribution computed in Step 5:

$$\Phi_{i,2}(\hat{O}) = \sum_{j \in [s]} p_{i,j} \, |o_{ij}|^2,$$

with $p_{i,j}$ given by (66). Note that $|o_{ij}|^2$ is indeed a homogeneous polynomial of degree 2 in the real coordinates of o_{ij} in both cases: it is o_{ij}^2 for $O(p)$, and $\Re(o_{ij})^2 + \Im(o_{ij})^2$ for $U(p)$.

Rescaling. Since $\Phi_{i,n}$ is homogeneous of degree n, we have $\Phi_{i,n}(\hat{O}) = p^{-n/2} \, \Phi_{i,n}(\sqrt{p}\,\hat{O})$. Define

$$\xi_i^{(p)} := p^{3/2}\Big(\mu_i^{(p)} - P_1(\alpha_i) - \sum_{j \in [s]} p_{i,j} \, |o_{ij}|^2\Big) = p^{3/2} \sum_{n=3}^{\infty} \Phi_{i,n}(\hat{O}) = \sum_{n=3}^{\infty} p^{(3-n)/2} \, \Phi_{i,n}\big(\sqrt{p}\,\hat{O}\big).$$

This allows us to rewrite (72) as

$$\mu_i^{(p)} = P_1(\alpha_i) + \frac{1}{p} \sum_{j \in [s]} p_{i,j} \big|\sqrt{p}\, o_{ij}^{(p)}\big|^2 + \frac{\xi_i^{(p)}}{p^{3/2}},$$

which is the claimed expansion for $\mu_i^{(p)}$. The proof of the corresponding expansion for $\mu_{r+j}^{(p)}$ is identical, with $q_{i,j}$ instead of $p_{i,j}$.

Convergence in law of $\xi_i^{(p)}$. We show that $\xi_i^{(p)}$ converges in law. By Proposition 2.2, $\sqrt{p}\,\hat{O}^{(p)} \xrightarrow{\text{law}} G$, where $G = (g_{ij})$ has i.i.d. entries (real $\mathcal{N}(0,1)$ in the orthogonal case, standard complex Gaussian in the unitary case). We show that $\xi_i^{(p)} \xrightarrow{\text{law}} \Phi_{i,3}(G)$.

Write $\xi_i^{(p)} = T_p + R_p$ where $T_p := \Phi_{i,3}\big(\sqrt{p}\,\hat{O}\big)$ and $R_p := \sum_{n=4}^{\infty} p^{(3-n)/2} \, \Phi_{i,n}\big(\sqrt{p}\,\hat{O}\big)$.

By the continuous mapping theorem applied to $\sqrt{p}\,\hat{O}$ and the continuous function $\Phi_{i,3}$, $T_p = \Phi_{i,3}\big(\sqrt{p}\,\hat{O}\big) \xrightarrow{\text{law}} \Phi_{i,3}(G)$.

We show $R_p \xrightarrow{\mathbb{P}} 0$. Since $\sum_n \Phi_{i,n}(\hat{O})$ converges absolutely for $\|\hat{O}\| < \delta$ for some $\delta > 0$, there exist constants $C_n > 0$ with $|\Phi_{i,n}(X)| \le C_n \, \|X\|^n$ for all X and $\sum_{n=2}^{\infty} C_n \, \delta^n < \infty$. Hence, on the event $\{\|\hat{O}\| < \delta\}$ (which has probability tending to 1, since $\|\hat{O}\| \xrightarrow{\text{a.s.}} 0$ by Lemma B.4),

$$|R_p| \le \sum_{n=4}^{\infty} p^{(3-n)/2} \, C_n \left(\sqrt{p}\,\|\hat{O}\|\right)^n \le p^{-1/2} \, D \left(\sqrt{p}\,\|\hat{O}\|\right)^4,$$

where $D := \sum_{n \ge 4} C_n \delta^{n-4} < \infty$ (we used $p^{(3-n)/2} \le p^{-1/2}$ for $n \ge 4$, and $\|\hat{O}\| < \delta$ to bound $\|\hat{O}\|^{n-4} \le \delta^{n-4}$). Given that $\sqrt{p}\,\|\hat{O}\| \xrightarrow{\text{law}} \|G\|$, the continuous mapping theorem yields $(\sqrt{p}\,\|\hat{O}\|)^4 = \mathrm{O}_{\mathbb{P}}(1)$, hence $p^{-1/2} D (\sqrt{p}\,\|\hat{O}\|)^4 \xrightarrow{\mathbb{P}} 0$. This gives $R_p \xrightarrow{\mathbb{P}} 0$.

As a result, by the Slutsky lemma,

$$\xi_i^{(p)} = T_p + R_p \xrightarrow{\text{law}} \Phi_{i,3}(G),$$

which is a $\mathbb{C}$-valued random variable. The same argument applies to $\zeta_j^{(p)}$. None of these steps use the field: they only use the homogeneity of $\Phi_{i,n}$, the a.s. decay of $\|\hat{O}\|$ (Lemma B.4) and the convergence in law of $\sqrt{p}\,\hat{O}$ (Proposition 2.2), all of which hold for the two groups.

Limiting law of the building blocks. Finally, by the continuous mapping theorem applied to $\sqrt{p}\,\hat{O}^{(p)} \xrightarrow{\text{law}} G$ and the map $X \mapsto (|X_{ij}|^2)_{i\in[r],\, j\in[s]}$, we get the joint convergence

$$\big\{|\sqrt{p}\, o_{ij}^{(p)}|^2 : i \in [r],\, j \in [s]\big\} \xrightarrow{\text{law}} \big\{|g_{ij}|^2 : i \in [r],\, j \in [s]\big\}.$$

The law of $|g_{ij}|^2$ depends on the group:

- Orthogonal case: the g_{ij} are i.i.d. real $\mathcal{N}(0,1)$, so $|g_{ij}|^2 = g_{ij}^2$ are i.i.d. χ_1^2;
- Unitary case: the g_{ij} are i.i.d. standard complex Gaussian, so $|g_{ij}|^2 = \Re(g_{ij})^2 + \Im(g_{ij})^2$ is a sum of two independent $\mathcal{N}(0,1/2)^2$ variables, which follows the exponential law $\mathrm{Exp}(1) = \frac{1}{2}\chi_2^2$.

This completes the proof. □

C Proof of Proposition 4.6

We first explain why resolvents and the Marchenko–Pastur transform enter the proof. Let λ_p denote the target population eigenvalue, including the noise level σ^2, and let $\widehat{\lambda}_p$ be the corresponding sample eigenvalue. Although $p/n \to 0$ gives consistency, the deterministic sampling shift is of order p/n, which need not be negligible at the $n^{-1/2}$ fluctuation scale.

After placing the target coordinate first, the sample matrix has the block form

$$S_p = \begin{pmatrix} \lambda_p q_p & b_p^* \\ b_p & C_p \end{pmatrix}, \qquad q_p = \frac{1}{n}\sum_{k=1}^{n} |z_k|^2,$$

where the z_k are independent standard real or circular complex Gaussian variables, C_p is the remaining sample covariance block, and b_p is the off-diagonal block. On an event of probability tending to one, a fixed interval around the target spike is separated from the spectrum of C_p. The Schur complement then gives the exact equation

$$\widehat{\lambda}_p = \lambda_p q_p + B_p(\widehat{\lambda}_p), \qquad B_p(t) = b_p^*(tI - C_p)^{-1} b_p.$$

The first term has a one-dimensional central limit theorem. The second term accounts for the interaction with the remaining coordinates. For fixed t, its conditional mean, given the remaining Gaussian rows, is

$$\frac{\lambda_p}{n}\operatorname{Tr}\{C_p(tI - C_p)^{-1}\}.$$

The Gaussian quadratic-form estimate in Step 5 below shows that replacing $B_p(t)$ by this mean has a uniform error $\mathrm{O}_{\mathbb{P}}(\sqrt{p}/n) = o_{\mathbb{P}}(n^{-1/2})$ on the fixed interval. Uniformity is needed because the equation is evaluated at the random point $t = \widehat{\lambda}_p$.

Let K be the fixed dimension of the spike block in Proposition 4.6, and put $c_p = (p-K)/n$, the aspect ratio of the pure-bulk block. Step 6 removes the other $K-1$ spike coordinates with an $\mathrm{O}(n^{-1})$ trace error on the same high-probability event. Lemma C.1 then approximates the pure-bulk trace by $h_{c_p}(t)$, defined in (57), with uniform error $o_{\mathbb{P}}(n^{-1/2})$. Lemma C.2 identifies the deterministic solution

$$\rho_p = \lambda_p\left(1 + c_p\frac{\sigma^2}{\lambda_p - \sigma^2}\right) = \lambda_p\{1 + h_{c_p}(\rho_p)\}.$$

Finally, the bound $h'_{c_p} = \mathrm{O}(c_p)$ allows us to deduce

$$\sqrt{n}(\widehat{\lambda}_p - \rho_p) = \lambda_p\sqrt{n}(q_p - 1) + o_{\mathbb{P}}(1).$$

The desired CLT is therefore reduced to the usual CLT for $n^{-1}\sum_k |z_k|^2$. Replacing $c_p = (p-K)/n$ by p/n changes the centering by $\mathrm{O}(n^{-1})$, which is negligible at this scale. We now state and prove the two lemmas, followed by the full proof.

Lemma C.1 (Pure-bulk resolvent trace estimate)**.** *Let $m = m(p)$ and $n = n(p)$ satisfy $m/n \to 0$. Let Z be an $m \times n$ real or complex standard Gaussian matrix and set*

$$C = C_{m,n} = \frac{\sigma^2}{n} ZZ^*.$$

Put $c = c_{m,n} = m/n$. Let I be a fixed compact interval contained in $(\sigma^2 + \varepsilon, \infty)$ for some $\varepsilon > 0$. Then

$$\sup_{t\in I}\left|\frac{1}{n}\operatorname{Tr}\big(C(tI - C)^{-1}\big) - h_c(t)\right| = o_{\mathbb{P}}(n^{-1/2}), \tag{74}$$

where h_c is defined in Appendix A.3. Moreover,

$$\sup_{t\in I}|h_c'(t)| = \mathrm{O}(c). \tag{75}$$

Proof. We give the proof for the real and complex cases at the same time. Constants may change from line to line. Write the columns of σZ as $x_1, \dots, x_n \in \mathbb{F}^m$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Thus

$$C = \frac{1}{n} \sum_{k=1}^{n} x_k x_k^*, \qquad x_k \sim N_{\mathbb{F}}(0, \sigma^2 I_m),$$

and the vectors x_k are independent.

For $t \notin \operatorname{Spec}(C)$, define

$$Q(t) = (tI - C)^{-1}, \qquad s_n(t) = \frac{1}{n} \operatorname{Tr} Q(t), \qquad r_n(t) = \frac{1}{n} \operatorname{Tr}\big(CQ(t)\big).$$

The identity $CQ(t) = tQ(t) - I_m$ gives

$$r_n(t) = t s_n(t) - c. \tag{76}$$

Fix $0 < \eta < \varepsilon/8$. Let

$$\Omega_n^{op} = \{\|C - \sigma^2 I_m\|_{\mathrm{op}} \le \eta\}, \qquad \Omega_n^{col} = \left\{ \max_{k \le n} \frac{\|x_k\|^2}{n} \le \eta \right\},$$

and put $\Omega_n = \Omega_n^{op} \cap \Omega_n^{col}$. By the Gaussian covariance bound and the chi-square tail estimate in Appendix A.2,

$$\mathbb{P}(\Omega_n) \to 1. \tag{77}$$

For each k, define the leave-one-out matrix

$$C^{(k)} = C - \frac{1}{n} x_k x_k^*, \qquad Q_k(t) = (tI - C^{(k)})^{-1},$$

and the event

$$\Omega_n^{(k)} = \{\|C^{(k)} - \sigma^2 I_m\|_{\mathrm{op}} \le 2\eta\}.$$

The event $\Omega_n^{(k)}$ is independent of x_k, and $\Omega_n \subset \Omega_n^{(k)}$. Indeed, on Ω_n,

$$\|C^{(k)} - \sigma^2 I_m\|_{\mathrm{op}} \le \|C - \sigma^2 I_m\|_{\mathrm{op}} + \frac{\|x_k\|^2}{n} \le 2\eta.$$

On Ω_n^{op}, every eigenvalue of C lies in $[\sigma^2 - \eta, \sigma^2 + \eta]$. Since $I \subset (\sigma^2 + \varepsilon, \infty)$,

$$\sup_{t \in I} \|Q(t)\|_{\mathrm{op}} \le C_0 \quad \text{on } \Omega_n^{op}. \tag{78}$$

Similarly,

$$\sup_{t \in I} \|Q_k(t)\|_{\mathrm{op}} \le C_0 \quad \text{on } \Omega_n^{(k)}. \tag{79}$$

Here C_0 depends only on ε. Put

$$q_k(t) = \frac{1}{n} x_k^* Q_k(t) x_k, \qquad s_k(t) = \frac{1}{n} \operatorname{Tr} Q_k(t).$$

On Ω_n, using (79),

$$\sup_{t \in I} \max_{k \le n} |q_k(t)| \le C_0 \max_{k \le n} \frac{\|x_k\|^2}{n} \le C_0 \eta. \tag{80}$$

By taking η smaller if needed, we may and do assume that the right-hand side is at most $1/6$. Also,

$$\sup_{t \in I} |s_n(t)| \le C_0 c, \qquad \sup_{t \in I} \max_{k \le n} |s_k(t)| \le C_0 c \tag{81}$$

on the relevant events. Since $c \to 0$, these quantities are small for all large n.

Let $\psi(u) = u/(1-u)$. To avoid problems away from the good event, choose a smooth function χ such that $\chi(u) = 1$ for $|u| \le 1/2$, $\chi(u) = 0$ for $|u| \ge 3/4$, and $0 \le \chi \le 1$. Set

$$\widetilde{\psi}(u) = \chi(u)\frac{u}{1-u}.$$

Then $\widetilde{\psi}$, $\widetilde{\psi}'$, and $\widetilde{\psi}''$ are bounded on $\mathbb{R}$, and $\widetilde{\psi} = \psi$ on $[-1/2, 1/2]$.

By the Sherman–Morrison formula in Appendix A.5, on Ω_n,

$$\frac{1}{n}x_k^* Q(t) x_k = \frac{q_k(t)}{1-q_k(t)} = \psi(q_k(t)).$$

Hence, still on Ω_n,

$$r_n(t) = \frac{1}{n}\sum_{k=1}^{n}\psi(q_k(t)) = \frac{1}{n}\sum_{k=1}^{n}\widetilde{\psi}(q_k(t)). \tag{82}$$

Define

$$E_n(t) = \frac{1}{n}\sum_{k=1}^{n}\widetilde{\psi}(q_k(t)) - \widetilde{\psi}(\sigma^2 s_n(t)).$$

By (82), (76), and the smallness of $s_n(t)$, on Ω_n,

$$E_n(t) = r_n(t) - \frac{\sigma^2 s_n(t)}{1-\sigma^2 s_n(t)} = F_t(s_n(t)), \tag{83}$$

where

$$F_t(s) = ts - c - \frac{\sigma^2 s}{1-\sigma^2 s}.$$

We next estimate $E_n(t)$. Let $\mathbb{E}_k$ denote expectation with respect to x_k, with all other columns fixed. Since $Q_k(t)$ is independent of x_k, the Gaussian quadratic-form formula in Appendix A.4 gives

$$\mathbb{E}_k q_k(t) = \sigma^2 s_k(t), \qquad \mathbb{E}_k|q_k(t) - \sigma^2 s_k(t)|^2 \le C\frac{m}{n^2} \tag{84}$$

on $\Omega_n^{(k)}$, uniformly in $t \in I$. Because $\Omega_n \subset \Omega_n^{(k)}$ and $\Omega_n^{(k)}$ is independent of x_k, Cauchy's inequality gives

$$\mathbb{E}\left[\mathbf{1}_{\Omega_n}|q_k(t) - \sigma^2 s_k(t)|\right] \le C\frac{\sqrt{m}}{n}. \tag{85}$$

Also, by the resolvent identity and the rank-one trace bound in Appendix A.5,

$$\sup_{t\in I}\max_{k\le n}|s_k(t) - s_n(t)| \le \frac{C}{n} \quad \text{on } \Omega_n. \tag{86}$$

Since $\widetilde{\psi}$ is Lipschitz, (85) and (86) imply, for each fixed $t \in I$,

$$\mathbb{E}\left[\mathbf{1}_{\Omega_n}|E_n(t)|\right] \le C\left(\frac{\sqrt{m}}{n} + \frac{1}{n}\right) = o(n^{-1/2}). \tag{87}$$

We need the estimate uniformly in t. On Ω_n, the functions above are C^1 in t, and

$$Q'(t) = -Q(t)^2, \qquad Q_k'(t) = -Q_k(t)^2.$$

Thus

$$q_k'(t) = -\frac{1}{n}x_k^* Q_k(t)^2 x_k, \qquad s_k'(t) = -\frac{1}{n}\operatorname{Tr} Q_k(t)^2.$$

The same quadratic-form estimate as above, with $Q_k(t)^2$ in place of $Q_k(t)$, gives

$$\mathbb{E}\left[\mathbf{1}_{\Omega_n}|q_k'(t) - \sigma^2 s_k'(t)|\right] \le C\frac{\sqrt{m}}{n}. \tag{88}$$

The same rank argument gives

$$\sup_{t\in I}\max_{k\le n}|s_k'(t) - s_n'(t)| \le \frac{C}{n} \quad \text{on } \Omega_n, \tag{89}$$

and $\sup_{t\in I}|s_n'(t)| \le C$ on Ω_n. Since $\widetilde{\psi}'$ is bounded and Lipschitz, differentiating $E_n(t)$ and using (85)–(89) gives

$$\sup_{t\in I}\mathbb{E}\left[\mathbf{1}_{\Omega_n}|E_n'(t)|\right] \le C\left(\frac{\sqrt{m}}{n} + \frac{1}{n}\right) = o(n^{-1/2}). \tag{90}$$

Fix $t_0 \in I$. By the fundamental theorem of calculus,

$$\sup_{t\in I}|E_n(t)| \le |E_n(t_0)| + \int_I |E_n'(u)|\,du.$$

Combining (87), (90), Fubini's theorem, and Lemma A.15, we obtain

$$\sup_{t\in I}|E_n(t)| = o_{\mathbb{P}}(n^{-1/2}). \tag{91}$$

Together with (83), this gives

$$\sup_{t\in I}|F_t(s_n(t))| = o_{\mathbb{P}}(n^{-1/2}). \tag{92}$$

Now compare $s_n(t)$ with the Marchenko–Pastur quantity $s_c(t)$ from Appendix A.3. For all large n, the right edge $b_c = \sigma^2(1+\sqrt{c})^2$ is at most $\sigma^2 + \varepsilon/2$, so $I \subset (b_c, \infty)$. By Proposition A.7,

$$F_t(s_c(t)) = 0, \qquad t \in I.$$

Moreover,

$$0 \le s_c(t) = c\int \frac{1}{t-x}\,\mu_{c,\sigma^2}(dx) \le \frac{2c}{\varepsilon}, \qquad t \in I.$$

On Ω_n, (81) gives $0 \le s_n(t) \le Cc$. Hence both $s_n(t)$ and $s_c(t)$ lie in $[0, \delta_n]$, where $\delta_n \to 0$. For $s \in [0, \delta_n]$ and $t \in I$,

$$F_t'(s) = t - \frac{\sigma^2}{(1-\sigma^2 s)^2} \ge \frac{\varepsilon}{2}$$

for all large n. The mean value theorem gives

$$\sup_{t\in I}|s_n(t) - s_c(t)| \le C\sup_{t\in I}|F_t(s_n(t))| = o_{\mathbb{P}}(n^{-1/2}).$$

Finally, by $r_n(t) = ts_n(t) - c$ and $h_c(t) = ts_c(t) - c$,

$$\sup_{t\in I}|r_n(t) - h_c(t)| \le (\max I)\sup_{t\in I}|s_n(t) - s_c(t)| = o_{\mathbb{P}}(n^{-1/2}).$$

This proves (74).

It remains only to prove (75). Since the support of μ_{c,σ^2} is contained in $[a_c, b_c]$, and for large n this support is separated from I, differentiation under the integral sign is valid and gives

$$h_c'(t) = -c\int \frac{x}{(t-x)^2}\,\mu_{c,\sigma^2}(dx).$$

For $t \in I$ and $x \in [a_c, b_c]$, the denominator is bounded below by a positive constant and x is bounded above by a constant depending only on σ^2 and ε. Hence $\sup_{t\in I}|h_c'(t)| = \mathrm{O}(c)$. □

Lemma C.2 (Outlier identity for the identity bulk)**.** *Let $\lambda > \sigma^2$ and let $c \geq 0$. Assume that*

$$\sqrt{c} < \frac{\lambda - \sigma^2}{\sigma^2}. \tag{93}$$

For $c > 0$, assume also $c < 1$. Define

$$\rho_c(\lambda) = \lambda \left(1 + c\frac{\sigma^2}{\lambda - \sigma^2}\right). \tag{94}$$

Then

$$\rho_c(\lambda) - \sigma^2(1 + \sqrt{c})^2 = \frac{\left(\lambda - \sigma^2 - \sigma^2\sqrt{c}\right)^2}{\lambda - \sigma^2}. \tag{95}$$

In particular, under (93)*, $\rho_c(\lambda) > b_c = \sigma^2(1 + \sqrt{c})^2$. Moreover,*

$$h_c(\rho_c(\lambda)) = c\frac{\sigma^2}{\lambda - \sigma^2}, \qquad \rho_c(\lambda) = \lambda\big(1 + h_c(\rho_c(\lambda))\big). \tag{96}$$

Proof. Put $d = \lambda - \sigma^2 > 0$ and $\rho = \rho_c(\lambda)$. First,

$$\begin{aligned}\rho - \sigma^2(1 + \sqrt{c})^2 &= d + \frac{c\sigma^2\lambda}{d} - 2\sigma^2\sqrt{c} - c\sigma^2 \\ &= d + \frac{c\sigma^4}{d} - 2\sigma^2\sqrt{c} = \frac{(d - \sigma^2\sqrt{c})^2}{d}.\end{aligned}$$

This proves (95). Hence $\rho > b_c$ under (93).

If $c = 0$, then $\rho_0(\lambda) = \lambda$ and $h_0 \equiv 0$, so the claim is immediate. Assume now $0 < c < 1$. Define

$$s = \frac{c}{d + c\sigma^2}.$$

Then $0 < s < 1/\sigma^2$, and

$$\rho s - c = \lambda \left(1 + c\frac{\sigma^2}{d}\right) \frac{c}{d + c\sigma^2} - c = \frac{c\sigma^2}{d}.$$

Also,

$$\frac{\sigma^2 s}{1 - \sigma^2 s} = \frac{c\sigma^2}{d}.$$

Thus

$$F_\rho(s) := \rho s - c - \frac{\sigma^2 s}{1 - \sigma^2 s} = 0.$$

We must identify this root with the Marchenko–Pastur root $s_c(\rho)$. By Proposition A.8, for $\rho > b_c$, the root corresponding to $s_c(\rho)$ is the root of $F_\rho(s) = 0$ at which $F_\rho'(s) > 0$. Now

$$F_\rho'(s) = \rho - \frac{\sigma^2}{(1 - \sigma^2 s)^2}.$$

Using $1 - \sigma^2 s = d/(d + c\sigma^2)$ and $\rho = \lambda(d + c\sigma^2)/d$, we get

$$F_\rho'(s) > 0 \quad \Longleftrightarrow \quad \lambda d > \sigma^2(d + c\sigma^2) \quad \Longleftrightarrow \quad d^2 > c\sigma^4.$$

This last condition is exactly (93). Hence $s = s_c(\rho)$. Therefore, using $h_c(t) = ts_c(t) - c$,

$$h_c(\rho) = \rho s - c = \frac{c\sigma^2}{d} = c\frac{\sigma^2}{\lambda - \sigma^2}.$$

This is the first identity in (96). The second follows from the definition of ρ, because

$$\frac{\rho}{\lambda} - 1 = c\frac{\sigma^2}{\lambda - \sigma^2} = h_c(\rho).$$

□

We can finally state the proof of Proposition 4.6:

Proof. We first give the main line of the proof. After that, we prove the only technical estimate used in the main line.

Step 1: Put the target spike in the first coordinate. Use a deterministic permutation to move coordinate ℓ to coordinate 1. This only conjugates Λ_p and S_p, so it does not change their eigenvalues. After the permutation, relabel the spike values again as $\lambda_{p,1}, \dots, \lambda_{p,K}$, with the target value in the first place. We write

$$\lambda_p = \lambda_{p,1}, \qquad \lambda_* = \lambda_\ell^*, \qquad \widehat{\lambda}_p = \widehat{\lambda}_{p,\ell}^{\det},$$

$$m = p - K, \qquad c_p = \frac{m}{n} = \frac{p-K}{n}.$$

At the end, we return to the original order of the coordinates.

By Lemma A.1, there are fixed numbers $\nu > 0$, $\varepsilon > 0$, an inner interval J, and a compact interval I such that, for all large p,

$$\lambda_p \in J \subset \operatorname{int} I, \qquad \operatorname{dist}(\lambda_p, \partial I) \geq \nu, \tag{97}$$

$$\operatorname{dist}\big(I, \{\sigma^2, \lambda_{p,2}, \dots, \lambda_{p,K}\}\big) \geq 4\nu, \qquad I \subset (\sigma^2 + \varepsilon, \infty). \tag{98}$$

Write the data matrix as

$$Y_p = \Lambda_p^{1/2} Z_p, \qquad Z_p = \begin{pmatrix} z \\ Z_0 \end{pmatrix},$$

where $z \in \mathbb{F}^{1\times n}$, $Z_0 \in \mathbb{F}^{(p-1)\times n}$, and z is independent of Z_0. Here $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Put

$$D_p = \operatorname{diag}(\lambda_{p,2}, \dots, \lambda_{p,K}, \sigma^2, \dots, \sigma^2).$$

Then

$$S_p = \begin{pmatrix} \lambda_p q_p & b_p^* \\ b_p & C_p \end{pmatrix}, \tag{99}$$

where

$$q_p = \frac{zz^*}{n} = \frac{1}{n}\sum_{k=1}^{n} |z_k|^2, \qquad b_p = \frac{\sqrt{\lambda_p}}{n} D_p^{1/2} Z_0 z^*, \qquad C_p = \frac{1}{n} D_p^{1/2} Z_0 Z_0^* D_p^{1/2}.$$

Let Z_b be the last m rows of Z_0, and set

$$C_b = \frac{\sigma^2}{n} Z_b Z_b^*.$$

Fix $0 < \eta \leq \nu$, and define

$$\begin{aligned} \Omega_p^{(0)} = \{&\|C_p - D_p\|_{\mathrm{op}} \leq \eta\} \\ &\cap \{\|C_b - \sigma^2 I_m\|_{\mathrm{op}} \leq \eta\}, \end{aligned} \tag{100}$$

$$\Omega_p^{(1)} = \{\|S_p - \Lambda_p\|_{\mathrm{op}} \leq \nu\}. \tag{101}$$

The first event depends only on Z_0. By Corollary A.3,

$$\mathbb{P}(\Omega_p^{(0)}) \to 1, \qquad \mathbb{P}(\Omega_p^{(1)}) \to 1. \tag{102}$$

On $\Omega_p^{(0)}$, Lemma A.10 and (98) give

$$\operatorname{dist}(I, \operatorname{Spec}(C_p)) \geq 3\nu, \qquad \sup_{t\in I} \|(tI - C_p)^{-1}\|_{\mathrm{op}} \leq \frac{1}{3\nu}. \tag{103}$$

Also, on this event,

$$\|C_p\|_{\mathrm{op}} \le \|D_p\|_{\mathrm{op}} + \eta \le C. \tag{104}$$

The eigenvalue in (47) was defined by its rank, so it is defined on the whole probability space. On $\Omega_p^{(1)}$, Lemma A.10 gives

$$|\widehat{\lambda}_p - \lambda_p| \le \nu.$$

Thus $\widehat{\lambda}_p \in I$. Every other population eigenvalue is at distance at least 4ν from I, so every other sample eigenvalue is at distance at least 3ν from I. Hence

$$\mathrm{Spec}(S_p) \cap I = \{\widehat{\lambda}_p\} \quad \text{on } \Omega_p^{(1)}. \tag{105}$$

Define the deterministic center

$$\rho_p = \lambda_p \left(1 + c_p \frac{\sigma^2}{\lambda_p - \sigma^2}\right). \tag{106}$$

Because $\lambda_p - \sigma^2 \ge \delta_0$, λ_p is bounded, and $c_p \to 0$, we have $\rho_p - \lambda_p \to 0$. Therefore $\rho_p \in I$ for all large p. Also, the condition of Lemma C.2 holds for all large p, and

$$\rho_p = \lambda_p\{1 + h_{c_p}(\rho_p)\}. \tag{107}$$

Step 2: State the key estimate. On $\Omega_p^{(0)}$, define, for $t \in I$,

$$R_p(t) = (tI - C_p)^{-1}, \qquad B_p(t) = b_p^* R_p(t) b_p.$$

Outside $\Omega_p^{(0)}$, set these quantities equal to zero. We will prove below that

$$\Delta_p := \mathbf{1}_{\Omega_p^{(0)}} \sup_{t \in I} |B_p(t) - \lambda_p h_{c_p}(t)| = o_{\mathbb{P}}(n^{-1/2}). \tag{108}$$

This estimate is uniform in t, so it may later be used at the random point $t = \widehat{\lambda}_p$.

Step 3: Use the Schur equation. Let

$$\Omega_p = \Omega_p^{(0)} \cap \Omega_p^{(1)}.$$

By (102), $\mathbb{P}(\Omega_p) \to 1$. On Ω_p, we have $\widehat{\lambda}_p \in I$ and $I \cap \mathrm{Spec}(C_p) = \varnothing$. The Schur complement formula in Lemma A.11 applied to (99) gives the exact equation

$$\widehat{\lambda}_p = \lambda_p q_p + B_p(\widehat{\lambda}_p) \quad \text{on } \Omega_p. \tag{109}$$

Define a random remainder on the whole probability space by

$$r_p = \begin{cases} B_p(\widehat{\lambda}_p) - \lambda_p h_{c_p}(\widehat{\lambda}_p), & \text{on } \Omega_p, \\ 0, & \text{on } \Omega_p^c. \end{cases}$$

Then $|r_p| \le \Delta_p$ on Ω_p, and therefore

$$r_p = o_{\mathbb{P}}(n^{-1/2}). \tag{110}$$

Subtracting (107) from (109) gives, on Ω_p,

$$\widehat{\lambda}_p - \rho_p = \lambda_p(q_p - 1) + r_p + \lambda_p\{h_{c_p}(\widehat{\lambda}_p) - h_{c_p}(\rho_p)\}. \tag{111}$$

Lemma C.1 gives

$$\sup_{t\in I}|h'_{c_p}(t)| \le Cc_p.$$

Since $\widehat{\lambda}_p, \rho_p \in I$ on Ω_p, the mean value theorem gives a random number ξ_p such that

$$h_{c_p}(\widehat{\lambda}_p) - h_{c_p}(\rho_p) = \xi_p(\widehat{\lambda}_p - \rho_p), \qquad |\xi_p| \le Cc_p$$

on Ω_p. Set $\xi_p = 0$ on Ω_p^c. Then (111) becomes

$$(1-\lambda_p\xi_p)(\widehat{\lambda}_p - \rho_p) = \lambda_p(q_p - 1) + r_p \quad \text{on } \Omega_p. \tag{112}$$

Because λ_p is bounded and $c_p \to 0$, for all large p,

$$|1-\lambda_p\xi_p| \ge \frac{1}{2} \quad \text{on } \Omega_p,$$

and $(1-\lambda_p\xi_p)^{-1} = 1 + o(1)$ uniformly on Ω_p. Also,

$$\sqrt{n}(q_p - 1) = O_{\mathbb{P}}(1), \qquad \sqrt{n}\, r_p = o_{\mathbb{P}}(1).$$

It follows from (112) that

$$\mathbf{1}_{\Omega_p}\left[\sqrt{n}(\widehat{\lambda}_p - \rho_p) - \lambda_p\sqrt{n}(q_p-1)\right] = o_{\mathbb{P}}(1).$$

Since $\mathbb{P}(\Omega_p^c) \to 0$, the indicator can be removed. Hence

$$\sqrt{n}(\widehat{\lambda}_p - \rho_p) = \lambda_p\sqrt{n}(q_p - 1) + o_{\mathbb{P}}(1). \tag{113}$$

Step 4: Apply the one-dimensional central limit theorem. In the real case, $|z_k|^2 = z_k^2$ and $z_k \sim \mathcal{N}(0,1)$. In the complex case, $|z_k|^2 \sim \mathrm{Exp}(1)$. Thus

$$\sqrt{n}(q_p - 1) = \frac{1}{\sqrt{n}}\sum_{k=1}^{n}(|z_k|^2 - 1) \xrightarrow{\text{law}} \mathcal{N}(0,\kappa).$$

Since $\lambda_p \to \lambda_*$, Slutsky's theorem and (113) give

$$\sqrt{n}(\widehat{\lambda}_p - \rho_p) \xrightarrow{\text{law}} \mathcal{N}(0, \kappa\lambda_*^2). \tag{114}$$

Finally,

$$\rho_p = \lambda_p + \frac{p-K}{n}\frac{\sigma^2\lambda_p}{\lambda_p - \sigma^2}.$$

The ratio on the right is bounded by (45)–(46). Therefore

$$\sqrt{n}\left|\rho_p - \lambda_p - \frac{p}{n}\frac{\sigma^2\lambda_p}{\lambda_p-\sigma^2}\right| \le \frac{CK}{\sqrt{n}} \to 0.$$

Together with (114), this proves (48).

It remains to prove (108).

Step 5: Remove the conditional quadratic-form fluctuation. On $\Omega_p^{(0)}$, put

$$A_p(t) = Z_0^* D_p^{1/2} R_p(t) D_p^{1/2} Z_0.$$

Then

$$B_p(t) = \frac{\lambda_p}{n^2} z A_p(t) z^*. \tag{115}$$

Conditionally on Z_0, the matrix $A_p(t)$ is deterministic and Hermitian, while z is an independent standard Gaussian row vector. By Lemma A.9,

$$\begin{aligned} G_p(t) &:= \mathbb{E}_z[B_p(t) \mid Z_0] \\ &= \lambda_p \frac{1}{n} \operatorname{Tr}\{C_p(tI - C_p)^{-1}\}. \end{aligned} \tag{116}$$

Here $\mathbb{E}_z$ means expectation with respect to z, with Z_0 fixed.

On $\Omega_p^{(0)}$, (103)–(104) give

$$\|D_p^{1/2} Z_0\|_{\mathrm{op}}^2 = n\|C_p\|_{\mathrm{op}} \le Cn.$$

Hence, uniformly for $t \in I$,

$$\|A_p(t)\|_{\mathrm{op}} \le Cn, \qquad \operatorname{rank} A_p(t) \le p - 1,$$

and therefore

$$\operatorname{Tr}(A_p(t)^2) \le Cpn^2.$$

Using the variance formula in Lemma A.9, we get

$$\mathbf{1}_{\Omega_p^{(0)}} \operatorname{Var}_z(B_p(t) \mid Z_0) \le C\frac{p}{n^2}.$$

Let $F_p(t) = B_p(t) - G_p(t)$. Since $\Omega_p^{(0)}$ depends only on Z_0, conditional Cauchy–Schwarz and the tower property give, for each fixed $t \in I$,

$$\mathbb{E}\left[\mathbf{1}_{\Omega_p^{(0)}} |F_p(t)|\right] \le C\frac{\sqrt{p}}{n} = o(n^{-1/2}). \tag{117}$$

We now make the estimate uniform in t. On $\Omega_p^{(0)}$,

$$R_p'(t) = -R_p(t)^2,$$

so

$$A_p'(t) = -Z_0^* D_p^{1/2} R_p(t)^2 D_p^{1/2} Z_0, \qquad B_p'(t) = \frac{\lambda_p}{n^2} z A_p'(t) z^*.$$

We do not need to exchange differentiation and conditional expectation. We can compute both sides directly from (116):

$$G_p'(t) = -\lambda_p \frac{1}{n} \operatorname{Tr}\{C_p R_p(t)^2\} = \mathbb{E}_z[B_p'(t) \mid Z_0].$$

Thus

$$F_p'(t) = B_p'(t) - \mathbb{E}_z[B_p'(t) \mid Z_0].$$

The same bounds as above, now with $R_p(t)^2$, give

$$\sup_{t \in I} \mathbb{E}\left[\mathbf{1}_{\Omega_p^{(0)}} |F_p'(t)|\right] \le C\frac{\sqrt{p}}{n}. \tag{118}$$

Fix $t_0 \in I$. The fundamental theorem of calculus gives, on $\Omega_p^{(0)}$,

$$\sup_{t \in I} |F_p(t)| \le |F_p(t_0)| + \int_I |F_p'(u)|\, du.$$

Using (117), (118), Fubini's theorem, and Lemma A.15, we obtain

$$\mathbf{1}_{\Omega_p^{(0)}} \sup_{t \in I} |B_p(t) - G_p(t)| = o_{\mathbb{P}}(n^{-1/2}). \tag{119}$$

Step 6: Remove the other $K-1$ spike coordinates. Write

$$Z_0 = \begin{pmatrix} Z_s \\ Z_b \end{pmatrix}, \qquad Z_s \in \mathbb{F}^{(K-1)\times n}, \qquad Z_b \in \mathbb{F}^{m\times n},$$

and embed C_b into a $(p-1)\times(p-1)$ matrix:

$$\widetilde{C}_b = \mathrm{diag}(0_{K-1}, C_b).$$

The matrix

$$E_p = C_p - \widetilde{C}_b$$

has nonzero entries only in the first $K-1$ rows or the first $K-1$ columns. Hence

$$\mathrm{rank}(E_p) \le 2(K-1).$$

Also, $\|E_p\|_{\mathrm{op}} \le C$ on $\Omega_p^{(0)}$. On the same event, the resolvents of both C_p and $\widetilde{C}_b$ are uniformly bounded for $t \in I$. Therefore Lemma A.14 gives

$$\mathbf{1}_{\Omega_p^{(0)}} \sup_{t\in I} \left| \frac{1}{n} \mathrm{Tr}\{(tI - C_p)^{-1} - (tI - \widetilde{C}_b)^{-1}\} \right| \le \frac{C}{n}. \tag{120}$$

Using

$$C(tI - C)^{-1} = t(tI - C)^{-1} - I$$

for the two matrices of the same size, the identity terms cancel. Since the zero block of $\widetilde{C}_b$ contributes nothing, (120) gives

$$\begin{aligned} \mathbf{1}_{\Omega_p^{(0)}} \sup_{t\in I} \left| \frac{1}{n} \mathrm{Tr}\{C_p(tI - C_p)^{-1}\} - \frac{1}{n} \mathrm{Tr}\{C_b(tI_m - C_b)^{-1}\} \right| \\ \le \frac{C}{n}. \end{aligned} \tag{121}$$

Now apply Lemma C.1 to C_b. We have $m/n = c_p \to 0$ and $I \subset (\sigma^2 + \varepsilon, \infty)$, so

$$\sup_{t\in I} \left| \frac{1}{n} \mathrm{Tr}\{C_b(tI_m - C_b)^{-1}\} - h_{c_p}(t) \right| = o_{\mathbb{P}}(n^{-1/2}).$$

Since $n^{-1} = o(n^{-1/2})$, (121) implies

$$\mathbf{1}_{\Omega_p^{(0)}} \sup_{t\in I} \left| \frac{1}{n} \mathrm{Tr}\{C_p(tI - C_p)^{-1}\} - h_{c_p}(t) \right| = o_{\mathbb{P}}(n^{-1/2}). \tag{122}$$

Combining (116), (119), and (122), and using the boundedness of λ_p, proves (108). This completes the proof.

□

Acknowledgments

This work was supported in part by JSPS KAKENHI Grant Numbers 25H00593, 23K03133 and 26K06828, and by a JSPS Bilateral Joint Research Project 120249936.

ChatGPT (OpenAI) was used to assist with language editing, literature searches, the examination of mathematical arguments during the preparation of this manuscript, and the LaTeX formatting and preparation of the submission files. The authors take full responsibility for the content of the paper.

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